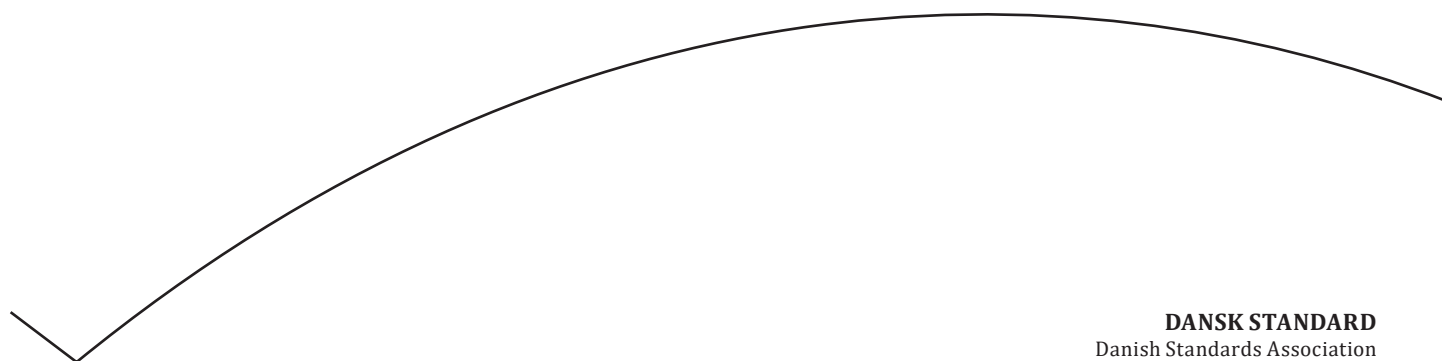




Vindenergisystemer – Del 12-1: Måling af elproducerende vindmøllers ydelse

Wind energy generation systems – Part 12-1: Power performance measurements of electricity producing wind turbines



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(IEC 61400-12-1:2022)**

Systèmes de génération d'énergie éolienne - Partie 12-1:
Mesurages de performance de puissance des éoliennes de
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(IEC 61400-12-1:2022)

Windenergieanlagen - Teil 12-1: Messung des
Leistungsverhaltens von Windenergieanlagen
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The text of document 88/822/CDV, future edition 3 of IEC 61400-12-1, prepared by IEC/TC 88 "Wind energy generation systems" was submitted to the IEC-CENELEC parallel vote and approved by CENELEC as EN IEC 61400-12-1:2022.

The following dates are fixed:

- latest date by which the document has to be implemented at national (dop) 2023-07-10 level by publication of an identical national standard or by endorsement
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Annex ZA **(normative)**

Normative references to international publications with their corresponding European publications

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

NOTE 1 Where an International Publication has been modified by common modifications, indicated by (mod), the relevant EN/HD applies.

NOTE 2 Up-to-date information on the latest versions of the European Standards listed in this annex is available here: www.cenelec.eu.

<u>Publication</u>	<u>Year</u>	<u>Title</u>	<u>EN/HD</u>	<u>Year</u>
IEC 60688	-	Electrical measuring transducers for converting AC and DC electrical quantities to analogue or digital signals	EN 60688	-
IEC 61400-2	-	Wind turbines - Part 2: Small wind turbines	EN 61400-2	-
IEC 61400-12-2	-	Wind energy generation systems - Part 12-2: Power performance of electricity producing wind turbines based on nacelle anemometry	EN IEC 61400-12-2	-
IEC 61400-12-3	-	Wind energy generation systems - Part 12-3: Power performance - Measurement based site calibration	EN IEC 61400-12-3	-
IEC 61400-12-5	-	Wind energy generation systems - Part 12-5: Power performance - Assessment of obstacles and terrain	EN IEC 61400-12-5	-
IEC 61400-50-1	-	Wind energy generation systems - Part 50-1: Wind Measurement - Application of Meteorological Mast, Nacelle and Spinner Mounted Instruments	EN IEC 61400-50-1 ¹	-
IEC 61400-50-2	-	Wind energy generation systems - Part 50-2: Wind measurement - Application of ground-mounted remote sensing technology	EN IEC 61400-50-2	-
IEC 61869-1	-	Instrument transformers - Part 1: General requirements	EN 61869-1	-
IEC 61869-2	-	Instrument transformers - Part 2: Additional requirements for current transformers	EN 61869-2	-
IEC 61869-3	-	Instrument transformers - Part 3: Additional requirements for inductive voltage transformers	EN 61869-3	-

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EN IEC 61400-12-1:2022 (E)

ISO 2533	-	Standard Atmosphere	-	-
ISO/IEC Guide 98-3 2008		Uncertainty of measurement - Part 3: Guide to the expression of uncertainty in measurement (GUM:1995)	-	-



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**Systèmes de génération d'énergie éolienne –
Partie 12-1: Mesurages de performance de puissance des éoliennes de
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**Wind energy generation systems –
Part 12-1: Power performance measurements of electricity producing wind
turbines**

**Systèmes de génération d'énergie éolienne –
Partie 12-1: Mesurages de performance de puissance des éoliennes de
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INTERNATIONAL ELECTROTECHNICAL COMMISSION

WIND ENERGY GENERATION SYSTEMS –

Part 12-1: Power performance measurements of electricity producing wind turbines

FOREWORD

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- 9) Attention is drawn to the possibility that some of the elements of this IEC Publication may be the subject of patent rights. IEC shall not be held responsible for identifying any or all such patent rights.

IEC 61400-12-1 has been prepared by IEC technical committee 88: Wind energy generation systems. It is an International Standard.

This third edition of IEC 61400-12-1 is part of a structural revision that cancels and replaces the performance standards IEC 61400-12-1:2017 and IEC 61400-12-2:2013. The structural revision contains no technical changes with respect to IEC 61400-12-1:2017 and IEC 61400-12-2:2013, but the parts that relate to wind measurements, measurement of site calibration and assessment of obstacle and terrain have been extracted into separate standards.

The purpose of the re-structure was to allow the future management and revision of the power performance standards to be carried out more efficiently in terms of time and cost and to provide a more logical division of the wind measurement requirements into a series of separate standards which could be referred to by other use case standards in the IEC 61400 series and subsequently maintained and developed by appropriate experts.

The text of this International Standard is based on the following documents:

Draft	Report on voting
88/822/CDV	88/867/RVC

Full information on the voting for its approval can be found in the report on voting indicated in the above table.

The language used for the development of this International Standard is English.

This document was drafted in accordance with ISO/IEC Directives, Part 2, and developed in accordance with ISO/IEC Directives, Part 1 and ISO/IEC Directives, IEC Supplement, available at www.iec.ch/members_experts/refdocs. The main document types developed by IEC are described in greater detail at www.iec.ch/standardsdev/publications.

A list of all parts in the IEC 61400 series, published under the general title *Wind energy generation systems*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under webstore.iec.ch in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

IMPORTANT – The "colour inside" logo on the cover page of this document indicates that it contains colours which are considered to be useful for the correct understanding of its contents. Users should therefore print this document using a colour printer.

INTRODUCTION

IEC TC 88 has made this third edition in order to reduce the complexity of the previous edition. Wind measurement procedures have been extracted from the performance standard, recognizing that wind measurements need to be referenced from other standards, such as in loads, noise and resource assessment measurements. IEC TC 88 recommends that the restructured standards gradually take over the previous standards before Maintenance Cycle Reports are written on the restructured standards introducing new technical requirements. Revision of the restructured documents should be proposed at the same time to incorporate such technical changes, recommendations, clarifications and simplifications.

The purpose of the IEC 61400-12 series is to provide a uniform methodology that will ensure consistency, accuracy and reproducibility in the measurement and analysis of power performance by wind turbines. This document has been prepared with the anticipation that it would be applied by:

- a) a wind turbine manufacturer striving to meet well-defined power performance requirements and/or a possible declaration system;
- b) a wind turbine purchaser in specifying such performance requirements;
- c) a wind turbine operator who can be required to verify that stated, or required, power performance specifications are met for new or refurbished units;
- d) a wind turbine planner or regulator who needs to be able to accurately and fairly define power performance characteristics of wind turbines in response to regulations or permit requirements for new or modified installations.

This document provides guidance in the measurement, analysis, and reporting of power performance testing for wind turbines. This document will benefit those parties involved in the manufacture, installation planning and permitting, operation, utilization, and regulation of wind turbines. The technically accurate measurement and analysis techniques recommended in this document should be applied by all parties to ensure that continuing development and operation of wind turbines is carried out in an atmosphere of consistent and accurate communication relative to wind turbine performance. This document presents measurement and reporting procedures expected to provide accurate results that can be replicated by others. Meanwhile, a user of this document should be aware of differences that arise from large variations in wind shear and turbulence. Therefore, a user should consider the influence of these differences and the data selection criteria in relation to the purpose of the test before contracting the power performance measurements.

The wind turbine power performance characteristics are determined by the measured power curve and the estimated annual energy production (AEP). The measured power curve, defined as the relationship between the wind speed and the wind turbine power output, is determined by collecting simultaneous measurements of meteorological variables (including wind speed), as well as wind turbine signals (including power output) at the test site for a period that is long enough to establish a statistically significant database over a range of wind speeds and under varying wind and atmospheric conditions. The AEP is calculated by applying the measured power curve to reference wind speed frequency distributions, assuming 100 % availability.

A key element of power performance testing is the measurement of wind speed. This document specifies the use of cup or sonic anemometers or remote sensing devices (RSD) in conjunction with anemometers to measure wind. Even though suitable procedures for calibration, validation and classification are adhered to, the nature of the measurement principle of these devices can potentially cause them to perform differently. These instruments are robust and have been regarded as suitable for this kind of test with the limitation of some of them to certain classes of terrain.

Recognizing that, as wind turbines become ever larger, a wind speed measured at a single height is increasingly unlikely to accurately represent the wind speed through the entire turbine rotor, this document introduces an additional definition of wind speed. Whereas previously wind speed was defined as that measured at hub height only, this can now be supplemented with a

so-called rotor equivalent wind speed (REWS) defined by an arithmetic combination of simultaneous measurements of wind speed at a number of heights spanning the complete rotor diameter between lower tip and upper tip. The power curves defined by hub height wind speed and REWS are not the same and indeed the wind speed distributions defined by hub height wind speed and REWS are also not the same such that the annual energy production (AEP) is defined by the combination of a measured power curve and wind speed distribution, both of which are based on an identical definition of wind speed.

The technical requirements in this document have been extracted from and are identical to those in IEC 61400-12-1:2017. The corrections in IEC 61400-12-1:2017/Cor.1:2019, IEC 61400-12-1:2017/Cor.2:2020 and IEC 61400-12-1:2017/Cor.3:2021 have been incorporated in this document. Specifically, technical corrections have been applied to Equations (E.8), (E.44) and (E.17). A further technical correction to Equation (E.45) has been made to correct inconsistent units in the components of the summation. Refer to IEC 61400-12:2022 for an overview of the re-structuring of the IEC 61400-12 series and the relationships between different parts of the standard.

Procedures for calibration, classification and uncertainty of cup anemometers and ultrasonic anemometers are given in IEC 61400-50-1. Procedures for calibration, classification and uncertainty of remote sensing devices are given in IEC 61400-50-2. Special care should be taken in the selection of the instruments chosen to measure the wind speed because it can influence the result of the test.

WIND ENERGY GENERATION SYSTEMS –

Part 12-1: Power performance measurements of electricity producing wind turbines

1 Scope

This part of IEC 61400 specifies a procedure for measuring the power performance characteristics of a single wind turbine and applies to the testing of wind turbines of all types and sizes connected to the electrical power network. In addition, this document defines a procedure to be used to determine the power performance characteristics of small wind turbines (as defined in IEC 61400-2) when connected to either the electric power network or a battery bank. The procedure can be used for performance evaluation of specific wind turbines at specific locations, but equally the methodology can be used to make generic comparisons between different wind turbine models or different wind turbine settings when site-specific conditions and data filtering influences are taken into account.

Considerations which can be of relevance to the assessment of uncertainty of power performance tests on multiple turbines are presented in Annex R on an informative basis.

This document defines a measurement methodology that requires the measured power curve and derived energy production figures to be supplemented by an assessment of uncertainty sources and their combined effects. Uncertainty sources of wind measurements are assessed from procedures described in the relevant wind measurement equipment standards while uncertainty of the power curve and annual energy production are assessed by procedures in this document.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60688, *Electrical measuring transducers for converting AC and DC electrical quantities to analogue or digital signals*

IEC 61400-2, *Wind turbines – Part 2: Small wind turbines*

IEC 61400-12-2, *Wind energy generation systems – Part 12-2: Power performance of electricity producing wind turbines based on nacelle anemometry*

IEC 61400-12-3, *Wind energy generation systems – Part 12-3: Power performance – Measurement based site calibration*

IEC 61400-12-5, *Wind energy generation systems – Part 12-5: Power performance – Assessment of obstacles and terrain*

IEC 61400-50-1, *Wind energy generation systems – Part 50-1: Wind measurement – Application of meteorological mast, nacelle and spinner mounted instruments*

IEC 61400-50-2, *Wind energy generation systems – Part 50-2: Wind measurement – Application of ground mounted remote sensing technology*

IEC 61869-1, *Instrument transformers – Part 1: General requirements*

IEC 61869-2, *Instrument transformers – Part 2: Additional requirements for current transformers*

IEC 61869-3, *Instrument transformers – Part 3: Additional requirements for inductive voltage transformers*

ISO 2533, *Standard Atmosphere*

ISO/IEC Guide 98-3:2008, *Uncertainty of measurement – Part 3: Guide to the expression of uncertainty in measurement (GUM:1995)*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

3.1

accuracy

closeness of the agreement between the result of a measurement and a true value of the measurand

3.2

annual energy production

AEP

estimate of the total energy production of a wind turbine over a one-year period by applying the measured power curve to different reference wind speed frequency distributions at hub height, assuming 100 % availability

3.3

atmospheric stability

measure of tendency of the wind to encourage or suppress vertical mixing

Note 1 to entry: Stable atmosphere is characterized by a high temperature gradient with altitude, high wind shear, possible wind veer and low turbulence relative to unstable conditions. Neutral and unstable atmosphere generally result in lower temperature gradients and low wind shear.

3.4

complex terrain

terrain surrounding the test site that features significant variations in topography and terrain obstacles (3.18) that can cause flow distortion

3.5

cut-in wind speed

lowest wind speed at which a wind turbine will begin to produce power

3.6

cut-out wind speed

wind speed at which a wind turbine cuts out from the grid due to high wind speed

3.7

data set

collection of data sampled over a continuous period

3.8

extrapolated power curve

extension of the measured power curve by estimating power output from the maximum measured wind speed to cut-out wind speed

3.9

flow distortion

change in air flow caused by obstacles, topographical variations, or other wind turbines that results in the wind speed at the measurement location being different from the wind speed at the wind turbine location

3.10

hub height

<of a wind turbine> height of the centre of the swept area of the wind turbine rotor above the ground at the tower

Note 1 to entry: For a vertical axis wind turbine the hub height is defined as the height of the centroid of the swept area of the rotor above the ground at the tower.

3.11

maximum power

P_{60}

maximum one-minute average power output a small wind turbine in normal steady-state operation will produce

Note 1 to entry: Peak instantaneous power output can be higher than maximum power.

Note 2 to entry: This definition applies only to small wind turbines as referred to in Annex H.

3.12

measured power curve

table and graph that represent the measured, corrected and normalized net power output of a wind turbine as a function of measured wind speed, measured under a well-defined measurement procedure

3.13

measurement period

period during which a statistically significant database has been collected for the power performance test

3.14

measurement sector

sector of wind directions from which data are selected for the measured power curve

3.15

method of bins

data reduction procedure that groups test data for a certain parameter into intervals (bins)

Note 1 to entry: The method of bins is normally used for wind speed bins but is also applicable to other parameters.

Note 2 to entry: For each bin, the number of data sets or samples and their sum are recorded, and the average parameter value within each bin is calculated.

3.16

net active electric power

measure of the wind turbine electric power output that is delivered to the electrical power network

3.17

normal maintenance

intervention which is done according to a defined regular maintenance program, independent from the fact that a power performance test is being done, for example oil change, blade washing (if due anyway, independent from the power performance test) and any intervention which is outside of the scope of the regular maintenance program (e.g. repair of a failed component) and which is not a machine configuration change

3.18

obstacle

obstruction that blocks the wind and creates distortion of the flow

Note 1 to entry: Buildings and trees are examples of obstacles.

3.19

pitch angle

angle between the chord line at a defined blade radial location (usually 100 % of the blade radius) and the rotor plane of rotation

3.20

power coefficient

ratio of the net electric power output of a wind turbine to the power available in the free stream wind over the rotor swept area

3.21

power performance

measure of the capability of a wind turbine to produce electric power and energy

3.22

rated power

quantity of power assigned, generally by a manufacturer, for a specified operating condition of a component, device or equipment

3.23

reference power

averaged power level in the 11 m/s wind speed bin

Note 1 to entry: This definition applies only to small wind turbines as referred to in Annex H.

3.24

rotor equivalent wind speed

wind speed corresponding to the kinetic energy flux through the swept rotor area when accounting for the variation of the wind speed with height

SEE: Equation (5)

3.25

special maintenance

intervention which is outside of the scope of the regular maintenance program and which is not a machine configuration change, i.e. any intervention which is done in order to improve the power performance during a test period, for example an unscheduled blade washing, any replacement of an essential component

3.26

standard uncertainty

uncertainty of the result of a measurement expressed as a standard deviation

3.27

swept area

<for a horizontal axis wind turbine> projected area of the moving rotor upon a plane normal to the axis of rotation

Note 1 to entry: For teetering rotors, it should be assumed that the rotor remains normal to the low-speed shaft. For a vertical axis wind turbine, the swept area is the projected area of the moving rotor upon a vertical plane.

3.28

test site

location of the wind turbine under test and its surroundings

3.29

uncertainty in measurement

parameter, associated with the result of a measurement, which characterizes the dispersion of the values that could reasonably be attributed to the measurand

3.30

wind measurement equipment

meteorological mast-mounted instruments or remote sensing device

3.31

wind shear

change of wind speed with height across the wind turbine rotor

3.32

wind shear exponent

α

exponent of the power law defining the variation of wind speed with height

Note 1 to entry: This parameter is used as a measure of the magnitude of wind shear for site calibration in Annex C and can be otherwise useful. The power law equation is

$$v_{zi} = v_h \left(\frac{z_i}{H} \right)^\alpha \quad (1)$$

where

v_h is the hub height wind speed;

H is the hub height (m);

v_{zi} is the wind speed at height z_i ;

α is the wind shear exponent.

3.33

wind veer

change of wind direction with height across the wind turbine rotor

4 Symbols, units and abbreviated terms

Symbol or abbreviated term	Description	Unit
A	swept area of the wind turbine rotor	[m ²]
A_i	area of the i -th wind turbine rotor segment	[m ²]
A_w	Weibull scale factor	[m/s]

Symbol or abbreviated term	Description	Unit
AEP	annual energy production	[Wh]
B	barometric pressure	[Pa]
$B_{10\text{min}}$	measured air pressure averaged over 10 min	[Pa]
C_P	assumption of constant power coefficient	
$C_{P,i}$	power coefficient in bin i	
$C_{P,\text{max}}$	maximum power coefficient	
c	sensitivity factor of a parameter (the partial differential)	
$c_{B,i}$	sensitivity factor of air pressure in bin i	[W/Pa]
$c_{d,i}$	sensitivity factor of data acquisition system in bin i	
$c_{k,i}$	sensitivity factor of component k in bin i	
$c_{l,i}$	sensitivity factor of component l in bin i	
$c_{l,j}$	sensitivity factor of component l in bin j	
$c_{m,i}$	sensitivity factor of v_{eq} on wind speed at height m in wind speed bin i	
$c_{n,i}$	sensitivity factor of v_{eq} on wind speed at height n in wind speed bin i	
$c_{\text{RH},i}$	sensitivity factor for relative humidity in bin i	
$c_{T,i}$	sensitivity factor of air temperature in bin i	[W/K]
$c_{V,i}$	sensitivity factor of wind speed in bin i	[Ws/m]
$c_{\rho,i}$	sensitivity factor of air density correction in bin i	[Wm ³ /kg]
D	rotor diameter	[m]
D_e	equivalent rotor diameter	[m]
D_n	rotor diameter of neighbouring and operating wind turbine	[m]
d	meteorological mast diameter	[m]
$F(V)$	the Rayleigh cumulative probability distribution function for wind speed	
f_i	the relative occurrence of wind speed in a wind speed interval bin i	
f_r	wind shear correction factor as calculated in 9.1.3.3	
$f_{r,\text{MM}}$	wind shear correction factor, measured using meteorological mast mounted instruments	
$f_{r,\text{RSD}}$	wind shear correction factor, measured using a remote sensing device	
$f_{r,X,i}$	wind shear correction factor in wind speed bin i	
$f(v)$	wind speed distribution within 10 min period	
H	hub height of wind turbine	[m]
h	height of obstacle	[m]
I_{ref}	reference turbulence intensity (Annex M)	
i	wind speed bin referring to bin averaging of power curve	
k	class number	
k	Weibull shape factor	
L	distance between the wind turbine and the wind measurement equipment	
L_e	distance between the wind turbine or the wind measurement equipment and an obstacle	[m]
L_n	distance between the wind turbine or the wind measurement equipment and a neighbouring and operating wind turbine	[m]

Symbol or abbreviated term	Description	Unit
M	number of uncertainty components in each bin	
M_A	number of category A uncertainty components	
M_B	number of category B uncertainty components	
N	number of bins	
N_h	number of hours in one year $\approx 8\,760$	[h]
N_i	number of 10 min data sets in wind speed bin i	
n	number of available measurement heights ($n \geq 3$)	
n_h	number of available measurement heights	
P_o	porosity of obstacle (0: solid, 1: no obstacle)	
P_i	normalized and averaged power output in bin i	[W]
$P_{I=0}(v)$	zero turbulence power curve	
P_{kin}	kinetic energy in the wind across the rotor height range	
P_n	normalized power output	
$P_{n,i,j}$	normalized power output of data set j in bin i	[W]
P_{rated}	rated power	
$P_{sim}(v)$	simulated 10 min average of measured power output	
P_w	vapour pressure	[Pa]
P_{10min}	measured power averaged over 10 min	[W]
P_{60}	maximum one-minute average power output	
$\overline{P(v)}$	10 min average of measured power output	
$\overline{P_{sim,I}(v)}$	simulated 10 min average of measured power output according to the above condition a): Equation (M.1) applied for the measured wind speed distribution	
$\overline{P_{sim,I_{ref}}(v)}$	simulated 10 min average of measured power output according to the above condition b): Equation (M.1) applied for the measured average wind speed and for the reference turbulence intensity I_{ref} by assuming a Gaussian wind speed distribution.	
R	rotor radius	[m]
R_0	gas constant of dry air (287,05)	[J/kgK]
R_d	distance to meteorological mast centre	[m]
REWS	rotor equivalent wind speed	
R_w	gas constant of water vapour (461,5)	[J/kgK]
RSD	remote sensing device	
r	correlation coefficient	
s	category A standard uncertainty component	
$s_{k,i}$	category A standard uncertainty of component k in bin i	
s_i	combined category A uncertainties in bin i	
$s_{P,i}$	category A standard uncertainty of power in bin i	[W]
s_{sc}	category A standard uncertainty of site calibration	[m/s]
$s_{sc,i}$	category A standard uncertainty of site calibration in bin i	
T	absolute temperature	[K]

Symbol or abbreviated term	Description	Unit
T_i	average temperature in bin i	
$T_{10\text{min}}$	measured absolute air temperature averaged over 10 min	[K]
t	time	[s]
U	wind speed	[m/s]
U_i	wind speed in bin i	[m/s]
u	category B standard uncertainty component	
u_{AEP}	combined standard uncertainty in the estimated annual energy production	[Wh]
$u_{\text{AD},i}$	uncertainty related to the air density bin i	
$u_{\text{AD,method},i}$	uncertainty related to the air density correction bin i	
$u_{\text{B},i}$	category B standard uncertainty of air pressure in bin i	[Pa]
$u_{\text{B,cal},i}$	uncertainty related to the calibration of the pressure sensor bin i	[Pa]
$u_{\text{B,mnt},i}$	uncertainty related to the mounting of the pressure sensor bin i	[Pa]
$u_{\text{c},i}$	combined standard uncertainty of the power in bin i	[W]
$u_{\text{d},i}$	standard uncertainty of the data acquisition system for the full range of a certain measurement channel bin i	
$u_{\text{d,SA},i}$	uncertainty due to system accuracy in bin i	
$u_{\text{d,SC},i}$	uncertainty due to signal conditioning in bin i	
$u_{\text{d,ST},i}$	uncertainty due to signal transmission in bin i	
$u_{\text{d,VT},i}$	uncertainty related to the data acquisition of the signals of the wind speed sensors according to IEC 61400-50-1 bin i	
$u_{\text{dVS},i}$	uncertainty related to the data acquisition of the wind speed signal bin i	
$u_{\text{dB},i}$	uncertainty related to the data acquisition of the signal of the pressure sensor bin i	[Pa]
$u_{\text{dP},i}$	uncertainty component related to the data acquisition of the power signal bin i	
$u_{\text{dRH},i}$	uncertainty related to the data acquisition of the signal from the humidity sensor bin i	[%]
$u_{\text{dT},i}$	uncertainty related to the data acquisition of the signal of the temperature sensor bin i	[°C]
$u_{\text{dWV},i}$	uncertainty related to the data acquisition of the signal from the wind direction sensor (wind vane or sonic anemometer) bin i	
$u_{\text{dWR},i}$	uncertainty related to the data acquisition of the signal from the wind direction sensor (remote sensing device) bin i	
u_i	combined category B uncertainties in bin i	
$u_{f_{\text{r},X},i}$	standard uncertainty of $f_{\text{r},X,i}$ in wind speed bin i	
$u_{k,i}$	category B standard uncertainty of component k in bin i	
$u_{l,i}$	standard uncertainty of component l in bin i	
$u_{l,j}$	standard uncertainty of component l in bin j	
$u_{\text{M},i}$	uncertainty related to the applied methodology bin i	
$u_{\text{M,cc},i}$	uncertainty related to the measurement in a cold climate bin i	
$u_{\text{M,shear},i}$	uncertainty related to missing information regarding shear across the whole rotor bin i	
$u_{\text{M,sfx},i}$	uncertainty related to the unquantifiable seasonal effects on the turbine bin i	

Symbol or abbreviated term	Description	Unit
$u_{M,ti,i}$	uncertainty related to missing information regarding turbulence when no hub-height meteorological mast wind speed signal is available bin i	
$u_{M,tinorm,i}$	uncertainty related to turbulence normalization bin i	
$u_{M,upflow,i}$	uncertainty related to missing information regarding upflow across the whole rotor bin i	
$u_{M,veer,i}$	uncertainty related to missing information regarding wind veer across the whole rotor bin i	
$u_{P,i}$	category B standard uncertainty of power in bin i	[W]
$u_{P,CT,i}$	uncertainty component due to the current transformer bin i	
$u_{P,PT,i}$	uncertainty component due to the power transducer bin i	
$u_{P,VT,i}$	uncertainty component due to the voltage transformer bin i	
$u_{REWS,i}$	uncertainty of rotor equivalent wind speed in bin i	
$u_{REWS.shear,i}$	uncertainty of the REWS wind speed based on the multiple wind speed measurements the REWS bin i	
$u_{REWS.veer,i}$	uncertainty related to the influence of veer on the REWS bin i	
$u_{RH,i}$	category B standard uncertainty of relative humidity in bin i	
$u_{RH,cal,i}$	uncertainty related to the calibration of the humidity sensor bin i	[%]
$u_{RH,mnt,i}$	uncertainty related to the mounting of the humidity sensor bin i	[%]
$u_{T,i}$	category B standard uncertainty of air temperature in bin i	[K]
$u_{T,cal,i}$	uncertainty component of the temperature sensor bin i	[°C]
$u_{T,mnt,i}$	uncertainty related to the mounting of the temperature sensor bin i	[°C]
$u_{T,shield,i}$	uncertainty related to the radiation shielding of the temperature sensor bin i	[°C]
$u_{V,i}$	category B standard uncertainty of wind speed in bin i	[m/s]
$u_{v_h,i}$	standard uncertainty of $v_{h,i}$ in wind speed bin i	
$u_{v_{eq},final,i}$	standard uncertainty of final rotor equivalent wind speed in bin i	
$u_{v_{eq},k,i}$	standard uncertainty sub-component k of v_{eq} in wind speed bin i	
$u_{v_{eq},l,i}$	standard uncertainty sub-component l of v_{eq} in wind speed bin i	
$u_{v_{eq},X,i}$	standard uncertainty of $v_{eq,X,i}$ in bin i	
$u_{v_{eq},X,k,i}$	standard uncertainty component k of rotor equivalent wind speed as measured with the remote sensing device in bin i , to be analysed according to Equation (E.32) bin i	
$u_{VHW,i}$	uncertainty on the hardware used and is one of $u_{VS,i}$, $u_{VR,i}$ or $u_{REWS,i}$ bin i	
$u_{v_{h,X,i}}$	standard uncertainty of $v_{h,X,i}$ bin i	
$u_{v_{h,X,k,i}}$	standard uncertainty component k of hub height wind speed as measured with the remote sensing device in bin i	
$u_{v_M,i}$	uncertainty related to the method applied bin i	
$u_{v_m,k,i}$	standard uncertainty sub-component k of wind speed measurement at height m in wind speed bin i	
$u_{v_n,k,i}$	standard uncertainty sub-component k of wind speed measurement at height n in wind speed bin i	

Symbol or abbreviated term	Description	Unit
$u_{VR,i}$	uncertainties in the wind speed measurement from RSD	
$u_{VR,class}$	uncertainty related to the classification of the RSD bin i	
$u_{VR,class,i}$	uncertainty related to the classification of the remote sensing device according to IEC 61400-50-2	
$u_{VR,flow,i}$	uncertainty related to the flow variation across the measurement volume of the RSD bin i	
$u_{VR,isc,i}$	uncertainty due to the in-situ test bin i	
$u_{VR,mnt,i}$	uncertainty related to the mounting of the RSD bin i	
$u_{VR,mon,i}$	uncertainty related to the monitoring of the RSD bin i	
$u_{VR,ver,i}$	uncertainty due to the verification test bin i	
$u_{VREWS,i}$	uncertainty of the rotor equivalent wind speed (REWS) bin i	
$u_{VREWS,shear,i}$	uncertainty due to the influence on REWS from the shear measurement across the rotor bin i	
$u_{VREWS,veer,i}$	uncertainty related to the classification of the RSD bin i	
u_{VS}	uncertainty related to wind speed (cup or sonic)	
$u_{VS,class,i}$	uncertainty related to the classification of the sensors bin i	
$u_{VS,igt,i}$	uncertainty related to the flow distortion from lightning finial bin i	
$u_{VS,mnt,i}$	uncertainty related to the mounting of the sensors bin i	
$u_{VS,postcal,i}$	uncertainty related to the flow distortion from the terrain bin i	
$u_{VS,precal,i}$	uncertainty on the hardware used and is one of $u_{VS,i}$, $u_{VR,i}$ or $u_{REWS,i}$ bin i	
$u_{VT,i}$	uncertainty related to the flow distortion from the terrain bin i	
$u_{VT,class,i}$	uncertainty related to the classification of the sensors for the site calibration according to IEC 61400-50-1	
$u_{VT,coc,i}$	uncertainty related to calibration of the sensors according to IEC 61400-12-3 before the test	
$u_{VT,coc,i,j}$	uncertainty related to the change of correction value between bins	
$u_{VT,igt,i}$	uncertainty related to a possible top-mounted lightning finial bin i	
$u_{VT,mnt,i}$	uncertainty related to the mounting of the sensors bin i	
$u_{VT,precal,i}$	uncertainty related to the calibration of the sensors before the test bin i	
$u_{VT,postcal,i}$	uncertainty related to the in-situ calibration (Annex K) and/or the post-calibration of the sensor (IEC 61400-50-1) during and/or after the site calibration bin i	
$u_{VT,rmv,i}$	uncertainty related to removal of the wind direction sensor between the site calibration and the power performance test bin i	
$u_{VT,sv,i}$	uncertainty related to seasonal variation between the site calibration and the power performance test bin i	
$u_{v,X,k,i}$	standard uncertainty of component k of either $v_{h,X,i}$ or $v_{eq,X,i}$ bin i	
$u_{WR,i}$	uncertainty related to the wind direction measured with an RSD bin i	
$u_{WR,align,i}$	uncertainty related to the alignment of the RSD bin i	
$u_{WR,fv,i}$	uncertainty related to the flow variation in different probe volumes bin i	
$u_{WR,class,i}$	uncertainty related to the classification of the RSD bin i	
$u_{WR,mda,i}$	uncertainty related to the magnetic declination angle bin i	

Symbol or abbreviated term	Description	Unit
$u_{WR,mon,i}$	uncertainty related to the in-situ monitoring of the signal from the wind direction monitoring by the RSD device bin i	
$u_{WR,ver,i}$	uncertainty related to the verification test of the RSD bin i	
$u_{WV,i}$	uncertainty related to the wind direction measured with a mast mounted wind direction sensor (wind vane or sonic anemometer) bin i	
$u_{WV,bo,i}$	uncertainty related to the boom orientation on which the wind direction sensor is mounted bin i	
$u_{WV,cal,i}$	uncertainty related to the calibration of the wind direction sensor bin i	
$u_{WV,mda,i}$	uncertainty related to the magnetic declination angle bin i	
$u_{WV,nm,i}$	uncertainty related to north marking of the wind direction sensor bin i	
$u_{WV,oe,i}$	uncertainty related to the influence of the mast on the free stream wind direction at the point of measurement bin i	
V	wind speed	[m/s]
V_{ave}	annual average wind speed at hub height	[m/s]
V_i	normalized and averaged wind speed in bin i	[m/s]
V_{eq}	measured equivalent wind speed	
V_n	normalized wind speed	[m/s]
V_{10min}	measured wind speed averaged over 10 min	[m/s]
v	transversal wind speed component	[m/s]
$\bar{v}$	mean flow air speed	[m/s]
v_{eq}	measured equivalent wind speed	[m/s]
$v_{eq,final}$	final rotor equivalent wind speed	[m/s]
$v_{eq,i}$	equivalent wind speed in bin i	
$v_{eq,MM}$	equivalent wind speed based on meteorological mast measurements	[m/s]
$v_{eq,RSD}$	equivalent wind speed based on remote sensing device measurements	[m/s]
$v_{eq,X,i}$	rotor equivalent wind speed as measured with the remote sensing device in bin i	
v_h	hub height wind speed	[m/s]
$v_{h,i}$	wind speed at hub height in bin i	
$v_{h,i}$	wind speed measured at hub height in bin i by cup or sonic anemometer bin i	
$v_{h,MM}$	wind speed measured at hub height with meteorological mast	[m/s]
v_{hn}	hub height wind speed normalized for a specific wind shear profile	[m/s]
$v_{h,RSD}$	wind speed measured at hub height by the remote sensing device	[m/s]
$v_{h,X,i}$	wind speed measured by remote sensing device in bin i	
v_i	wind speed measured at height bin i	[m/s]
$v_{m,i}$	wind speed at height m in wind speed bin i	
v_{rated}	rated wind speed	
v_{zi}	wind speed at height zi	[m/s]
WME	wind measurement equipment	
w	vertical wind speed component	[m/s]
w_i	weighting function to define deviation envelope	

Symbol or abbreviated term	Description	Unit
z	height above ground	[m]
z_i	height of the i -th wind turbine rotor segment	[m]
z_m	wind speed measurements of successive measurement heights	[m]
α	wind shear exponent from power law	[°]
κ	von Karman constant 0,4	
ρ	air density	[kg/m ³]
ρ_i	average air density in bin i	
$\rho_{k,l,i}$	correlation coefficient between the uncertainty component k height m and n in wind speed bin i	
$\rho_{k,l,i,j}$	correlation coefficient between uncertainty component k in bin i and uncertainty component l in bin j	
$\rho_{m,n,k,i}$	correlation coefficient between the uncertainty component k height m and n in wind speed bin i	
$\rho_{u_{v_{eq},X,i},u_{v_{h,X,i}}}$	correlation coefficient between uncertainty components $u_{v_{eq},X,i}$ and $u_{v_{h,X,i}}$ in bin i	
$\rho_{u_{v_{eq,X}},u_{v_{h,X},k,i}}$	correlation coefficient between standard uncertainty components $u_{v_{eq,X}}$ and $u_{v_{h,X},k,i}$ in bin i	
ρ_0	reference air density	[kg/m ³]
ρ_{10min}	derived air density averaged over 10 min	[kg/m ³]
σ	standard uncertainty	
$\sigma_{P,i}$	standard deviation of the normalized power data in bin i	[W]
σ_{10min}	standard deviation of parameter averaged over 10 min	
Φ	relative humidity (range 0 % to 100 %)	
Φ_i	average relative humidity (range 0 % to 100 %) in bin i	
$\Phi_{m,i}$	wind direction at m -th height relative to wind direction at hub height in wind speed bin i	
ϕ_i	angle difference between the wind direction at hub height and segment i	

5 Power performance method overview

The wind shear and wind veer can vary significantly over the rotor height of large wind turbines for atmospheric stability conditions and these are also dependent on topography at the site. The occurrence of extreme atmospheric stability conditions is a site-specific issue, and if these occur during a power performance test the power curve can vary significantly.

The power performance measurement method used in this document is based on a definition of the power curve that expresses power produced versus the wind speed that represents effectively the kinetic energy flux in the wind flowing across the swept area of the rotor.

The kinetic energy flux (referring to a certain point in time or period of time, typically 10 min, assuming that the wind speed does not change within this time across the vertical capture area) is in general terms expressed as:

$$P_{\text{kin}} = \int_A \frac{1}{2} \rho V^3 dA \quad (2)$$

NOTE 1 If the wind speed changes (i.e. if the turbulence intensity is greater than zero) during a certain time period, then the kinetic power (averaged over this time period) is higher than in the case of a constant wind speed, whereas a wind turbine has only a limited possibility to transform this additional kinetic power into additional electric power. This issue is not taken into further consideration here. As a simplification, the Equations (2), (3), (4) are considered valid here, even in the case of a turbulence intensity greater than zero. The impact of wind speed changes on the time averaged kinetic power and the associated impact on the wind turbine power curve are treated by the turbulence normalization procedure as included in Annex M.

Here the wind speed V , measured at a point in space over the rotor area, is the horizontal wind speed.

NOTE 2 Wind turbine power seems to correlate better with the horizontal wind speed definition than with a vector wind speed definition for a one point hub height wind speed measurement.

The horizontal wind speed is defined as the average magnitude of the horizontal component of the instantaneous wind velocity vector, including only the longitudinal and lateral (but not the vertical) components. When we consider a horizontal axis wind turbine, the wind veer is also taken into account and the kinetic energy in the wind is corrected according to the wind direction at hub height:

$$P_{\text{kin}} = \int_A \frac{1}{2} \rho (V \cos(\varphi - \varphi_{\text{hub}}))^3 dA \quad (3)$$

Here φ_{hub} is the wind direction at hub height. The wind veer can vary significantly over the rotor height of large wind turbines for extreme atmospheric stability conditions and it is also dependent on topography at the site.

In this document we do not consider wind shear and wind veer in the horizontal plane. Thus, the energy equivalent wind speed that corresponds to the kinetic energy in the wind as derived from the expression of kinetic energy in Equation (3) in general is described as:

$$V_{\text{eq}} = \left(\frac{1}{A} \int_i (V_i \cos(\varphi_i - \varphi_{\text{hub}}))^3 dA_i \right)^{1/3} \quad (4)$$

Here the index i refers to the height within the rotor area.

NOTE 3 However, when wind speed is mentioned in this document it is by default referring to the hub height wind speed definition unless specifically stated to be this energy equivalent wind speed definition.

Although horizontal wind speed is considered to be the influential wind speed parameter, on sites with significant non-horizontal flow (upflow or downflow), there is additional uncertainty associated with both the measurement of the horizontal wind speed and the response of the wind turbine.

At sites with low and homogeneous wind shear and wind veer over the rotor (and for turbines with small rotor diameters in possibly more complex wind flow conditions), the wind speed measured at hub height can be a good representation of the kinetic energy to be captured by the rotor. Hub height wind speed is the wind speed upon which power curves have historically been defined in all previous editions of IEC 61400-12-1. For that reason, the wind speed measured at hub height is the default definition of wind speed and shall always be measured and reported, even when more comprehensive measurements of wind speed are available over the rotor height.

At sites and seasons where extreme atmospheric stability conditions are expected to be frequent, it is recommended to measure wind shear.

If wind shear and wind veer are not measured over the full height of the rotor there is added uncertainty in the equivalent wind speed. This uncertainty decreases as more wind speed and wind direction measurement heights are used. If measurements are limited to hub height only and there is no measurement of wind shear over the most significant parts of the rotor, then this implies an uncertainty in determination of the equivalent wind speed.

For small wind turbines, where the influence of the wind shear and wind veer are insignificant, the wind speed shall be represented by a hub height wind speed measurement alone without adding uncertainty due to lack of wind shear and wind veer measurements.

NOTE 4 For small turbines refer to IEC 61400-2.

For vertical axis wind turbines, where the influence of the wind veer is not present, the wind veer shall be neglected.

As the wind conditions at the position of the test turbine and at the position of the wind measurement can differ significantly if the test turbine or the wind measurement is located in wakes of any wind turbines, such situations shall be excluded from the test.

The air density ρ also varies over the height of a large wind turbine rotor. However, this variation is small. For practical implementation of the power performance measurement method, it is sufficient to define and determine the air density at hub height only. The power curve is normalized to the average air density at the measurement site over the measurement period or to a pre-defined reference air density.

Power curves are also influenced by the turbulence at the test site, and turbulence can vary over the rotor. In this document only the site turbulence at hub height is considered. High turbulence increases the radius of curvature of the power curve at cut-in and at the start of power regulation at nominal power while low turbulence will make these corners of the power curve sharper. Site turbulence shall be measured and presented as a supplement to the power curve. If needed, a normalization to a specified turbulence can be done using the method of Annex M.

In summary, the power curve according to this document is a climate-specific power curve, where:

- a) the wind speed at a point in space is defined as the horizontal wind speed;
- b) the wind speed of a power curve is defined as the hub height wind speed. This definition may be supplemented with the equivalent wind speed, as defined in Equation (4), taking account of vertical wind shear and wind veer;

NOTE 5 For vertical axis wind turbines the wind veer is omitted in Equation (3) (setting $\varphi_i = \varphi_{\text{hub}}$).

- c) air density is measured at hub height and the power curve is normalized to a site average air density during the measurement period or to a pre-defined reference air density;
- d) turbulence is measured at hub height and the power curve is presented without a turbulence normalization;
- e) the power curve can be normalized to a broader range of climatic conditions (e.g. specific air density, turbulence intensity, vertical shear and veer).

NOTE 6 The power curve normalization is only valid for limited ranges of climatic conditions from the actual site conditions.

In this document, all necessary procedures for measurements, calibration, classification, data correction, data normalization and determination of uncertainties are either provided within this document or referred to in the relevant wind measurement standards. However, if not all parameters are sufficiently measured, then uncertainty shall be applied due to the lack of

measurement. This applies, for example, to the measurement of a power curve of a large wind turbine with only a hub height wind speed sensor. In this case an uncertainty shall be applied for the variability of the wind shear and of the wind veer.

The best results from the use of this document are achieved by measurement of all required parameters and use of all relevant procedures. However, if this is not possible, there are options both for the measurement setup and for the use of the procedures. These options are described in Table 1. The options refer to the use of wind measurement equipment, the applied normalizations, and additional uncertainties connected to the lack of measurements.

Table 1 – Overview of wind measurement configurations for power curve measurements that meet the requirements of this document

Wind measurement configuration	1. Meteorology mast at hub height and remote sensing at all heights	2. Meteorology mast below hub height and remote sensing at all heights	3. Meteorology mast above hub height	4. Meteorology mast at hub height
Typical application	Large wind turbines ^a in flat terrain (see Annex B)	Large wind turbines in flat terrain (see Annex B)	Large and small wind turbines in all types of terrain	Large and small wind turbines in all types of terrain
Wind measurement sensors	IEC 61400-50-1 for cup and sonic anemometers and IEC 61400-50-2 for remote sensing	IEC 61400-50-1 for cup and sonic anemometers and IEC 61400-50-2 for remote sensing	IEC 61400-50-1 for cup and sonic anemometers	IEC 61400-50-1 for cup and sonic anemometers
Normalization procedures for climate-specific power curve determination	Air density, wind shear; 9.1.5 and 9.1.3.4	Air density, wind shear; 9.1.5 and 9.1.3.4	Air density, wind shear; 9.1.5 and 9.1.3.4	Air density; 9.1.5
Additional uncertainty due to lack of wind shear measurement	No additional uncertainty dependent on measurement height coverage; E.11.2.2	No additional uncertainty dependent on measurement height coverage; E.11.2.2	No additional uncertainty dependent on measurement height coverage; E.11.2.2	Additional gross uncertainty for large wind turbines due to lack of vertical wind shear; E.11.2.2
Optional normalization procedures ^b	Turbulence, wind veer and upflow angle; 9.1.6 and 9.1.4	Turbulence, wind veer and upflow angle; 9.1.6 and 9.1.4	Turbulence, wind veer and upflow angle; 9.1.6 and 9.1.4. Meteorological mast flow distortion; 9.1.2. Site calibration; Annex C.	Turbulence and upflow angle; 9.1.6. Site calibration; Annex C.
^a Refer to IEC 61400-2 for definition of large and small turbines.				
^b Upflow influences the power curve and can be measured with 3D sonic anemometers or upflow vanes. If an upflow angle normalization is applied then the method should be documented (uncertainty on upflow is considered in Annex E). However no specific procedure is described in this document on how to normalize for upflow angle.				

6 Preparation for performance test

6.1 General

The specific test conditions related to the power performance measurement of the wind turbine shall be well-defined and documented in the test report, as detailed in Clause 10.

6.2 Wind turbine and electrical connection

As detailed in Clause 10, the wind turbine and electrical connection shall be described and documented to identify uniquely the specific machine configuration that is tested.

6.3 Test site

6.3.1 General

At the test site wind measurement equipment shall be set up in the neighbourhood of the wind turbine to determine the wind speed that drives the wind turbine.

Wind shear and atmospheric stability characteristics of the site can have significant influences on the wind measurement and the actual power performance of the wind turbine. Often there is a diurnal cycle of atmospheric stability, with stable atmosphere forming at night and neutral or unstable atmosphere forming during the day as the sun heats the ground, increasing turbulence and mixing in the boundary layer. Wind shear, wind veer, and turbulence are all a function of atmospheric stability and impact the relationship between the hub height wind speed to the rotor equivalent wind speed and unusual profiles can impact a wind turbine's energy conversion. In addition, flow distortion effects can cause the wind speed at the position of the wind speed measurement and the wind speed at the wind turbine to be different, though correlated.

The test site shall be assessed for sources of wind flow distortion in order to:

- a) choose the position of the wind measurement equipment;
- b) define a suitable measurement sector;
- c) determine if a site calibration is required then determine the appropriate flow corrections by measurement according to Annex C;
- d) evaluate the uncertainty due to wind flow distortion.

The following factors shall be considered, in particular:

- 1) topographical variations and roughness;
- 2) other wind turbines;
- 3) obstacles (buildings, trees, etc.).

The test site shall be documented as detailed in Clause 10.

6.3.2 Location of the wind measurement equipment

Care needs to be taken in locating the wind measurement equipment. The wind measurement equipment shall not be located too close to the wind turbine, since the wind speed will be influenced in front of the wind turbine. Also, it shall not be located too far from the wind turbine, since the correlation between wind speed and electric power output will be reduced. The wind speed measurement instrumentation shall be positioned at a distance from the wind turbine of between 2 and 4 times the rotor diameter D of the wind turbine. A distance of 2,5 times the rotor diameter D is recommended. In the case of a vertical axis wind turbine, refer to Clause H.3.

Prior to carrying out the power performance test and in selecting the location for the wind measurement equipment, account should be taken of the requirement to exclude measurements from all sectors in which either the wind measurement equipment or the wind turbine will be subject to flow disturbance and hence the wind measurement equipment location should be chosen to ensure that valid sectors remain.

In most cases, the best location for the wind measurement equipment will be upwind of the wind turbine in the direction from which most valid wind is expected to come during the test. In other cases, however, it might be more appropriate to place the wind measurement equipment alongside the wind turbine as the wind conditions will be more similar, for example for a wind turbine sited on a ridge.

6.3.3 Measurement sector

The measurement sector(s) shall exclude directions having significant obstacles and other wind turbines, as seen from both the wind turbine under test and the wind measurement equipment.

For all neighbouring wind turbines and significant obstacles, the directions to be excluded due to wake effects shall be determined using the procedure in IEC 61400-12-5. The disturbed sectors to be excluded due to the wind measuring equipment being in the wake of the wind turbine under test are shown in Figure 1 for distances of $2D$, $2,5D$ and $4D$. Reasons to reduce the measurement sector(s) might be special topographic conditions or unexpected measurement data from directions with complicated structures. All reasons for reducing the measurement sector shall be clearly documented.

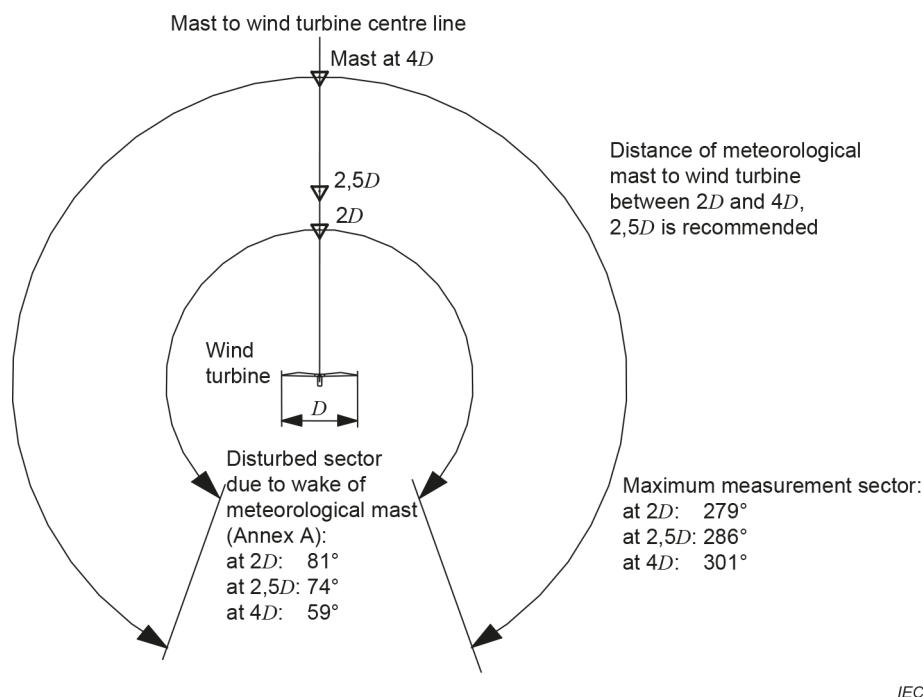


Figure 1 – Requirements as to distance of the wind measurement equipment and maximum allowed measurement sectors

6.3.4 Correction factors and uncertainty due to flow distortion originating from topography

The test site shall be assessed for sources of wind flow distortion due to topographical variations. The assessment in Annex B shall identify whether the power curve can be measured without a site calibration. If the criteria of Annex B are met, the wind flow regime of the site does not need a site calibration. However, in assuming that no flow correction is necessary, the applied uncertainty due to flow distortion of the test site shall be a minimum of 2 % of the measured wind speed if the wind measurement equipment is positioned at a distance between 2 and 3 times the rotor diameter of the wind turbine and 3 % or greater if the distance is between 3 and 4 times the rotor diameter, unless objective evidence can be provided quantifying a different uncertainty.

NOTE These uncertainties were derived from a WAsP (Wind Atlas Analysis and Application Program, DTU Wind Energy) analysis of a Gaussian hill meeting the terrain requirements of Annex B.

If the criteria of Annex B are not met, or a smaller uncertainty due to flow distortion of the test site is desired, then an experimental site calibration shall be undertaken in accordance with Annex C. The measured flow correction factors for each sector shall be used.

7 Test equipment

7.1 Electric power

The net electric power of the wind turbine shall be measured using a power measurement device (e.g. power transducer) and be based on measurements of current and voltage on each phase.

The class of the current transformers shall meet the requirements of IEC 61869-2 and the class of the voltage transformers, if used, shall meet the requirements of IEC 61869-3. They shall be of class 0,5 or better.

The accuracy of the power measurement device, if it is a power transducer, shall meet the requirements of IEC 60688 and shall be class 0,5 or better. If the power measurement device is not a power transducer, then the accuracy should be equivalent to class 0,5 power transducers. The operating range of the power measurement device shall be set to measure all positive and negative instantaneous power peaks generated by the wind turbine. As a guide for megawatt-size active control regulated wind turbines, the full-scale range of the power measurement device should be set to –25 % to +125 % of the wind turbine rated power. In other cases, a higher range can be necessary. This shall be checked on an individual case basis. All data shall be periodically reviewed during the test to ensure that the range limits of the power measurement device have not been exceeded. The power transducer shall be calibrated to traceable standards. The power measurement device shall be mounted between the wind turbine and the electrical connection to ensure that only the net active electric power (i.e. reduced by self-consumption) is measured. It shall be stated whether the measurements are made on the wind turbine side or the network side of the transformer.

7.2 Wind speed

7.2.1 General

The wind speed measured at hub height only (HH) is the default wind speed definition and shall always be used. This can be considered the limiting case of the rotor equivalent wind speed where there is only one measurement height and there is additional uncertainty due to the lack of a wind shear or wind veer profile measurement (see E.11.2.2). It is recommended that the hub height wind speed measurement be supplemented with wind shear measurements in the lower half of the rotor to reduce the wind speed uncertainty. To further reduce the wind speed uncertainty, the rotor equivalent wind speed (REWS), see 9.1.3.2 and Annex Q, should be used as the wind speed input variable to the power curve.

The wind speed measurement configurations are summarized in Table 2, which takes account of the current limitations of each measurement technology with respect to the terrain complexity classification. Remote sensing devices that assume horizontal flow uniformity through the scanned volume limit the application of these technologies to non-complex terrain conditions for power performance testing. Thus, only configurations based on Table 2 shall be applied.

Table 2 – Wind speed measurement configurations (X indicates allowable configuration)

Wind speed measurement	HH	HH	REWS	REWS
Terrain type	Non-complex	Complex	Non-complex	Complex
Hub height meteorological mast	X	X		
Hub height meteorological mast + RSD	X	X	X	
RSD + non-hub height meteorological mast	X		X	
Meteorological mast covering heights above HH + 2/3 R	X	X	X	X

The different possible sensor configurations comprise meteorological mast top-mounted and side-mounted anemometers (see IEC 61400-50-1) and remote sensing devices (see IEC 61400-50-2). These provide measurements of hub height wind speed, rotor equivalent wind speed and wind shear profile. IEC 61400-50-1 describes the general, common requirements and procedures for mast top-mounted and side-mounted cup and sonic anemometers wind measurements. IEC 61400-50-2 describes the general common requirements and procedures for remote sensing wind measurements.

7.2.2 General requirements for meteorological mast mounted anemometers

The following requirements apply to all cup and sonic anemometer applications as described in IEC 61400-50-1.

The arrangement of anemometers for power performance tests shall include a primary hub height anemometer and a control anemometer.

The anemometer shall meet the requirements in Annex I and IEC 61400-50-1 for cup and sonic anemometers. For power performance measurements an anemometer with a class better than 1,7A or 1,7C shall be used. Additionally, in terrain that requires a site calibration, it is recommended that a class better than class 2,5B, 2,5D or 1,7S be used; see Annex I, Annex J and IEC 61400-50-1 regarding classification classes.

The anemometer shall be calibrated according to Annex F and IEC 61400-50-1 before the measurement campaign and, if required, calibrated again after the measurement campaign (post-calibration). It is required to check and document that the anemometer maintains the validity of its calibration throughout the measurement period. This can be achieved by comparing the initial calibration results with the outcome of the post-calibration; alternatively, the in-situ anemometer comparison following the procedure in Annex K and IEC 61400-50-1 is permissible.

Where a post-calibration is carried out, the difference between the regression lines of calibration and post-calibration shall be within $\pm 0,1$ m/s in the range 4 m/s to 12 m/s. Only the calibration before the measurement campaign shall be used for the performance test. Calibration of the anemometer shall be made according to the procedure in Annex F and IEC 61400-50-1. If the maximum difference between the regression lines of calibration and post-calibration is outside of $\pm 0,1$ m/s in the range of 4 m/s to 12 m/s, then the standard uncertainty of the anemometer calibration $u_{VS,precal,i}$ shall be increased (at least to this maximum difference, but not to more than $\pm 0,2$ m/s). If the difference is above $\pm 0,2$ m/s then the in-situ anemometer comparison described in Annex K and IEC 61400-50-1 is to be used to identify the point in time when the deviation in the data occurred and the subsequent faulty data shall be rejected. If the in-situ test cannot determine the point at which the deviation began, then the post-calibration difference is added as an uncertainty.

As an alternative, the in-situ calibration procedure described in Annex K and IEC 61400-50-1 shall be used to check the anemometer integrity throughout the measurement period. In this procedure a control anemometer is used to monitor the primary anemometer. Where a cup anemometer is used as the primary anemometer, then either a cup anemometer or a sonic anemometer may be used as the control anemometer. Where a sonic anemometer is used as the primary anemometer, then the control anemometer shall be a cup anemometer. In the case where a REWS derived power curve is obtained from taller than hub height meteorological mast measurements, there shall be a side-mounted primary anemometer at hub height on the mast with an associated control anemometer satisfying the mounting requirements in Annex G and IEC 61400-50-1.

The uncertainty in wind speed measurement derives from several sources of uncertainty as specified in IEC 61400-50-1. Specifically, the uncertainties in calibration, uncertainty due to operational characteristics (classification of anemometry) and uncertainties due to mounting effects shall all be derived from IEC 61400-50-1.

7.2.3 Top-mounted anemometers

Where wind speed measurements are made with a top-mounted anemometer, the requirements given in IEC 61400-50-1 with respect to mounting apply. The installed height of the sensor above ground level shall be verified by measurement and the measurement method and its standard uncertainty documented. For the purpose of defining ground level, an estimate of the mean elevation over a radius of 2 m around the mast base or 5 m radius around the turbine base can be made. The sensor height measurement uncertainty should exclude the uncertainty

of the ground level estimate. For offshore conditions, ground level should be considered as mean sea level.

NOTE The height measurement can be performed by means of measurement device with a traceable calibration, for example a theodolite able to derive heights from an angle measurement in the vertical plane.

The standard uncertainty of the measurement of the height of the wind speed sensor above the estimated ground level shall be less than or equal to 0,2 m. The control anemometer shall be mounted according to the requirements in IEC 61400-50-1. Where a lightning finial is present and meets the requirements of IEC 61400-50-1 with regard to separation from the top-mounted anemometer and wake interference, then no additional uncertainty due to flow distortion from the finial need be considered.

7.2.4 Side-mounted anemometers

The mounting shall follow the requirement for side-mounted anemometers in IEC 61400-50-1. The installed height of the side-mounted anemometers above ground level (see 7.2.3 for ground level definition and height measurement guidance) shall be verified by measurement and the measurement method and its uncertainty documented. The height measurement standard uncertainty shall be less than or equal to 0,2 m.

An anemometer operating in the wake of the meteorological mast is highly disturbed. Measurements obtained in this condition shall not be used in the power performance analysis.

Correction of side-mounted anemometers for meteorological mast flow distortion is permitted and further described in IEC 61400-50-1. The technical basis for the correction and the effect of the correction shall be documented. The booms shall have identical orientations to ensure similarity of flow distortion between different heights. The meteorological mast and boom design should have similar flow distortion effect at the sensor with a maximum allowed difference in wind speed distortion of 1 % between all different heights. The meteorological mast cross-sectional dimensions should be consistent at each elevation; thus in the case of free-standing meteorological masts where the meteorological mast cross-sectional area is larger at the lower elevations, special care should be taken to follow the recommendations in IEC 61400-50-1. An alternative option is to mount a second anemometer at each measurement height on a separate boom and to limit the measurement sector such that the wind speed measurements do not deviate by more than 1 %.

7.2.5 Remote sensing device (RSD)

Remote sensing devices that assume horizontal flow uniformity through the scanned volume limit the application of these technologies to non-complex terrain conditions for power performance testing as defined by Annex B. The RSD shall be verified before the measurement campaign or in situ according to IEC 61400-50-2. The remote sensing device can be used to measure hub height wind speed, wind shear profile, wind veer and/or the rotor equivalent wind speed based on measurements at more than one height (see 7.2.8). In any case the RSD shall be simultaneously compared with a top-mounted anemometer on a meteorological mast at a height not less than the minimum of the wind turbine rotor lower tip-height or 40 m as defined in IEC 61400-50-2. Requirements on the top-mounted anemometer are identical to those specified in 7.2.3.

The uncertainty of the RSD wind speed measurements shall be derived according to IEC 61400-50-2.

7.2.6 Rotor equivalent wind speed measurement

If the wind speed is measured at three or more heights across the wind turbine rotor as defined in 7.2.8, then the rotor equivalent wind speed can be calculated according to 9.1.3. Note that more than three measurement heights are recommended. There are three options for measuring the rotor equivalent wind speed as described in a) to c) as follows.

- a) Where a hub height top-mounted anemometer satisfying the requirements of 7.2.3 is used together with an RSD satisfying the requirements of 7.2.5 and the terrain meets the requirements of Annex B, then the measurements from the hub height anemometer and RSD are combined to determine the rotor equivalent wind speed according to 9.1.3.
- b) Where an anemometer not at hub height but otherwise satisfying the requirements for top-mounted anemometers of 7.2.3 is used with an RSD satisfying the requirements of 7.2.5 and the terrain meets the requirements in IEC 61400-12-5:2022, Clause 9, then the RSD measurements are used directly to determine the rotor equivalent wind speed according to 9.1.3.
- c) Where a taller than hub height meteorological mast is used with side-mounted measurements distributed across the rotor height, including an anemometer at hub height, then the side-mounted anemometer wind speed measurements may be used directly to measure the rotor equivalent wind speed according to 9.1.3.

7.2.7 Hub height wind speed measurement

There are three options for measuring hub height wind speed as described in a) to c) as follows.

- a) Where a hub height meteorological mast is used, the hub height wind speed measurements shall meet the requirements described in 7.2.3.
- b) If the terrain meets the requirements in IEC 61400-12-5:2022, Clause 9, then the hub height wind speed can be measured with an RSD meeting the requirements of 7.2.5 and noting specifically the requirement to compare the RSD against a simultaneous top-mounted anemometer.
- c) A meteorological mast that is taller than the hub height may be used to better capture the wind speeds across the rotor area. In this case the hub height wind speed shall be measured with a side-mounted sensor on a boom following the requirements described in 7.2.4.

For the hub height definition of wind speed, the lack of knowledge of the vertical wind shear or wind veer across the wind turbine rotor shall be accounted for by adding an uncertainty term according to Annex E based on the estimated or measured wind shear or wind veer. Where only a hub height wind speed measurement is available, an estimated wind shear or wind veer based on site characteristics (e.g. roughness) or prior measurement or modelling at the site (e.g. during a resource assessment campaign) shall be used as input to the uncertainty analysis. Where the hub height wind speed is determined using an RSD or taller than hub height meteorological mast with side-mounted wind speed measurements across the rotor or where below hub height side-mounted instruments are present and satisfying the minimum requirements described in 7.2.8, then wind shear or wind veer derived from the RSD or side-mounted instruments shall be used as input to the uncertainty assessment.

7.2.8 Wind shear measurements

Where wind speed measurements are available over a range of heights, wind shear shall be measured and used for the rotor equivalent wind speed or for wind shear exponent determination.

Wind shear measurements shall either be performed using side-mounted anemometers as described in 7.2.4 or by a single remote sensing instrument as described in 7.2.5. Further specifications on wind shear measurement using remote sensing instruments or meteorological mast measurements are given in IEC 61400-50-2 and IEC 61400-50-1, respectively.

The rotor equivalent wind speed measurement shall include wind speed measurements above hub height. To apply a measurement-based wind shear correction, there shall be at least three wind speed measurements distributed over the rotor swept area. However, to minimize wind speed uncertainty, it is recommended to have as many measurement heights as possible. Measurement heights should be distributed symmetrically around hub height and evenly over the vertical range of the rotor swept area.

The measurement heights shall include the following heights as a minimum:

- a) $H \pm 1,0 \%$;
- b) between $H - R$ and $H - (2/3R)$;
- c) between $H + (2/3R)$ and $H + R$,

where H is the hub height of the wind turbine and R the radius of the rotor swept area, see Figure 2.

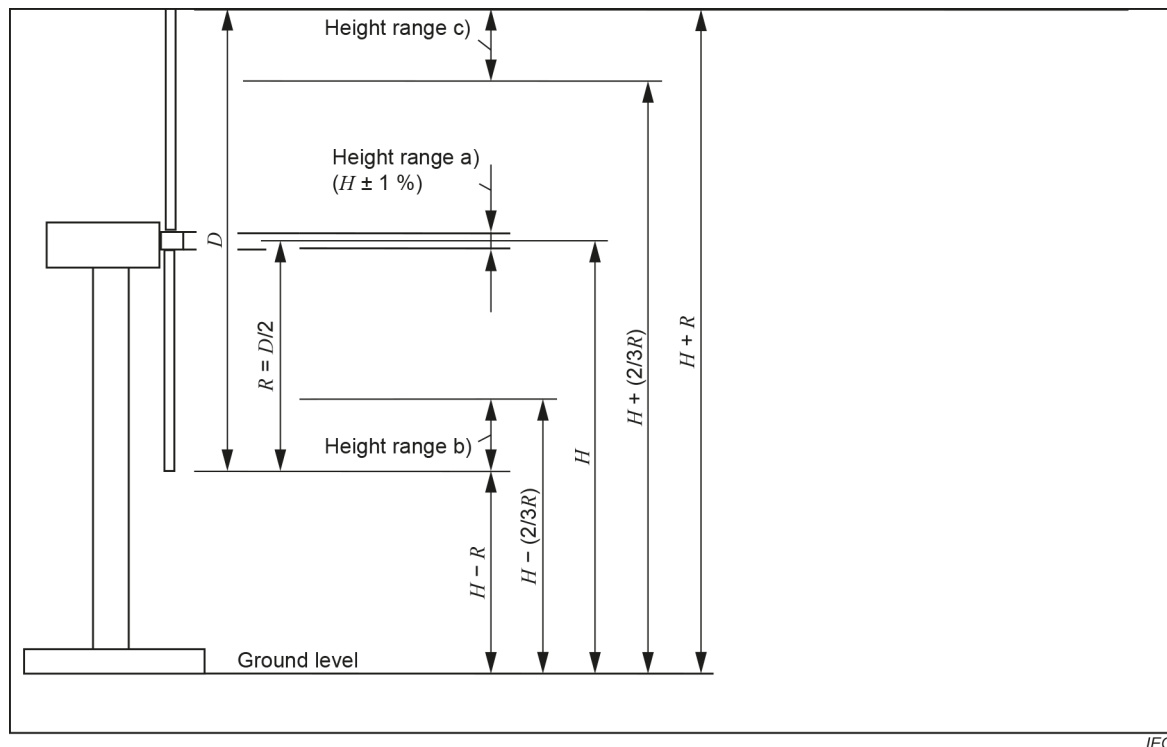


Figure 2 – Wind shear measurement heights appropriate to measurement of rotor equivalent wind speed

If the meteorological mast is at hub height or a little above, then it is possible that no wind speed measurements above hub height will be available for wind shear measurement. In that case the measurements used to derive wind shear shall include at least the following heights:

- a) a side-mounted anemometer as near to hub height as possible whilst still satisfying the requirements in IEC 61400-50-1 for separation from the top-mounted anemometer;
- b) between $H - R$ and $H - (2/3R)$ and satisfying the requirements in IEC 61400-50-1 for side mounted anemometers;

where H is the hub height of the wind turbine and R the radius of the rotor swept area, see Figure 3.

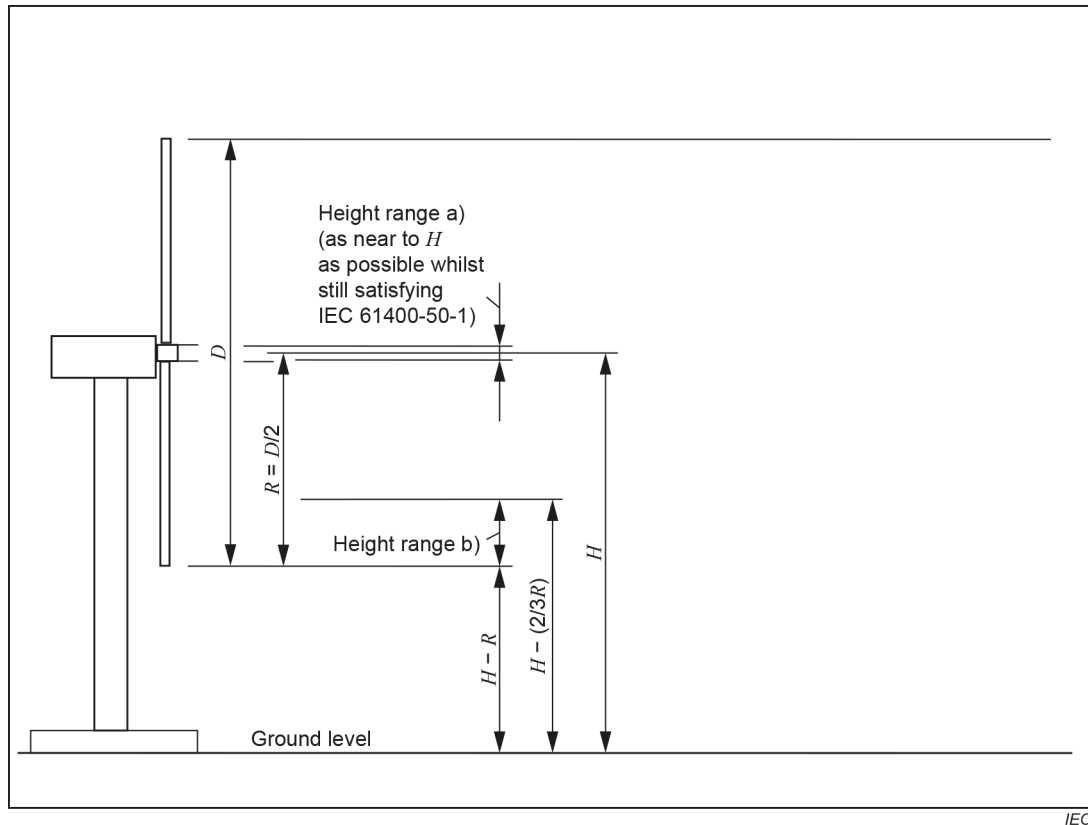


Figure 3 – Wind shear measurement heights when no wind speed measurements above hub height are available (for wind shear exponent determination only)

7.3 Wind direction

Wind direction measurements are used as an input to the site calibration, for filtering data to the valid direction sector and for determining wind veer. Wind direction shall be measured with a wind direction sensor. This may be a wind vane or a 2D or 3D sonic anemometer or an RSD. Where a sonic anemometer is used, it shall be used in conjunction with a conventional wind vane as a control. If an RSD is used, it should be subjected to a verification test on the wind direction according to IEC 61400-50-2.

The instantaneous horizontal wind direction shall be determined and averaged over 10 min. Vector averaging (averaging of cosine and sine components of instantaneous wind direction values taking arctan of the average values and adjusted to the 0° to 360° scale) is one method for deriving the average wind direction. Another method is to extend the wind direction scale for values above 360° and calculating the 10 min average, then adjusting the average value to the 0° to 360° range. Data measured within the dead band of a wind vane, usually at the north mark of the wind direction sensor body, are usually not defined (open circuit or short circuit) and shall be excluded. The combined calibration, operation, and orientation standard uncertainty of the wind direction measurement shall be less than 5°. The wind direction sensor shall be calibrated according to Annex N and IEC 61400-50-1.

7.4 Air density

Air density shall be derived from measurement of air temperature, air pressure and relative humidity. As an alternative to the humidity measurement, an assumed value of 50 % relative humidity may be used if humidity is not measured. The air density shall be calculated using Equation (12) in 9.1.5.

Except in the case where a meteorological mast shorter than hub height is being used in conjunction with a remote sensing device, temperature, barometric pressure and relative

humidity sensors shall be located within 10 m of the hub height on the meteorological mast at a minimum of 1,5 m below the primary anemometer whilst meeting the mounting requirements for other instruments defined in IEC 61400-50-1.

The temperature sensor shall be mounted in a radiation shield.

The barometric pressure sensor may be mounted in a weatherproof box. However, care should be taken to ensure that the box is properly vented so that pressure readings are not influenced by the pressure distribution around the box. Air pressure measurements shall always be corrected to the appropriate hub height according to ISO 2533.

The humidity sensor should be mounted within 10 m of hub height to represent the humidity at the wind turbine rotor centreline.

If the meteorological mast available for the power performance test is shorter than hub height (specifically when a shorter meteorological mast is used in conjunction with a remote sensing device as defined in Annex L), temperature, humidity and pressure sensors shall be located at a height within 1,5 m to 10 m of the primary anemometer. Atmospheric pressure shall be adjusted to hub height in accordance with this document. Furthermore, the air temperature shall be adjusted to hub height assuming that the atmosphere varies according to ISO 2533. Alternatively, a temperature sensor shall be mounted on the wind turbine nacelle. The sensor shall be mounted at least 1 m above the nacelle and upwind of any existing ventilation systems.

7.5 Rotational speed and pitch angle

Rotational speed and pitch angle should be measured throughout the test if there is a specific need for it. For example, if there is a need to apply the measurements in connection with acoustic noise tests. If measured, the measurements shall be reported according to Clause 10.

7.6 Blade condition

The condition of the blades can influence the power curve particularly for stall regulated wind turbines. It can be useful in understanding the characteristics of the wind turbine to monitor the factors that affect blade condition including precipitation, icing and bug and dirt accretion.

7.7 Wind turbine control system

Sufficient status signals shall be identified, verified and monitored to allow the rejection criteria of 8.4 to be applied. Obtaining these parameters from the wind turbine controller's data system is adequate. The definition of each status signal shall be reported.

NOTE A status signal on generator cut-in is adequate to verify cut-out hysteresis control algorithm.

7.8 Data acquisition system

A digital data acquisition system having a sampling rate per channel of at least 1 Hz shall be used to collect measurements and store either sampled data or statistics of the data sets as described in 8.3.

The calibration and accuracy of the data system chain (transmission, signal conditioning and data recording) shall be verified by injecting known signals from a traceable, calibrated source at the transducer ends and comparing these inputs against the recorded readings. As a guideline, the uncertainty of the data acquisition system should be negligible compared with the uncertainty of the sensors.

8 Measurement procedure

8.1 General

The objective of the measurement procedure is to collect data that meet a set of clearly defined criteria to ensure that the data are of sufficient quantity and quality to determine the power performance characteristics of the wind turbine accurately. The measurement procedure shall be documented, as detailed in Clause 10, so that every procedural step and test condition can be reviewed and, if necessary, repeated.

Accuracy of the measurements shall be expressed in terms of standard uncertainty, as described in Annex D. During the measurement period, data should be periodically validated to ensure high quality. Test logs shall be maintained to document all important events during the power performance test.

8.2 Wind turbine operation

During the measurement period, the wind turbine shall be in normal operation, as specified in the wind turbine operations manual, and the machine configuration shall not be changed. The operational status of the wind turbine shall be reported as described in Clause 10. Normal maintenance of the wind turbine shall be carried out throughout the measurement period, but such work shall be noted in the test log. Any special maintenance actions, such as frequent blade washing, which ensure good performance during the test shall in particular be noted. Such special maintenance actions should be avoided.

8.3 Data collection

Data shall be collected continuously at a sampling rate of 1 Hz or higher. Air temperature, air pressure and humidity and precipitation, if measured, may be sampled at a slower rate, but at least once per minute.

The data acquisition system shall store either sampled data or statistics of data sets as follows:

- a) mean value;
- b) standard deviation;
- c) maximum value;
- d) minimum value.

Selected data sets shall be based on 10 min periods derived from contiguous measured data. Data shall be collected until the requirements defined in 8.5 are satisfied.

8.4 Data rejection

To ensure only data obtained during normal operation of the wind turbine are used in the analysis, and to ensure data are not corrupted, data sets shall be excluded from the database under the following circumstances:

- a) external conditions other than wind speed are out of the operating range of the wind turbine;
- b) the wind turbine cannot operate because of a wind turbine fault condition;
- c) the wind turbine is manually shut down or in a test or maintenance operating mode;
- d) failure or degradation (e.g. due to icing) of measurement equipment;
- e) wind direction outside the measurement sector(s) as defined in 6.3.3;
- f) wind directions outside valid (complete) site calibration sectors;
- g) any special atmospheric condition filtered during the site calibration shall also be filtered during the power curve test.

Any other rejection criteria shall be clearly reported.

The effect on the power curve of a large hysteresis loop in the cut-out control algorithm can be considerable. This effect shall not be included in the power curve, and all data sets where the wind turbine has stopped generating power due to cut-out at high wind speed shall be excluded. If cut-out behaviour has been reached during the measurement period, the measurements can be presented in a special database that includes all data points in the database. The power curve shall capture the effect of hysteresis at the cut-in control algorithm, as well as the effect of parasitic losses below cut-in. Cut-out hysteresis affects the higher wind speed bins and neglecting it can therefore lead to overestimation of energy production especially for scenarios with higher annual average wind speeds.

Subsets of the database collected under special operational conditions (e.g. high blade roughness due to dust, salt, insects and ice or if grid conditions vary significantly) or atmospheric conditions (e.g. precipitation, wind shear) that occur during the measurement period may be selected as special databases.

8.5 Database

After data normalization (see 9.1) the selected data sets shall be sorted using the "method of bins" procedure, see 9.2. The wind speed range shall be divided into 0,5 m/s contiguous bins centred on multiples of 0,5 m/s. The selected data sets shall at least cover a wind speed range extending from 1 m/s below cut-in to 1,5 times the wind speed at 85 % of the rated power of the wind turbine. Alternatively, the wind speed range shall extend from 1 m/s below cut-in to a wind speed at which "AEP-measured" is greater than or equal to 95 % of "AEP-extrapolated", see 9.3, where "AEP-measured" and "AEP-extrapolated" are defined using the appropriate, consistent definitions of wind speed (i.e. the hub height wind speed derived power curve and wind speed distribution and, in the case where REWS is derived, by the REWS derived power curve and wind speed distribution). For active pitch controlled wind turbines, the power curve may also be considered complete when the rated power has been reached and the average power does not change by more than the larger of 0,5 % of the power or 5 kW for three consecutive wind speed bins and there is not a trend of increasing power across these three bin-averages. The report shall state which one of the three wind speed range criteria has been used.

The database shall be considered complete when it has met the following criteria:

- a) each bin includes a minimum of 30 min of sampled data;
- b) the database includes a minimum of 180 h of sampled data.

If a single incomplete bin is preventing completion of the test, then that bin value can be estimated by linear interpolation from the two adjacent complete bins.

The database shall be presented in the test report as detailed in Clause 10.

9 Derived results

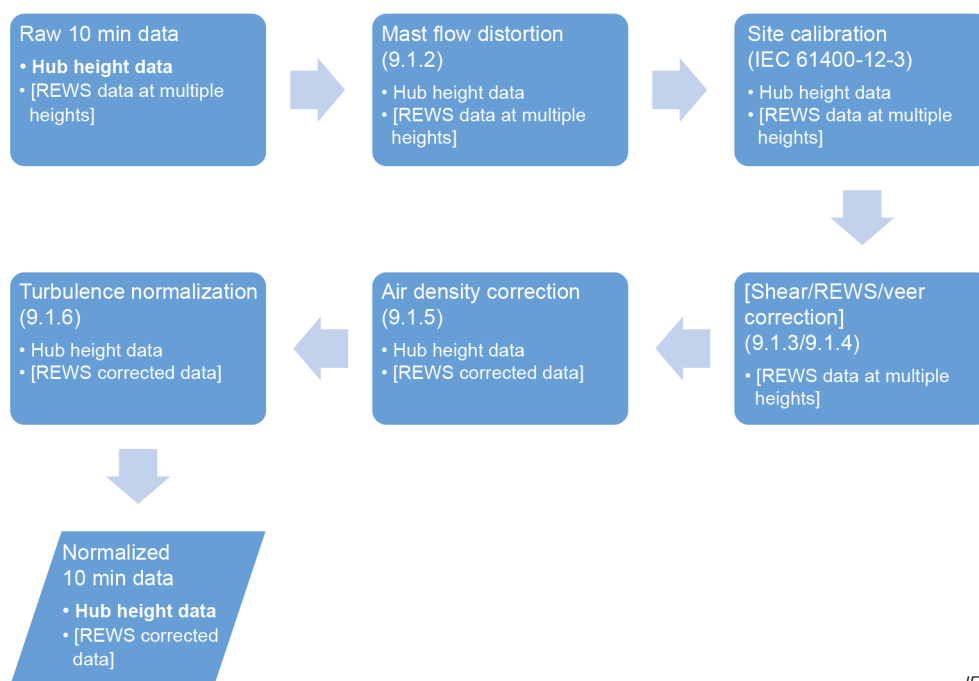
9.1 Data normalization

9.1.1 General

In 9.1.3 to 9.1.6, three methods of normalization are described for the main atmospheric drivers on the power curve results: air density, wind shear, wind veer, and turbulence intensity.

The purpose of these normalizations is to improve the accuracy of the results by means of concrete formulations for each variable. This will to some extent allow the comparison of results from different data sets bringing them to similar scale.

The various normalizations should be applied as indicated in the flow chart in Figure 4.



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NOTE Where the optional REWS measurements are available, the REWS data follow a normalization path parallel to the default hub height data normalizations as indicated by the terms in square brackets. The normalization of the REWS data is carried out separately at each measurement height for all normalization steps up to the step where the results from each height are combined into the wind shear, REWS and veer correction. A single REWS corrected value is derived for each 10 min data point at each subsequent step in parallel to the corrections applied to each hub height data point.

Figure 4 – Process of application of the various normalizations

9.1.2 Correction for meteorological mast flow distortion of side-mounted anemometer

Correction of the wind speeds from side mounted anemometers for meteorological mast flow distortion is permitted (the requirement in Annex S and IEC 61400-50-1 for a maximum of 1 % meteorological mast flow distortion before application of the correction is still required). Any correction method shall be documented and reported following the requirements of Clause 10.

The impact of the meteorological mast flow distortion on the wind shear measurement may be minimized by reducing the measurement sector where meteorological mast flow distortion is below a certain limit. The technical basis for any such reduction in sector width shall be documented. IEC 61400-50-1 gives a possible method for determining the flow distortion correction for a lattice mast.

9.1.3 Wind shear correction (when REWS measurements available)

9.1.3.1 General

If the wind speed over the wind turbine rotor area is constant, the wind speed at hub height would be representative of the wind speed over the wind turbine rotor and the use of the hub height wind speed would be justified. However, the assumption of a point wind speed such as at hub height representing the wind speed over the wind turbine rotor area is not necessarily representative for large wind turbines. It is therefore necessary to introduce corrections that account for the wind speed at hub height and the variations introduced by the wind shear over the wind turbine rotor. Three quantities are defined:

- the rotor equivalent wind speed;
- the wind shear correction factor;
- the wind shear corrected wind speed.

The wind shear correction factor can be used to derive a climate specific power curve as explained in Annex P. However, this correction is based on the assumption that a wind turbine is able to convert all of the available kinetic energy.

9.1.3.2 Rotor equivalent wind speed

The rotor equivalent wind speed is the wind speed corresponding to the kinetic energy flux through the swept rotor area, when accounting for the vertical wind shear. Where the wind speeds for at least three measurement heights are available (see 7.2.6), the rotor equivalent wind speed is defined as:

$$v_{eq} = \left(\sum_{i=1}^{n_h} v_i^3 \frac{A_i}{A} \right)^{1/3} \quad (5)$$

where

n_h is the number of available measurement heights ($n_h \geq 3$);

v_i is the wind speed measured at height i ;

A is the complete area swept by the rotor (i.e. πR^2 with radius R);

A_i is the area of the i -th segment, i.e. the segment for which the wind speed v_i is representative, derived from Equation (6).

The segments (with areas A_i) shall be chosen in such a way that the horizontal separation line between two segments lies in the middle of two measurement points. The segment areas are then derived according to Equation (6):

$$A_i = \int_{z_i}^{z_{i+1}} c(z) dz = g(z_{i+1}) - g(z_i) \quad (6)$$

where

z_i is the height of the i -th segment separation line ($H - R < z_i < H + R$), numbered in the same order as v_i (either top down or bottom up).

The rotor width at height z is:

$$c(z) = 2\sqrt{R^2 - (z - H)^2} \quad (7)$$

where

R is the rotor radius;

H is the hub height.

The integrated function is:

$$g(z) = (z - H) \sqrt{R^2 - (z - H)^2} + R^2 \arctan \left(\frac{z - H}{\sqrt{R^2 - (z - H)^2}} \right) \quad (8)$$

The following is an example of REWS calculation for one 10 min wind speed profile.

In this example, a wind turbine is assumed to have a hub height of 80 m and a rotor diameter of 100 m. The wind speed was measured at five heights with a meteorological mast. If the heights could have been chosen, ideally they would have been evenly distributed (40 m, 60 m, 80 m, 100 m and 120 m). This example shows a case where the heights were fixed independently from the REWS evaluation purpose. The segment limits were set in the middle of two consecutive measurements. The resulting REWS is equal to 9,38 m/s, see Table 3.

Table 3 – Example of REWS calculation

Measurement heights	Wind speed	Segment weighting ^a	Segment inferior limit height (z_i)	Segment superior limit height (z_{i+1})	Segment height
m	m/s	%	m	m	m
116	11,46	16,31	108	130	22
100	10,43	21,04	90	108	18
80	9,24	25,3	70	90	20
60	7,81	23,16	50	70	20
40	6,05	14,24	30	50	20

^a Segment weighting is defined as the ratio between the segment area and the total swept rotor area.

9.1.3.3 Wind shear correction factor

9.1.3.3.1 Case 1: Hub height meteorological mast with remote sensing device or remote sensing device with short (below hub height) meteorological mast

A wind shear correction factor measured using a remote sensing device is defined as the ratio of the rotor equivalent wind speed relative to the wind speed measured at hub height according to Equation (9):

$$f_{r,RSD} = v_{eq,RSD} / v_{h,RSD} \quad (9)$$

where

$v_{eq,RSD}$ is the rotor equivalent wind speed measured by the remote sensing device, as defined in Equation (5);

$v_{h,RSD}$ is the wind speed measured at hub height by the remote sensing device.

9.1.3.3.2 Case 2: Meteorological mast above hub height

A wind shear correction factor measured using a meteorological mast is defined as the ratio of the rotor equivalent wind speed to the wind speed measured at hub height according to Equation (10):

$$f_{r,MM} = v_{eq,MM} / v_{h,MM} \quad (10)$$

where

$v_{eq,MM}$ is the rotor equivalent wind speed measured by the anemometers on the meteorological mast, as defined in Equation (5);

$v_{h,MM}$ is the wind speed measured by the anemometer at hub height.

NOTE $f_{r,MM}$ is for reporting purposes only.

9.1.3.4 Wind shear correction of wind speed

If the hub height wind speed and the wind shear are measured with the same type of WME, the rotor equivalent wind speed is calculated according to Equation (5).

If the hub height wind speed is measured using an anemometer mounted on a meteorological mast and the wind shear measurement with an RSD, the final rotor equivalent wind speed is calculated according to Equation (11):

$$v_{eq,final} = f_{r,RSD} v_{h,MM} \quad (11)$$

9.1.4 Wind veer correction

As explained in Annex Q, the change of wind direction over the rotor height range (wind veer) can have a significant impact on the wind turbine power output. In the case of large wind turbine rotors, it is recommended to apply the extended definition of equivalent wind speed including wind veer.

9.1.5 Air density normalization

The air density shall be determined from measured air temperature, air pressure and relative humidity according to Equation (12):

$$\rho_{10min} = \frac{1}{T_{10min}} \left(\frac{B_{10min}}{R_0} - \Phi P_w \left(\frac{1}{R_0} - \frac{1}{R_w} \right) \right) \quad (12)$$

where

ρ_{10min} is the derived 10 min averaged air density;

T_{10min} is the measured absolute air temperature averaged over 10 min [K];

B_{10min} is the air pressure corrected to hub height averaged over 10 min [Pa];

R_0 is the gas constant of dry air, 287,05 [J/kgK];

Φ is the relative humidity (range 0 % to 100 %);

R_w is the gas constant of water vapour, 461,5 [J/kgK];

P_w is the vapour pressure equal to $0,000\,020\,5 \exp(0,063\,184\,6\,T_{10min})$ [Pa];

Vapour pressure P_w depends on mean air temperature over the 10 min.

The selected data sets shall be normalized to at least one reference air density. The reference air density shall be the average of the measured air density of the valid collected data at the site during the test period (see 8.4), or alternatively a pre-defined nominal air density for the site. The average measured air density shall be rounded to the nearest 0,01 kg/m³ and reported in accordance with Clause 10.

For a stall-regulated wind turbine with constant pitch and constant rotational speed, data normalization shall be applied to the measured power output according to Equation (13):

$$P_n = P_{10\min} \left(\frac{\rho_0}{\rho_{10\min}} \right) \quad (13)$$

where

P_n is the normalized power output;

$P_{10\min}$ is the measured power averaged over 10 min;

ρ_0 is the reference air density.

$\rho_{10\min}$ is the air density derived according to Equation (12) averaged over 10 min.

For a wind turbine with active power control, the normalization shall be applied to the wind speed according to Equation (14):

$$V_n = V_{10\min} \left(\frac{\rho_{10\min}}{\rho_0} \right)^{1/3} \quad (14)$$

where

V_n is the normalized wind speed;

$V_{10\min}$ is the measured wind speed averaged over 10 min.

9.1.6 Turbulence normalization

Wind turbine power curve measurements are influenced by the turbulence intensity. A significant part of the turbulence intensity effect is caused by the averaging of the measured power output and the measured wind speed over 10 min periods. It is recommended to remove this effect from the measurements by normalizing the power curve data to a reference turbulence intensity according to Annex M. The reference turbulence intensity shall be defined prior to the power curve test. It may be defined as a function of the wind speed at hub height. If not defined otherwise, a reference turbulence intensity of 10 % shall be applied. The uncertainty of the turbulence normalization shall be accounted for. If no turbulence normalization of the power curve data is performed, the uncertainty of the power curve due to turbulence effects shall be estimated. As a recommendation use the uncertainty method described in Annex M. Turbulence intensity is measured differently by cup anemometers, remote sensing devices and ultrasonic anemometers. This should be considered when interpreting the results.

9.2 Determination of the measured power curve

The power curve shall be determined based on the hub height wind speed and, if measured, the rotor equivalent wind speed. However, the uncertainty in a power curve derived from hub height measurements only is subject to higher uncertainty due to the lack of knowledge of other influential wind condition parameters. Therefore, to account for the vertical wind shear, wind veer and turbulence intensity and to reduce power curve measurement uncertainty, it is recommended to consider the rotor equivalent wind speed as the representative wind speed and the turbulence normalized power output according to Annex M as the relevant power output. The derivation of power curves based on wind speed normalized for a specific wind shear profile (see Annex P) and the turbulence normalized power output are options that are recommended if a comparison between different power curves is wanted or if the power curves are applied for wind resource assessment.

To indicate single 10 min averages a further index j should be added to the respective wind speeds.

The measured power curve is determined by applying the "method of bins" for the normalized data sets, using 0,5 m/s wind speed bins and by calculation of the mean values of the normalized wind speed and normalized power output for each wind speed bin according to Equations (15) and (16):

$$V_i = \frac{1}{N_i} \sum_{j=1}^{N_i} V_{n,i,j} \quad (15)$$

$$P_i = \frac{1}{N_i} \sum_{j=1}^{N_i} P_{n,i,j} \quad (16)$$

where

V_i is the normalized and averaged wind speed in bin i ;

$V_{n,i,j}$ is the normalized wind speed of data set j in bin i ;

P_i is the normalized and averaged power output in bin i ;

$P_{n,i,j}$ is the normalized power output of data set j in bin i ;

N_i is the number of 10 min data sets in bin i .

The measured power curve shall be presented as detailed in Clause 10 and where REWS power curves are measured, the hub height wind speed power curve shall also be presented. The hub height wind speed power curve shall always be presented for comparison whenever a REWS power curve is measured.

9.3 Annual energy production (AEP)

The AEP shall be calculated in two ways, one designated "AEP-measured", the other "AEP-extrapolated". If the measured power curve does not include data up to cut-out wind speed, the power curve shall be extrapolated from the maximum complete measured wind speed up to cut-out wind speed.

Furthermore, AEP can be defined as generic AEP or as site-specific AEP. Generic AEP is estimated by applying the measured power curve to different reference wind speed frequency distributions. For a specific development, nominal site conditions specifying the wind climate of the site might be known. If so, a site-specific AEP can, additionally, be reported and computed based on this site-specific information.

A power curve derived from hub height wind speed measurements shall only be combined with a wind speed frequency distribution based on the hub height wind speed definition to derive AEP whereas a power curve derived from REWS measurements shall only be combined with a REWS frequency distribution to derive AEP. The AEP derived from combining a REWS power curve with a hub height wind speed frequency distribution (and vice versa) is not a valid calculation. A Rayleigh distribution, which is identical to a Weibull distribution with a shape factor of 2, shall be used as the reference wind speed frequency distribution. AEP estimations shall be made for hub height annual average wind speeds of 4 m/s, 5 m/s, 6 m/s, 7 m/s, 8 m/s, 9 m/s, 10 m/s and 11 m/s according to Equation (17):

$$AEP = N_h \sum_{i=1}^N [F(V_i) - F(V_{i-1})] \left(\frac{P_{i-1} + P_i}{2} \right) \quad (17)$$

where

- AEP is the annual energy production;
- N_h is the number of hours in one year $\approx 8\,760$;
- N is the number of bins;
- V_i is the normalized and averaged wind speed in bin i ;
- P_i is the normalized and averaged power output in bin i .

$$F(V) = 1 - \exp\left(-\frac{\pi}{4}\left(\frac{V}{V_{\text{ave}}}\right)^2\right) \quad (18)$$

where

- $F(V)$ is the Rayleigh cumulative probability distribution function for wind speed;
- V_{ave} is the annual average wind speed;
- V is the wind speed.

The summation is initiated by setting V_{i-1} equal to $V_i - 0,5$ m/s and P_{i-1} equal to 0,0 kW.

For a specific site, nominal site conditions specifying the wind climate of the site might be known. If so, a site-specific AEP can, additionally, be reported and computed based on this site-specific information. If the site-specific wind distribution is known and is present in tabular form, the site-specific AEP can be calculated by first transforming the wind distribution to the corresponding cumulative distribution. The number of hours up to each measured wind speed of the measured power curve will be derived by linear interpolation between two adjacent values of the cumulative distribution. The hours will be divided by the sum of hours in the table, which is typically the nominal number of hours within a calendar year. Finally, the AEP will be calculated by using Equation (17). The tabular wind speed distribution shall use the same definition of wind speed (hub height or REWS) as the measured power curve, otherwise a valid AEP cannot be derived.

If the site-specific wind distribution is presented as a Weibull distribution with known shape and scale factors, then the AEP can be calculated with Equation (17) and substituting the cumulative Rayleigh distribution Equation (18) with the Weibull distribution Equation (19):

$$F(V) = 1 - e^{-\left(\frac{V}{A_w}\right)^k} \quad (19)$$

where

- $F(V)$ is now the Weibull cumulative probability distribution function for wind speed;
- V is the wind speed;
- A_w is the Weibull scale factor;
- k is the Weibull shape factor.

AEP-measured shall be obtained from the measured power curve by assuming zero power for all wind speeds above and below the range of the measured power curve and carrying out the summation according to Equation (17).

AEP-extrapolated shall be obtained from the measured power curve by assuming zero power for all wind speeds below the lowest wind speed in the measured power curve and constant power for wind between the highest wind speed in the measured power curve and the cut-out wind speed. The constant power used for the extrapolated AEP shall be the power value from the bin at the highest wind speed in the measured power curve.

AEP-measured and AEP-extrapolated shall be presented in the test report, as detailed in Clause 10. For all AEP calculations, the availability of the wind turbine shall be set to 100 %. For given annual average wind speeds, estimations of AEP-measured shall be labelled as "incomplete" when calculations show that the AEP-measured is less than 95 % of the AEP-extrapolated.

Estimations of measurement uncertainty in terms of standard uncertainty of the AEP according to Annex D shall be reported for the AEP-measured for all the used wind speed distributions.

The uncertainties in AEP, described above, only deal with uncertainties originating from the power performance test and do not take into account uncertainties due to other important factors relating to actual energy production for a given installation.

9.4 Power coefficient

The power coefficient, C_P , of the wind turbine shall be added to the test results and presented as detailed in Clause 10. C_P shall be determined from the measured power curve according to Equation (20):

$$C_{P,i} = \frac{P_i}{\frac{1}{2} \rho_0 A V_i^3} \quad (20)$$

where

- $C_{P,i}$ is the power coefficient in bin i ;
- V_i is the normalized and averaged wind speed in bin i ; (matching the defined wind speed as either rotor equivalent wind speed or hub height wind speed);
- P_i is the normalized and averaged power output in bin i ;
- A is the swept area of the wind turbine rotor;
- ρ_0 is the reference air density.

10 Reporting format

The test report shall contain the following information.

- a) Identification and description of the specific wind turbine configuration under test (see 6.2), including:
 - 1) wind turbine make, type, serial number, production year;
 - 2) rotor diameter and a description of the verification method used or reference to rotor diameter documentation;
 - 3) rotor speed or rotor speed range;
 - 4) rated power and rated wind speed;
 - 5) blade data: make, type, serial numbers, number of blades, fixed or variable pitch, and pitch angle(s);

- 6) hub height and tower type;
 - 7) description of the control system (device and software version) and documentation of status signals being used for data reduction;
 - 8) description of grid conditions at the wind turbine, i.e. voltage, frequency and their tolerances, and a drawing indicating where the power transducer is connected, specifically in relation to an internal or external transformer and self-consumption of power.
- b) A description of the test site (see 6.3), including:
- 1) photographs of all measurement sectors preferably taken from the wind turbine at hub height;
 - 2) a test site map showing the surrounding area covering a radial distance of at least 20 times the wind turbine rotor diameter and indicating the topography, location of the wind turbine, wind measurement equipment, significant obstacles, other wind turbines, and measurement sector;
 - 3) results of site assessment, i.e. the limits of the valid measurement sector(s);
 - 4) if site calibration is undertaken, the limits of the final measurement sector(s) shall also be reported, including the rationale for any changes from the results of the site assessment;
 - 5) a table of the coordinates and elevation of the test wind turbine, wind measurement equipment and any significant obstacles considered in the assessment of obstacles.
- c) A description of the test equipment (see Clause 7):
- 1) identification of all sensors, wind measurement equipment and data acquisition system, including documentation of calibrations of sensors, wind measurement equipment, transmission lines, and data acquisition system;
 - 2) sketch of the arrangement of the meteorological mast showing dimensions of the mast and instrument mounting, to document conformance with IEC 61400-50-1;
 - 3) description of method to ensure the calibration of the wind measurement equipment was maintained over the duration of the measurement period and documentation of results that show that the calibration is maintained.
- d) A description of the measurement procedure (see Clause 8):
- 1) documentation of the procedural steps, test conditions, sampling rate, averaging time, and measurement period, including:
 - documentation of the calibrations, corrections or transfer functions applied by the data logger and/or by post-processing;
 - 2) a test log book that records all important events during the power performance test, including:
 - a listing of all maintenance activities that occurred during the test;
 - a listing of any special actions (such as blade washing) that were completed to ensure good performance;
 - 3) a complete list of all filter criteria used to produce the reported result, including:
 - the parameter, measurement or time period or combination of parameters being filtered on;
 - the range or logical criteria for the filter;
 - the justification for the filter;
 - the order that the filters were applied shall be reported with the number of points removed each iteration. Alternatively, the number of data points the filter would remove from the database by itself;
 - the starting number of datasets in the database and the final number of datasets after all filters have been applied.
- e) Presentation of measured data (see 8.3 to 8.5). Data from each selected data set shall be presented in both tabular and graphical formats, providing statistics of measured power

output as a function of wind speed and of important meteorological parameters including (where REWS measurements are available, the measured data shall be presented separately for both the hub height data and the REWS data):

- 1) scatter plots of mean, standard deviation, maximum, and minimum power output as a function of wind speed (plots shall include information on sample frequency). An example is shown in Figure 5;
 - 2) scatter plots of mean wind speed and turbulence intensity as a function of wind direction;
 - 3) scatter plots of the turbulence intensity as a function of wind speed, and the average turbulence intensity in each wind speed bin shall be presented;
 - 4) special databases consisting of data collected under special operational or atmospheric conditions should also be presented as described above;
 - 5) if measured, rotational speed and pitch angle shall be presented with a scatter plot including binned values versus wind speed and a table with the binned values;
 - 6) definition of status signals and plots of status signals during the measurement period;
 - 7) scatter plots of air density as a function of wind direction and wind speed including average per bin;
 - 8) scatter plot of the wind shear exponent versus time of day and as function of wind speed. The wind shear exponent in the lower rotor half and in the upper rotor half shall be presented separately. In addition, the average values of both wind shear exponents per wind speed bin shall be represented;
 - 9) the average wind shear exponent or equivalent representation of the site wind shear conditions during the test;
 - 10) wind shear correction factor according to 9.1.3.3, if applicable based on measurement set up (use Equation (9) or Equation (10) depending on measurement configuration);
 - 11) the average air density measured during the test.
- f) Presentation of measured power curve for the reference air density (see 9.1 and 9.2) for the hub height wind speed derived power curve and also the REWS derived power curve if measured:
- 1) the power curve shall be presented in a table similar to Table 4. For each wind speed bin, the table shall list:
 - normalized and averaged wind speed;
 - normalized and averaged power output;
 - number of data sets;
 - calculated C_p value;
 - standard uncertainties of category A (see Annex D and Annex E);
 - standard uncertainties of category B (see Annex D and Annex E);
 - combined standard uncertainty (see Annex D and Annex E);

In addition to presenting the measured power curve in a table according to f) 1), the measured power curve can be presented as bin-centre values. The recommended method is to use a cubic spline for interpolation between the bin-values in the table to represent the measured wind speed, power and power coefficient values at the bin centre.
 - 2) the power curve shall be presented in a graph similar to Figure 6. The graph shall show the following as a function of normalized and averaged wind speed:
 - normalized and averaged power output;
 - combined standard uncertainty;
 - 3) the C_p curve shall be presented in a graph similar to Figure 7;
 - 4) both the graph and the table shall state the reference air density used for the normalization;

- 5) if cut-out wind speed has been reached during the measurement period, the power curve and C_p curve – or the parts of the curves influenced by the cut-out hysteresis – can be presented in a similar way to items f) 1), f) 2), f) 3) and f) 4).
- g) Presentation of measured power curves collected under special operational and atmospheric conditions:
 - 1) power curves derived from subsets of the database for special operational or atmospheric conditions might also be reported. If this is the case, a power curve should be reported in the same way as for the reference air density but with the density derived from the average of the measured air density of the subsets of the database or alternatively using a pre-defined nominal air density for the site.
- h) Presentation of estimated annual energy production for the reference air density (see 9.3) for the hub height wind speed derived power curve and also the REWS derived power curve if measured:
 - 1) a table similar to Table 5 for each annual average hub height wind speed shall include:
 - AEP-measured;
 - standard uncertainty of AEP-measured (see Annex D and Annex E);
 - AEP-extrapolated;
 - 2) the table shall also state:
 - reference air density;
 - cut-out wind speed;
 - 3) if at any annual average wind speed AEP-measured is less than 95 % of AEP-extrapolated, the table shall also include the label "incomplete" in the column of values of AEP-measured;
- i) Presentation of measured power coefficient (see 9.4):
 - 1) measured power coefficient should be presented as a function of wind speed in a table and a graph in which the swept area of the rotor shall be indicated.
- j) Presentation of results of site calibration (see IEC 61400-12-3 for reporting requirements):
 - 1) if a site calibration is undertaken, it shall be presented in the report as a table;
 - 2) the table shall for each wind direction bin present:
 - minimum and maximum wind direction limits;
 - the bin-averaged wind direction;
 - the characteristics of the wind speed correction;
 - number of hours of data;
 - combined standard uncertainty of the wind speed ratio for 6 m/s, 10 m/s and 14 m/s;
 - 3) the report shall furthermore include graphs and tables as required in IEC 61400-12-3.
- k) Uncertainty of measurement (see Annex D):
 - 1) uncertainty assumptions on all uncertainty components shall be provided (mast wind measurement uncertainty shall be derived according to IEC 61400-50-1 and remote sensing wind measurements shall be derived according to IEC 61400-50-2).
- l) Deviations from the procedure:
 - 1) any deviations from the requirements of this document shall be clearly documented in a separate clause. Each deviation shall be supported with the technical rationale and an estimate of its effect on test results.

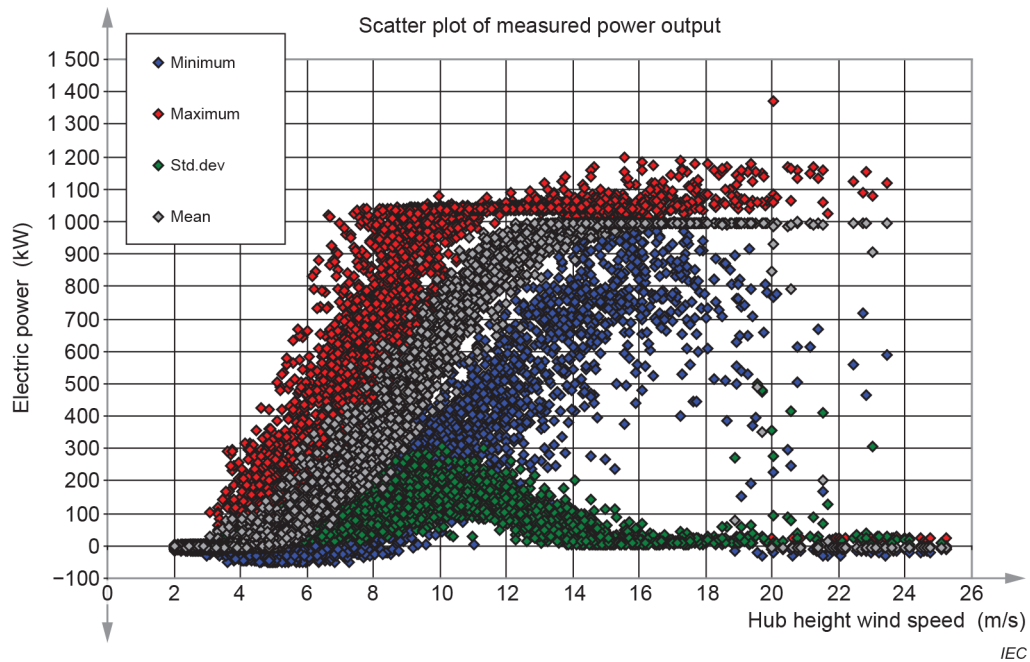


Figure 5 – Presentation of example database: power performance test scatter plot sampled at 1 Hz (mean values averaged over 10 min)

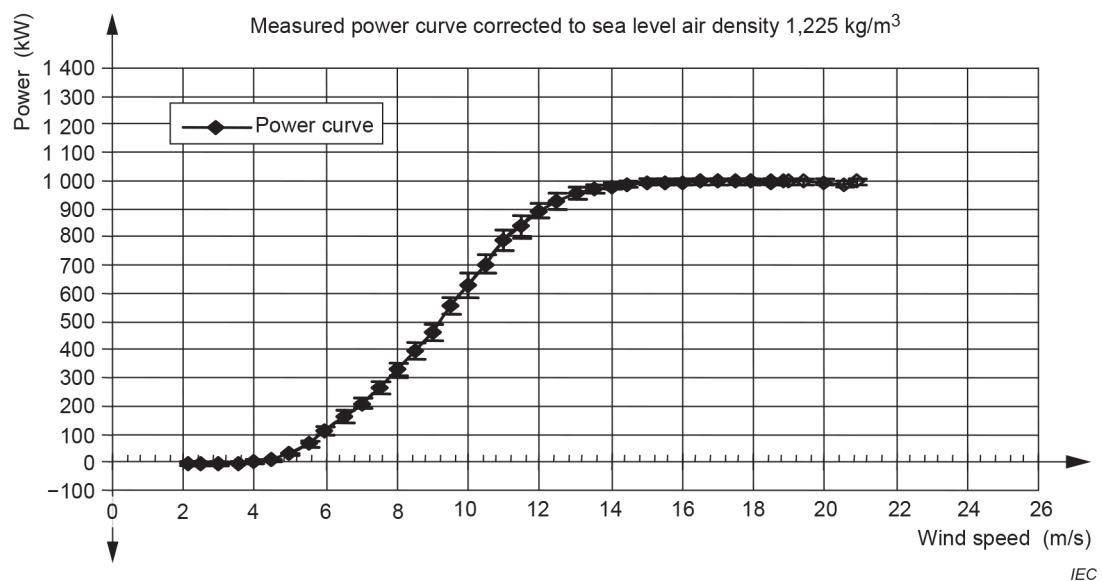


Figure 6 – Presentation of example measured power curve

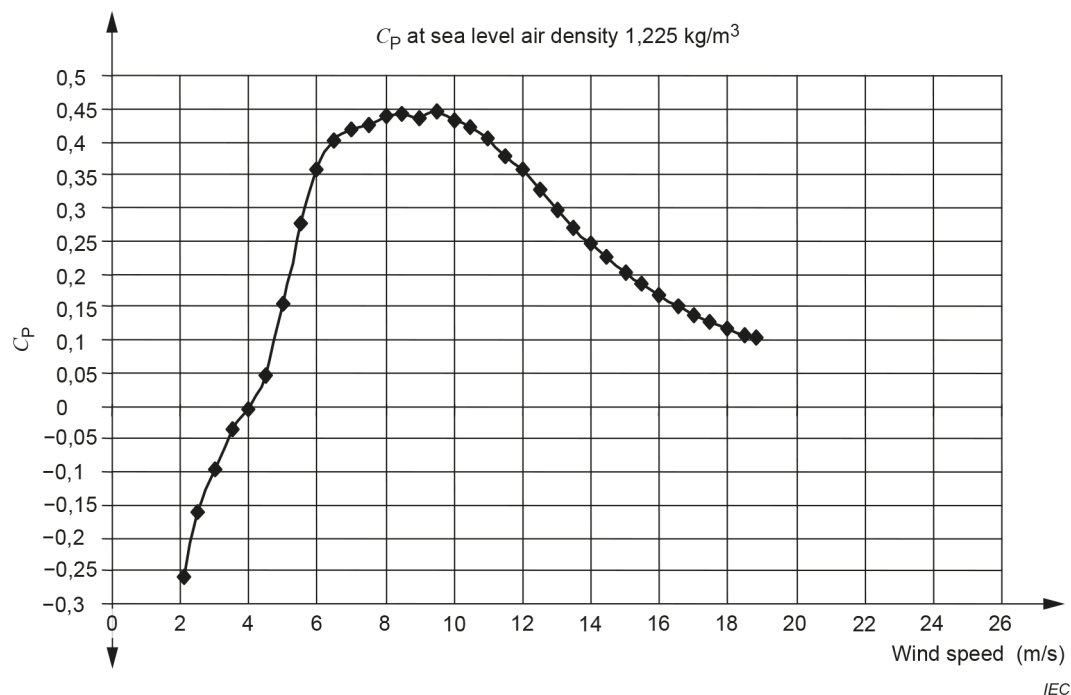


Figure 7 – Presentation of example C_p curve

Table 4 – Example of presentation of a measured power curve

Measured power curve							
Reference air density: 1,225 kg/m ³					Category A	Category B	Combined uncertainty
Bin no.	Hub height wind speed m/s	Power output kW	C_p	No. of data sets (10 min avg.)	Standard uncertainty s_i kW	Standard uncertainty u_i kW	Standard uncertainty u_{ci} kW
4	2,1	–3,6	–0,26	138	0,05	6,3	6,3
5	2,5	–3,6	–0,16	275	0,04	6,3	6,3
6	3,0	–3,8	–0,10	270	0,13	6,3	6,3
7	3,5	–2,2	–0,03	320	0,56	6,3	6,3
8	4,0	–0,4	0,00	347	0,56	6,3	6,3
9	4,5	6,0	0,05	362	0,67	6,3	6,4
10	5,0	27,7	0,15	333	1,09	6,8	6,9
11	5,5	67,4	0,28	285	1,65	10,9	11,0
12	6,0	111,3	0,36	262	2,26	16,1	16,3
13	6,5	160,9	0,40	265	3,08	20,1	20,3
14	7,0	209,4	0,42	286	3,22	20,4	20,7
15	7,5	262,0	0,43	287	3,23	20,7	20,9
16	8,0	327,6	0,44	248	3,28	23,3	23,5
17	8,5	395,2	0,44	215	4,38	28,6	28,9
18	9,0	462,0	0,44	179	4,94	29,8	30,2
19	9,5	556,1	0,45	183	5,02	29,9	30,3
20	10,0	629,8	0,43	133	5,83	41,5	41,9

Measured power curve							
Reference air density: 1,225 kg/m ³					Category A	Category B	Combined uncertainty
Bin no.	Hub height wind speed m/s	Power output kW	C_P	No. of data sets (10 min avg.)	Standard uncertainty s_i kW	Standard uncertainty u_i kW	Standard uncertainty u_{ci} kW
21	10,5	703,1	0,42	127	6,82	32,8	33,5
22	11,0	786,5	0,41	119	6,75	36,1	36,7
23	11,5	836,5	0,38	101	6,65	36,5	37,1
24	12,0	893,5	0,36	94	7,27	25,2	26,2
25	12,5	928,6	0,33	74	5,59	28,8	29,3
26	13,0	956,4	0,30	70	6,38	19,5	20,5
27	13,5	971,3	0,27	63	4,66	16,5	17,1
28	14,0	980,9	0,25	71	3,19	13,5	13,8
29	14,5	988,2	0,22	77	2,53	12,2	12,4
30	15,0	993,5	0,20	64	1,37	11,9	11,9
31	15,5	993,7	0,18	47	0,84	11,6	11,6
32	16,0	995,7	0,17	54	0,83	11,3	11,3
33	16,5	996,2	0,15	33	0,42	11,4	11,4
34	17,0	996,4	0,14	23	0,23	11,3	11,3
35	17,5	996,5	0,13	30	0,24	11,3	11,3
36	18,0	996,5	0,12	13	0,18	11,3	11,3
37	18,5	995,7	0,11	11	0,21	11,3	11,3
38	19,0	996,6	0,10	14	0,59	11,3	11,3
39	19,4	996,1	0,09	10	0,21	11,3	11,3
40	20,0	994,1	0,09	5	0,41	11,3	11,3
41	20,5	987,4 ^a	0,08	2	2,67	11,4	11,7
42	20,9	996,9	0,08	3	3,38	11,8	12,3

^a The dataset in bin 41 is incomplete (less than three data sets); therefore, the power value in bin 41 for AEP calculation is interpolated as 995,7.

Table 5 – Example of presentation of estimated annual energy production

Estimated annual energy production Reference air density: 1,225 kg/m³ Cut-out wind speed 25 m/s (extrapolation by constant power from last bin)					
Hub height annual average wind speed (Rayleigh) m/s	AEP-measured (measured power curve) MWh	Standard uncertainty in AEP MWh	Standard uncertainty in AEP %	AEP- extrapolated (extrapolated power curve) MWh	
4	480	82	17	480	
5	1 081	113	10	1 081	
6	1 824	138	8	1 824	
7	2 595	155	6	2 603	
8	3 305	163	5	3 342	
9	3 889	165	4	3 995	
10	4 318	162	4	4 536	
11	4 592	157	3	4 954	Incomplete

NOTE The standard uncertainty figures in Table 4 and Table 5 are based on a coverage factor of 1. This implies that the level of confidence (percentage of times in repeated power curve measurements the intervals will contain the "true" AEP value) is in the order of 58 % to 68 %. The level of confidence is only an estimate since detailed knowledge of the probability distribution of the measurand is normally not known. The upper value (68 %) applies to normal distributions and the lower value (58 %) applies to rectangular distributions.

Annex A
(normative)

Assessment of obstacles

Refer to IEC 61400-12-5:2022, Clause 6.

Annex B
(normative)

Assessment of terrain at the test site

Refer to IEC 61400-12-5:2022, Clause 9.

Annex C
(normative)

Site calibration procedure

Refer to IEC 61400-12-3.

Annex D (normative)

Evaluation of uncertainty in measurement

Annex D addresses the requirements for the determination of uncertainty in measurement. The theoretical basis for determining the uncertainty using the method of bins, with a worked example of estimating uncertainties, can be found in Annex E.

The measured power curve shall be supplemented with an estimate of the uncertainty of the measurement. The estimate shall be based on ISO/IEC Guide 98-3:2008.

Following ISO/IEC Guide 98-3:2008, there are two categories of uncertainties: category A, the magnitude of which can be deduced from measurements, and category B, which are estimated by other means. In both categories, uncertainties are expressed as standard deviations and are denoted standard uncertainties.

a) The measurands

The measurands are the power curve, determined by the measured and normalized bin values of electric power and wind speed (see 9.1 and 9.2), and the estimated annual energy production (see 9.3). Uncertainties in the measurements are converted to uncertainty in the measurand by means of sensitivity factors.

b) Uncertainty components

Table D.1 provides a minimum list of uncertainty parameters that shall be included in the uncertainty analysis. Some of the uncertainty components shall be found in standards that are connected to this document, as shown in the left column in Table D.1.

Table D.1 – List of uncertainty components

Measured parameter	Uncertainty component	Uncertainty category
Electric power (in accordance with IEC 61869-1, IEC 61869-2, IEC 61869-3 and IEC 60688)	Current transformers	B
	Voltage transformers	B
	Power transducer or power measurement device	B
	Data acquisition system (see Measured parameter = data acquisition system below)	B
	Variability of electric power	A
Wind speed (cup and sonic anemometer, in accordance with IEC 61400-50-1)	Wind tunnel calibration	B
	Classification	B
	Mast flow distortion	B
	Boom flow distortion	B
	Data acquisition system (see Measured parameter = data acquisition system below)	B
	In-situ test	B
Wind speed (remote sensing device, in accordance with IEC 61400-50-2)	Verification test for RSD	B
	In-situ test	B
	RSD classification	B
	Mounting effects	B
	Flow variation across probe volume at same height	B
	Monitoring test	B
Rotor equivalent wind speed	Wind shear measurement	B
	Wind veer measurement	B
Wind direction (vane or sonic, in accordance with IEC 61400-50-1)	Calibration	B
	North mark	B
	Boom orientation	B
	Operation (influence of mast)	B
	Magnetic declination angle	B
	Data acquisition system (see Measured parameter = data acquisition system below)	B
Wind direction (remote sensing device, in accordance with IEC 61400-50-2)	Verification test	B
	Classification	B
	Monitoring test	B
	Flow variation across probe volume at same height	B
	Alignment	B
	Magnetic declination angle	B
	Data acquisition system (see Measured parameter = data acquisition system below)	B
Air temperature	Temperature sensor	B
	Radiation shielding	B
	Mounting effects	B
	Data acquisition system (see Measured parameter = data acquisition system below)	B
Air pressure	Pressure sensor	B
	Mounting effects	B
	Data acquisition system (see Measured parameter = data acquisition system below)	B

Measured parameter	Uncertainty component	Uncertainty category
Relative humidity	Humidity sensor	B
	Mounting effects	B
	Data acquisition system (see Measured parameter = data acquisition system below)	B
Data acquisition system	Signal transmission	B
	System accuracy	B
	Signal conditioning	B
Terrain (no site calibration)	Flow distortion due to terrain	B
Terrain (with site calibration, in accordance with IEC 61400-12-3)	Anemometer calibration before test	B
	Post-calibration / In-situ calibration	B
	Anemometer classification	B
	Mounting effects	B
	Standard mounting	B
	Alternative mounting	B
	Side mounted	B
	Lightning finial	B
	Data acquisition system (see Measured parameter = data acquisition system above)	B
	Change of correction (adjacent wind direction bins)	B
	Removal of wind direction sensor between site calibration and power curve measurement	B
	Seasonal variation	B
	Statistical variability in site calibration	A
Method	Air density correction	B
	Wind conditions – missing shear information	B
	Wind conditions – missing veer information	B
	Wind conditions – missing upflow information	B
	Wind conditions – missing turbulence information	B
	Seasonal effects	B
	Turbulence normalization (or lack of turbulence normalization)	B
	Cold climate measurements	B

NOTE The implicit assumption of the hub height wind speed power curve method of this document is that the 10 min mean power yield from a wind turbine is fully explained by the simultaneous 10 min mean wind speed measured at hub height and the air density. This is not the case. Other flow variables affect power yield and thus identical wind turbines will yield different power at different sites even if the hub height wind speed and air density are the same. These other variables include turbulence fluctuations of wind speed (in three directions), the inclination of the flow vector relative to horizontal, scale of turbulence and wind shear of mean wind speed over the rotor. Currently, analytical tools offer little help in identification of the impact of some of these variables and experimental methods encounter equally serious difficulties. The result is that the power curve will vary from one site to the next which will appear as uncertainty.

This uncertainty stems from differences in observed power yield under different topographical and climatic conditions, i.e. when comparing an AEP measured in homogeneous terrain with an AEP measured at a non-homogeneous wind farm site. Quantification of this apparent uncertainty is difficult. Depending on site conditions and climate, the uncertainty can amount to several percent. In general terms, the uncertainty can be expected to increase with increasing complexity of topography and with increasing frequency of non-neutral atmospheric conditions.

However, this document presents methods to account for at least some of these influential variables (for example wind shear and turbulence), the assessment of the uncertainty and, based on the implicit assumption that these variables are significant, the uncertainty increase in cases where a correction is not applied.

Annex E (informative)

Theoretical basis for determining the uncertainty of measurement using the method of bins

E.1 General

In Annex E all the uncertainty statements made throughout the standard have been combined and added to in order to make a consistent uncertainty evaluation of a power curve measurement, covering both the power curve and the calculated AEP.

In order to achieve a clear structure to introduce this rather complex topic, this Annex E has been structured as shown below. The main structure is based on a separate section for each category of uncertainty components; as wind speed has many uncertainty components these have been given their own clauses, as seen below:

- | | |
|-------------|---|
| Clause E.2 | A general mathematical outline and a table with an overview of all uncertainty components and their default magnitudes. |
| Clause E.3 | Category A uncertainties |
| Clause E.4 | Category B uncertainties – Introduction and data acquisition |
| Clause E.5 | Category B – Power output |
| Clause E.6 | Category B – Wind speed – Introduction and sensors |
| Clause E.7 | Category B – Wind speed – RSD |
| Clause E.8 | Category B – Wind speed – REWS |
| Clause E.9 | Category B – Wind speed – Terrain |
| Clause E.10 | Category B – Air density |
| Clause E.11 | Category B – Method |
| Clause E.12 | Category B – Wind direction |
| Clause E.13 | Combining uncertainties. |

In this Annex E a calculation method is given for aggregating uncertainties, as well as relevant tables for correlation factors and a (non-exhaustive) numerical example.

For all the category B uncertainties a description is given as to what that uncertainty component covers, the symbol used for each component, references to documents or other annexes where the component is discussed and a discussion of the default magnitude.

E.2 Combining uncertainties

E.2.1 General

In its most general form, the combined standard uncertainty of the power in bin i , $u_{c,i}$ can be expressed by:

$$u_{c,i}^2 = \sum_{k=1}^M \sum_{l=1}^M c_{k,i} u_{k,i} c_{l,i} u_{l,i} \rho_{k,l,i,j} \quad (\text{E.1})$$

where

- $c_{k,i}$ is sensitivity factor of component k in bin i ;
- $u_{k,i}$ is the standard uncertainty of component k in bin i ;
- M is the number of uncertainty components in each bin;
- $\rho_{k,l,i,j}$ is the correlation coefficient between uncertainty component k in bin i and uncertainty component l in bin j (in the expression the components k and l are both in bin i).

The uncertainty component is the individual input quantity to the uncertainty of each measured parameter. The combined standard uncertainty in the estimated annual energy production, u_{AEP} , can in its most general form be expressed by

$$u_{\text{AEP}}^2 = N_h^2 \sum_{i=1}^N \sum_{j=1}^N \sum_{k=1}^M \sum_{l=1}^M f_i c_{k,i} u_{k,i} f_j c_{l,j} u_{l,j} \rho_{k,l,i,j} \quad (\text{E.2})$$

where

- f_i is the relative occurrence of wind speed within bin i ;
- N is the number of bins;
- N_h is the number of hours in one year $\approx 8\,760$.

It is seldom possible to deduce explicitly all the values of the correlation coefficients $\rho_{k,l,i,j}$ and normally significant simplifications are necessary. To allow the above expressions of combined uncertainties to be simplified to a practical level, the following assumptions may be made:

- a) uncertainty components are either fully correlated ($\rho_{k,l,i,j} = 1$, implying linear summation to obtain the combined standard uncertainty) or independent ($\rho_{k,l,i,j} = 0$, implying quadratic summation, i.e. the combined standard uncertainty is the square root of summed squares of the uncertainty components);
- b) category A uncertainties are uncorrelated across wind speed bins, while category B uncertainties are fully correlated across wind speed bins;
- c) only relevant for site calibration uncertainties: category A uncertainties are uncorrelated across wind direction bins, while category B uncertainties are fully correlated across wind direction bins.

The correlation of the same type of uncertainty across different wind measurement heights when calculating the uncertainty of the rotor equivalent wind speed or a wind shear and wind veer normalized wind speed shall be assessed individually for each component and each case. Each uncertainty may first be accumulated across the measurement heights for a certain bin of the rotor equivalent wind speed or wind shear and wind veer normalized wind speed. Afterwards the uncertainty components of the rotor equivalent wind speed or the wind shear and wind veer normalized wind speed can be treated as described in bullet points a) and b).

Using these assumptions, the combined uncertainty of the power within a bin, $u_{c,i}$, can be expressed by:

$$u_{c,i}^2 = \sum_{k=1}^{M_A} c_{k,i}^2 s_{k,i}^2 + \sum_{k=1}^{M_B} c_{k,i}^2 u_{k,i}^2 = s_i^2 + u_i^2 \quad (\text{E.3})$$

where

M_A is the number of category A uncertainty components;

M_B is the number of category B uncertainty components;

$s_{k,i}$ is the category A standard uncertainty of component k in bin i ;

s_i is the combined category A uncertainties in bin i ;

u_i is the combined category B uncertainties in bin i .

It should be noted that $u_{c,i}^2$ is not independent of bin size due to the dependency of $s_{p,i}$ on the number of data sets in the bin (see Equation (E.10)).

The assumptions imply that the combined standard uncertainty in energy production, u_{AEP} , is:

$$u_{AEP}^2 = N_h^2 \sum_{i=1}^N f_i^2 \sum_{k=1}^{M_A} c_{k,i}^2 s_{k,i}^2 + N_h^2 \sum_{k=1}^{M_B} \left(\sum_{i=1}^N f_i c_{k,i} u_{k,i} \right)^2 \quad (\text{E.4})$$

The significance of the second term in Equation (E.4) is that each individual category B uncertainty component progresses through to the corresponding AEP uncertainty, applying the assumption of full correlation across bins for the individual components. Finally, the cross-bin combined uncertainty components are added quadratically into a resulting AEP uncertainty.

Furthermore, certain components of category A uncertainty cannot necessarily be conveniently derived or estimated on a bin-wise basis; for example, the site calibration method category A components (IEC 61400-12-3), which could have been derived by a sensitivity analysis on the AEP calculation. In that case, these components should be added quadratically into the resulting AEP uncertainty. Refer to Equation (E.8) for an example of this.

In practice, it might not be convenient to sum category B uncertainty components across the bins before they are individually combined. An approximation, allowing the category B uncertainty components to be combined within bins before they are combined across bins (i.e. s_i and u_i can be used), leads to the more convenient expression:

$$\begin{aligned} u_{AEP}^2 &= N_h^2 \sum_{i=1}^N f_i^2 \sum_{k=1}^{M_A} c_{k,i}^2 s_{k,i}^2 + N_h^2 \left(\sum_{i=1}^N f_i \sqrt{\sum_{k=1}^{M_B} c_{k,i}^2 u_{k,i}^2} \right)^2 \\ &= N_h^2 \sum_{i=1}^N f_i^2 s_i^2 + N_h^2 \left(\sum_{i=1}^N f_i u_i \right)^2 \end{aligned} \quad (\text{E.5})$$

The u_{AEP} , obtained by Equation (E.5) is always equal to or larger than that obtained using Equation (E.4).

E.2.2 Expanded uncertainty

The combined standard uncertainties of the power curve and the AEP can additionally be expressed by expanded uncertainties. Referring to ISO/IEC Guide 98-3 and assuming normal distributions, intervals having levels of confidence shown in Table E.1 can be found by multiplying these combined standard uncertainties by a coverage factor also shown in Table E.1.

Table E.1 – Expanded uncertainties

Level of confidence %	Coverage factor
68,27	1
90	1,645
95	1,960
95,45	2
99	2,576
99,73	3

E.2.3 Basis for the uncertainty assessment

The calculation methodology goes through an estimate of the category A and B uncertainties for each bin of a measured power curve. The uncertainty of the power curve is derived, and finally the uncertainty of the annual energy production (AEP) is estimated. Note that a fully worked out example has not been included, due to the differences in calculations for the various measurement approaches. Where possible a worked-out example for part of the calculation has been included to show how the equations and estimates of default values for uncertainty components have been applied.

The calculation methodology follows ISO/IEC Guide 98-3 and the assumptions made above. Using the combination of the category B uncertainty components according to Equation (E.5), all uncertainty components within each bin can be combined first to express the combined category B uncertainty of each measured parameter, as for example for the wind speed:

$$u_{V,i}^2 = u_{V1,i}^2 + u_{V2,i}^2 + \dots \quad (\text{E.6})$$

where uncertainty components refer to the uncertainty components in Table E.2, using symbols and indices as in Table E.2.

Secondly, the standard uncertainties of the measurands can be expressed by the uncertainties of the measurement parameters in bin i :

$$u_{C,i}^2 = s_{P,i}^2 + u_{P,i}^2 + u_{M,Tinorm,i}^2 + c_{V,i}^2 u_{V,i}^2 + c_{T,i}^2 u_{T,i}^2 + c_{B,i}^2 u_{B,i}^2 + c_{RH,i}^2 u_{RH,i}^2 + c_{V,i}^2 u_{M,i}^2 \quad (\text{E.7})$$

where

- $s_{P,i}$ is the category A standard uncertainty of power in bin i ;
- $u_{P,i}$ is the category B standard uncertainty of power in bin i ;
- $U_{M,Tinorm,i}$ is the uncertainty related to turbulence normalization;
- $u_{V,i}$ is the category B standard uncertainty of wind speed in bin i ;

$u_{T,i}$	is the category B standard uncertainty of temperature in bin i ;
$u_{B,i}$	is the category B standard uncertainty of pressure in bin i ;
$u_{RH,i}$	is the category B standard uncertainty of relative humidity in bin i ;
$u_{M,i}$	is the category B standard uncertainty of the method in bin i ;
$c_{V,i}$	is the sensitivity factor for wind speed in bin i ;
$c_{T,i}$	is the sensitivity factor for temperature in bin i ;
$c_{B,i}$	is the sensitivity factor for pressure in bin i ;
$c_{RH,i}$	is the sensitivity factor for relative humidity in bin i .

This also gives

$$u_{AEP}^2 = N_h^2 \left(\sum_{i=1}^N f_i^2 (s_{P,i}^2) + \left(\sum_{i=1}^N f_i^2 \sqrt{u_{P,i}^2 + u_{M,Tinorm,i}^2 + c_{V,i}^2 u_{V,i}^2 + c_{T,i}^2 u_{T,i}^2 + c_{B,i}^2 u_{B,i}^2 + c_{RH,i}^2 u_{RH,i}^2 + c_{V,i}^2 u_{M,i}^2} \right)^2 \right) \quad (E.8)$$

In Equation (E.8) the uncertainties due to the data acquisition system are part of the uncertainty of each measurement parameter.

The measured power curve, shown in Figure 6 and Table 4, is used in the uncertainty calculation in this Annex E. The results of the uncertainty analysis in the example are also shown in Figure 6, Table 4 and Table 5. All sensitivity factors are listed in Table E.10, and category B uncertainties are listed in Table E.11.

Table E.2 – List of category A and B uncertainties

Category B: Instruments	Note	Uncertainty	Sensitivity	Magnitude
Power output		$u_{P,i}$	$c_{P,i} = 1$	
Current transformers	a	$u_{P,CT,i}$		0,75 %
Voltage transformers	a	$u_{P,VT,i}$		0,5 %
Power transducer or power measurement device	a	$u_{P,PT,i}$		0,5 %
DAQ	d	$u_{dP,i}$		0,1 % to 0,2 %
Wind speed		$u_{V,i}$		
Wind speed measurement		$u_{VHW,i}$		
Wind speed (cup or sonic)		$u_{VS,i}$	For the AEP uncertainty (and for the last bin of the power curve uncertainty) use:	From calibration
Calibration	b	$u_{VS,precal,i}$		
Post-calibration / In-situ calib.	b	$u_{VS,postcal,i}$	$c_{V,i} \approx \frac{P_i - P_{i-1}}{V_i - V_{i-1}}$	0,2 m/s
Classification	c	$u_{VS,class,i}$		1,0 %
Mounting effects			For the power curve uncertainty, use:	
Single top-mounted	d	$u_{VS,mnt,i}$	$c_{V,i} \approx \frac{1}{2} \cdot \left(\frac{(P_{i+1} - P_i)}{(V_{i+1} - V_i)} + \frac{(P_i - P_{i-1})}{(V_i - V_{i-1})} \right)$	0,5 %
Side-by-side mounted	d	$u_{VS,mnt,i}$		1,0 %
Side-mounted	d	$u_{VS,mnt,i}$		1,5 %
Lightning finial	d	$u_{VS,lgt,i}$		0,1 % to 0,2 %
DAQ	d	$u_{dVS,i}$		0,1 % to 0,2 %
Wind speed RSD		$u_{VR,i}$	$c_{V,i}$ (see above)	
Calibration	b, c	$u_{VR,ver,i}$		2 % to 3 %
In-situ check	c	$u_{VR,isc,i}$		See E.7.3
Classification	c	$u_{VR,class,i}$		1,0 % to 1,5 %
Mounting	c	$u_{VR,mnt,i}$		0,1 %
Flow variation in different probe volumes at same height	d	$u_{VR,flow,i}$		2 % to 3 % see E.7.6
Monitoring test	c	$u_{VR,mon,i}$		0,5 %
Rotor equivalent wind speed		$u_{REWS,i}$	$c_{V,i}$ (see above)	
Wind shear	c	$u_{REWS,shear,i}$		See E.8.2
Wind veer	c	$u_{REWS,veer,i}$		See E.8.3
Wind speed – Terrain effects		$u_{VT,i}$	$c_{V,i}$ (see above)	
Without site calibration				
Flow distortion due to terrain	d	$u_{VT,i}$	$c_{V,i}$ (see above)	2 % or 3 % (offshore 1 % or 2 %)
With site calibration				
Anemometer calibration	b	$u_{VT,precal,i}$		From calibration
Post-calibration / In-situ calib.	b	$u_{VT,postcal,i}$		0,2 m/s
Anemometer classification	c	$u_{VT,class}$		1,0 %

Mounting effects				
Single top-mounted	d	$u_{VT,mnt,i}$		0,5 %
Side-by-side mounted	d	$u_{VT,mnt,i}$		1,0 %
Side-mounted	d	$u_{VT,mnt,i}$		1,5 %
Lightning finial	d	$u_{VT,igt,i}$		0,1 % to 0,2 %
DAQ	d	$u_{dVT,i}$		0,1 % to 0,2 %
Change of correction (adjacent wind direction bins)	c	$u_{VT,coc,i}$		See E.9.8
Removal of wind direction sensor between site calibration and power curve measurement	c	$u_{VT,rmv,i}$		See E.9.9
Seasonal variation	c	$u_{VT,sv,i}$	See E.9.10	
Air density		$u_{AD,i}$		
Temperature		$u_{T,i}$	$c_{T,i}$ (see E.10.2)	
Temperature sensor	a, b	$u_{T,cal,i}$		0,4 K to 0,6 K
Radiation shielding	c, d	$u_{T,shield,i}$		1,5 K to 2,5 K
Mounting effects	c, d	$u_{T,mnt,i}$		0,25 K to 0,4 K
DAQ	c	$u_{dT,i}$		0,1 % to 0,2 %
<u>Air pressure</u>		$u_{B,i}$	$c_{B,i}$ (see E.10.7)	
Pressure sensor	a, b	$u_{B,cal,i}$		2 hPa to 4 hPa
Mounting effects	a	$u_{B,mnt,i}$		10 % of correction
DAQ	c	$u_{dB,i}$		0,1 % to 0,2 %
<u>Relative humidity</u>		$u_{RH,i}$	$c_{RH,i}$ (see E.10.11)	
Humidity sensor	a, b	$u_{RH,cal,i}$		1 % to 2 % (RH)
Mounting effects	c, d	$u_{RH,mnt,i}$		0,1 % to 0,2 %
DAQ	c	$u_{dRH,i}$		0,1 % to 0,2 %
<u>Method</u>	c, d	$u_{AD,method,i}$	$c_{AD,method,i} = c_{V,i}$ (for actively regulated turbines) $c_{AD,method,i} = 1$ (for stall regulated turbines)	0,2 % to 0,3 % on wind speed
Method		$u_{M,i}$	$c_{V,i}$ (see above)	
Wind conditions	d	$u_{M,wc,i}$		See E.11.2.1 bullet a) or b).
Wind conditions – wind shear	d	$u_{M,hear,i}$		See E.11.2.2
Wind conditions – wind veer	d	$u_{M,veer,i}$		See E.11.2.3
Wind conditions – upflow	d	$u_{M,upflow,i}$		See E.11.2.4
Wind conditions – turbulence	d	$u_{M,ti,i}$		0,3 % to 0,5 % (only for RSD with lower meteorological mast) See E.11.2.5
Seasonal effects	d	$u_{M,sfx,i}$		0,7 % See E.11.3
Turbulence normalization	d	$u_{M,tinorm,i}$		See E.11.4
Cold climate	d	$u_{M,cc,i}$		0,5 % to 1 %

Wind direction				
Wind direction vane		$u_{WV,i}$	None (see E.12.2.1)	
Calibration	b	$u_{WV,cal,i}$		See E.12.2.1
North mark	c	$u_{WV,nm,i}$		See E.12.2.2
Boom orientation	d	$u_{WV,bo,i}$		See E.12.2.3
Operation (influence of mast)	c	$u_{WV,oe,i}$		See E.12.2.4
Magnetic declination angle	c	$u_{WV,mda,i}$		See E.12.2.5
DAQ	c	$u_{dWV,i}$		See E.12.2.6
Wind direction RSD		$u_{WR,i}$	None (see E.12.3)	
Verification	a, b	$u_{WR,ver,i}$		See E.12.3.1
Classification	c	$u_{WR,class,i}$		See IEC 61400-50-2
Monitoring test	c	$u_{WR,mon,i}$		See E.12.3.2
Flow variation in different probe volumes at same height	c	$u_{WR,fv,i}$		See E.12.3.3
Alignment	d	$u_{WR,align,i}$		See E.12.3.4
Magnetic declination angle	c	$u_{WR,mda,i}$		See E.12.3.5
DAQ	c	$u_{dWR,i}$		See E.12.3.6
Category A: Statistical				
Electric power	e	$s_{P,i}$	$c_{P,i} = 1$	
Site calibration	e	s_{sc}	$c_{V,i}$ (see above)	
NOTE Identification of uncertainties: a = reference to standard; b = calibration; c = other "objective" method; d = "guesstimate"; e = statistics.				

The ranges for the uncertainty components given in Table E.2 should be considered. Where available, actual values from specific measurements should be used. Uncertainty components should not be estimated with a value of zero, in accordance with guidance in ISO/IEC Guide 98-3, unless the specific uncertainty components are not relevant for the specific test methodology applied (see Clause E.14).

Also, it is important to note that the wind direction uncertainty does not influence directly the uncertainty of the power curve or the AEP (unless a wind veer normalization is applied). However, the wind direction uncertainty is included here as it is important to understand how accurate the measurement sector filtering is being implemented as well as how accurate the site calibration factors are applied to data from the right sectors. As the wind direction uncertainty shall be reported, Table E.2 gives the minimum uncertainty components that shall be considered for the wind direction uncertainty. An estimate of the magnitude of these components is not given, but these should be included in a reported power curve.

Note that some calibration reports report an uncertainty for a coverage factor of two instead of one. In order to combine uncertainties correctly, the uncertainties should be converted to a coverage factor consistent with all uncertainty inputs (by default, use a coverage factor of 1).

E.3 Category A uncertainties

E.3.1 General

Category A uncertainties in measured and normalized electric power and due to climatic variations and the site calibration (if conducted) need to be considered.

E.3.2 Category A uncertainty in electric power

The standard deviation of the distribution of normalized power data in each bin is calculated by Equation (E.9):

$$\sigma_{P,i} = \sqrt{\frac{1}{N_i - 1} \sum_{j=1}^{N_i} (P_i - P_{n,i,j})^2} \quad (\text{E.9})$$

where

$\sigma_{P,i}$ is the standard deviation of the normalized power data in bin i ;

N_i is the number of 10 min data sets in bin i ;

P_i is the normalized and averaged power output in bin i ;

$P_{n,i,j}$ is the normalized power output of data set j in bin i .

The standard uncertainty of the normalized and averaged power in the bin is estimated by Equation (E.10):

$$s_{P,i} = \frac{\sigma_{P,i}}{\sqrt{N_i}} \quad (\text{E.10})$$

where

$s_{P,i}$ is the category A standard uncertainty of power in bin i ;

$\sigma_{P,i}$ is the standard deviation of the normalized power data in bin i ;

N_i is the number of 10 min data sets in bin i .

E.3.3 Category A uncertainties in the site calibration

The residuals between the site calibration corrected and measured wind speed at the wind turbine meteorological mast are used to define the category A uncertainty of the site calibration, s_{sc} . Refer to IEC 61400-12-3 for the method of derivation.

E.4 Category B uncertainties: Introduction and data acquisition system

E.4.1 Category B uncertainties: Introduction

The category B uncertainties are assumed to be related to the instruments, the data acquisition system, the terrain surrounding the power performance test site, the wind conditions and the uncertainty related to the applied method. If the uncertainties are expressed as uncertainty limits, or have implicit, non-unity coverage factors, the standard uncertainty shall be estimated or they shall be properly converted into standard uncertainties.

Consider an uncertainty expressed as an uncertainty limit $\pm U$. If a rectangular probability distribution is assumed, the standard uncertainty is:

$$\sigma = \frac{U}{\sqrt{3}} \quad (\text{E.11})$$

If a triangular probability distribution is assumed, the standard uncertainty is:

$$\sigma = \frac{U}{\sqrt{6}} \quad (\text{E.12})$$

E.4.2 Category B uncertainties: Data acquisition system

The uncertainties related to the data acquisition have been included under the treatment for the specific signals. The uncertainty of the electrical power includes a component for data acquisition and the same for wind speed and the other measured signals.

We assume in this Annex E the data acquisition system to have a standard uncertainty $u_{d,i}$ of 0,1 % of the full range of each measurement channel. However, for the reporting of a specific power performance test the data acquisition uncertainty shall be estimated for the specific test setup, including at least the contributions to the data acquisition uncertainty mentioned in E.4.2.

There can be uncertainties from transmission, signal conditioning, analogue-to-digital conversion, and data processing in the data acquisition system. The uncertainties can be different for each measurement channel. The standard uncertainty of the data acquisition system for the full range of a certain measurement channel, $u_{d,i}$, can be expressed as:

$$u_{d,i} = \sqrt{u_{d,ST,i}^2 + u_{d,SA,i}^2 + u_{d,SC,i}^2} \quad (\text{E.13})$$

where

$u_{d,ST,i}$ is the uncertainty due to signal transmission in bin i ;

$u_{d,SA,i}$ is the uncertainty due to system accuracy in bin i ;

$u_{d,SC,i}$ is the uncertainty due to signal conditioning in bin i .

Although the assumption of a standard uncertainty of 0,1 % of the full range of the measurement channel is reasonable in many cases, specific hardware and conditions can cause this to be much higher. In order to ensure that the uncertainty of the data acquisition system is indeed negligible compared to the uncertainty of the sensors (which as a guidance should be interpreted as a factor 10 in magnitude when comparing standard uncertainties), an assessment of the actual uncertainty of the data acquisition system used shall be done.

E.5 Category B uncertainties: Power output

E.5.1 General

The category B uncertainties regarding the power output are based on five different uncertainty components which are introduced here.

The symbol for this uncertainty component is $u_{P,i}$.

The uncertainty of the power measurement has uncertainty contributions from the current and voltage transformers and from the power transducer (or other power measurement device). Uncertainties of these sub-components are normally stated by their classification.

Further, an uncertainty component due to the dynamic behaviour of the power output of the wind turbine and the fact that the power measurement devices are classified under static conditions shall be added and finally the uncertainty related to the data acquisition of the power signal shall be added.

E.5.2 Category B uncertainties: Power output – Current transformers

This uncertainty component covers the uncertainty due to the current transformer as covered by the classification of the sensor.

The symbol for this uncertainty component is $u_{P,CT,i}$.

In this Annex E, the current transformers and the power transducer are all assumed to be of class 0,5, which is the minimum acceptable class for a power curve measurement as required by this document.

The current transformers of class 0,5 (nominal loads of the current transformers are here designed to match the nominal power, 2 000 kW, and not 125 % of nominal power). They have uncertainty limits, referring to IEC 61869-2, of $\pm 0,5$ % of the current at 100 % load. At 20 % and 5 % loads, though, the uncertainty limits are increased to $\pm 0,75$ % and $\pm 1,5$ % of the current, respectively. For power performance measurements on wind turbines, the most important energy production is produced at a reduced power. Thus, we anticipate the uncertainty limits of $\pm 0,75$ % of the current at 20 % load to be a good average.

The uncertainty distribution for a current transformer is assumed to be rectangular. If current transformers are not operated within their secondary loop operational load limits, additional uncertainties shall be added.

It is assumed that the uncertainties of the three current transformers are caused by external influence factors such as air temperature, grid frequency, etc. They are therefore assumed fully correlated (an exception from the general assumption) and are summed linearly. As each current transformer contributes by one-third to the power measurement, it follows that the uncertainty of all current transformers is proportional to the power as follows:

$$u_{P,CT,i} = \frac{0,75 \% \cdot P_i [\text{kW}]}{\sqrt{3}} \frac{1}{3} = 0,43 \% \cdot P_i [\text{kW}] \quad (\text{E.14})$$

E.5.3 Category B uncertainties: Power output – Voltage transformers

This uncertainty component covers the uncertainty due to the voltage transformer as covered by the classification of the sensor.

The symbol for this uncertainty component is $u_{P,VT,i}$.

In Annex E, the voltage transformers and the power transducer are all assumed to be of class 0,5, which is the minimum acceptable class for a power curve measurement as required by this document.

The voltage transformers of class 0,5 have uncertainty limits, referring to IEC 61869-3, of $\pm 0,5$ % of the voltage at all loads. The uncertainty distribution is assumed to be rectangular.

Voltage transformers are not used in all measurements and the value for this uncertainty component may be set to zero when not used.

If the voltage transformers are not operated within their secondary loop operational load limits, additional uncertainties shall be added.

The grid voltage is normally rather constant and independent of the wind turbine power. The uncertainties of the three voltage transformers are as for the current transformers assumed to be caused by external influence factors such as air temperature, grid frequency, etc. They are therefore assumed fully correlated (an exception from the general assumption) and are summed linearly. As each voltage transformer contributes by one-third to the power measurement, it follows that the uncertainty of all voltage transformers is proportional to the power as follows:

$$u_{P,VT,i} = \frac{0,5 \% \cdot P_i [\text{kW}]}{\sqrt{3}} \cdot \frac{1}{3} \cdot 3 = 0,29 \% \cdot P_i [\text{kW}] \quad (\text{E.15})$$

E.5.4 Category B uncertainties: Power output – Power transducer or other power measurement device

This uncertainty component covers the uncertainty due to the power transducer (or other power measurement device) as covered by the classification of the sensor.

The symbol for this uncertainty component is $u_{P,PT,i}$.

The power transducer of class 0,5, referring to IEC 60688, with a nominal power of 2 500 kW (125 % of the nominal power, 2 000 kW, of the wind turbine) has an uncertainty limit of 12,5 kW. The uncertainty distribution is assumed to be rectangular.

The uncertainty of the power transducer is thus:

$$u_{P,PT,i} = \frac{12,5 \text{ kW}}{\sqrt{3}} = 7,2 \text{ kW} \quad (\text{E.16})$$

E.5.5 Category B uncertainties: Power output – Data acquisition

This uncertainty component covers the uncertainty related to the data acquisition of the power signal.

The symbol for this uncertainty component is $u_{dP,i}$.

The magnitude of this component has an assumed default value of 0,1 % to 0,2 % of the full range of the data acquisition system.

Considering the electric power range of the measurement channel for a 2 MW turbine to be 3 000 kW and an uncertainty of the data acquisition system of 0,1 % of this range gives a resulting uncertainty of 3 kW.

E.6 Category B uncertainties: Wind speed – Introduction and sensors

E.6.1 Category B uncertainties: Wind speed – Introduction

The wind speed uncertainty consists of three components, each of which again consists of multiple sub-components. These three components are:

- a) the uncertainty related to use of sensing hardware (cup anemometers, sonic anemometers and remote sensing devices (RSD));
- b) the uncertainty related to the flow distortion due to the terrain;
- c) the uncertainty related to the methods applied.

The symbol for this uncertainty component is $u_{V,i}$.

These uncertainty components as well as the sub-components will be introduced in E.6.2 to E.11.

The sensitivity factor, $c_{V,i}$, is defined in Table E.2.

E.6.2 Category B uncertainties: Wind speed – Hardware

The wind speed uncertainty related to hardware consists of three components, each of which again consists of multiple sub-components. These three components are:

- a) the uncertainty related to use of sensors placed in a meteorological mast (cup anemometers and sonic anemometers);
- b) the uncertainty related to a remote sensing device (RSD);
- c) the uncertainty related to a rotor equivalent wind speed (REWS);

The symbol for this uncertainty component is $u_{VHW,i}$.

E.6.3 Category B uncertainties: Wind speed – Meteorological mast mounted sensors

E.6.3.1 General

This uncertainty component covers the uncertainty related to the use of cup anemometers and sonic anemometers in meteorological masts (either top mounted or side mounted).

The symbol for this uncertainty component is $u_{VS,i}$ (V stands for wind speed and S stands for sensors).

This uncertainty component has six sub-components, all of which should be derived according to IEC 61400-50-1:

- a) the uncertainty related to the calibration of the sensor before start of the power performance test;
- b) the uncertainty related to the calibration of the sensor during or after the power performance test;
- c) the uncertainty related to the operational characteristics as determined by the classification of the sensor;
- d) the uncertainty related to the mounting of the sensor;
- e) the uncertainty related to the mounting of the lightning finial;
- f) the uncertainty related to the data acquisition of the signal from the sensor.

E.6.3.2 Category B uncertainties: Wind speed – Met mast sensors – Pre-calibration

This uncertainty component covers the uncertainty related to calibration of the sensor before the test according to IEC 61400-50-1. This includes the variability of repeated tests for one test facility as well as the variability of repeated tests between various facilities.

The symbol for this uncertainty component is $u_{VS,precal,i}$.

For a specific power performance test, the values as indicated on the calibration for the sensors employed shall be used for the uncertainty calculation.

E.6.3.3 Category B uncertainties: Wind speed – Met mast sensors – Post-calibration

This uncertainty component covers the uncertainty related to the in-situ calibration according to IEC 61400-50-1, and/or the post-calibration of the sensor during and/or after the test according to IEC 61400-50-1.

The symbol for this uncertainty component is $u_{VS,postcal,i}$.

This uncertainty is also discussed in 7.2.2 and IEC 61400-50-1.

E.6.3.4 Category B uncertainties: Wind speed – Met mast sensors – Classification

This uncertainty component covers the uncertainty related to the operational characteristics of the sensor as determined by the classification of the sensor according to IEC 61400-50-1.

The symbol for this uncertainty component is $u_{VS,class,i}$.

This uncertainty is discussed in detail in IEC 61400-50-1.

A reference to the classification report shall be included in the power performance report.

The operational uncertainty of the reference anemometer (from a possible site calibration according to IEC 61400-12-3) shall be included in this uncertainty component. This can be none, some, or all of the reference anemometer uncertainty from the site calibration, depending on if the measured range of influence parameters experienced by the reference sensor on the permanent meteorological mast during the site calibration is significantly different from the range of influence parameters experienced by the reference sensor on the permanent meteorological mast during the power curve test.

By default, half of the operational uncertainty of the reference anemometer and all of operational uncertainty of the turbine anemometer during the site calibration and half of the operational uncertainty of the anemometer during the power curve test shall be included. These shall be added using the root-sum-square approach.

E.6.3.5 Category B uncertainties: Wind speed – Met mast sensors – Mounting

This uncertainty component covers the uncertainty related to the mounting of the sensor according to IEC 61400-50-1.

The symbol for this uncertainty component is $u_{VS,mnt,i}$.

This uncertainty is discussed in 7.2.4 and 9.1.2 as well as in IEC 61400-50-1.

This uncertainty component has three default values corresponding to the three mounting arrangements allowed by IEC 61400-50-1 (single top-mounted anemometer, side-by-side top-mounted anemometer or side-mounted anemometer). Refer to IEC 61400-50-1 for details.

E.6.3.6 Category B uncertainties: Wind speed – Met mast sensors – Lightning finial

This uncertainty component covers the uncertainty related to a possible top-mounted lightning finial and its influence on a top-mounted anemometer when the requirements in IEC 61400-50-1 for the mounting of the lightning finial cannot be met.

The symbol for this uncertainty component is $u_{VS,lgt,i}$.

Where the requirements of IEC 61400-50-1 are not met, the default magnitude for this uncertainty component is 0,1 % to 0,2 % of the wind speed signal.

E.6.3.7 Category B uncertainties: Wind speed – Met mast sensors – Data acquisition

This uncertainty component covers the uncertainty related to the data acquisition of the wind speed signal.

The symbol for this uncertainty component is $u_{d,VS,i}$.

This uncertainty is also discussed in E.4.2.

The default magnitude for this uncertainty component is 0,1 % to 0,2 % of the full range of the measured wind speed signal.

Considering a wind speed range of 30 m/s of the measurement channel and an uncertainty of the data acquisition system of 0,1 % of this range, the standard uncertainty from data acquisition is 0,03 m/s.

E.7 Category B uncertainties: Wind speed – RSD

E.7.1 General

This uncertainty component covers the uncertainty related to the use of remote sensing devices for wind speed measurements (lidar and sodar) according to IEC 61400-50-2.

The symbol for this uncertainty component is $u_{VR,i}$ (V stands for wind speed and R stands for remote sensing device).

This uncertainty component has six sub-components:

- a) uncertainty related to the verification of the device;
- b) uncertainty related to the in-situ check of the device;
- c) uncertainty related to the operational characteristics of the device as determined by the classification of the device;
- d) uncertainty related to the mounting of the device;
- e) uncertainty related to flow variation in different probe volumes at the same height;
- f) uncertainty related to the monitoring test.

E.7.2 Category B uncertainties: Wind speed – RSD – Calibration

This uncertainty component covers the uncertainty related to the calibration of the remote sensing device according to IEC 61400-50-2.

The symbol for this uncertainty component is $u_{VR,ver,i}$.

Prior to the measurement, a calibration according to IEC 61400-50-2 shall be undertaken and its uncertainties evaluated ($u_{VR,vrf,i}$). If the uncertainties for the exact measurement height are not available, the uncertainties should be interpolated. The default magnitude for this uncertainty component is 1,0 % to 3,0 % of the measured wind speed.

E.7.3 Category B uncertainties: Wind speed – RSD – In-situ check

This uncertainty component covers the uncertainty related to the in-situ verification of the remote sensing device according to IEC 61400-50-2, only applied for an RSD with a lower than hub height meteorological mast.

The symbol for this uncertainty component is $u_{VR,isc,i}$ ("isc" stands for in-situ check).

The in-situ test shall be evaluated as described in IEC 61400-50-1, where the RSD is the main measurement and the anemometer on the monitoring mast is the control anemometer with the same uncertainty limits.

E.7.4 Category B uncertainties: Wind speed – RSD – Classification

This uncertainty component covers the uncertainty related to the classification of the remote sensing device according to IEC 61400-50-2.

The symbol for this uncertainty component is $u_{VR,class,i}$.

The uncertainty due to the operational characteristics comes from the classification report of the RSD which provides slopes and ranges of environmental variables. There are three ways of calculating the uncertainty due to these variables as explained in IEC 61400-50-2.

According to IEC 61400-50-2, $u_{VR,class}$ becomes equal to zero if the verification test includes the same data as the power curve test.

The default magnitude for this uncertainty component is 1,0 % to 1,5 % of the measured wind speed.

E.7.5 Category B uncertainties: Wind speed – RSD – Mounting

This uncertainty component covers the uncertainty related to the mounting of the remote sensing device according to IEC 61400-50-2.

The symbol for this uncertainty component is $u_{VR,mnt,i}$.

The uncertainty of the remote sensing device due to non-ideal levelling of the device shall be estimated. The uncertainty is highly dependent on the type of instrument being used.

The default magnitude for this uncertainty component is 0,1 % of the measured wind speed.

E.7.6 Category B uncertainties: Wind speed – RSD – Flow variation

This uncertainty component covers the uncertainty related to the flow variation across the measurement volume of the remote sensing device according to IEC 61400-50-2.

The symbol for this uncertainty component is $u_{VR,flow,i}$.

The user should consult with the remote sensing device manufacturer for the best method for evaluating the uncertainty for their specific device at the test site.

E.7.7 Category B uncertainties: Wind speed – RSD – Monitoring test

This uncertainty component covers the uncertainty related to the result of the monitoring of the remote sensing device.

The symbol for this uncertainty component is $u_{VR,mon,i}$.

The derivation of this uncertainty is detailed in IEC 61400-50-2.

The default magnitude for this uncertainty component is 0,5 % of the measured wind speed.

E.8 Category B uncertainties: Wind speed – REWS

E.8.1 General

A number of uncertainty components are not related to the wind speed measurement (at least not when applied to a hub height wind speed) but to the use and definition of the rotor equivalent wind speed (REWS). The uncertainty components related to REWS will be covered in Clause E.8.

The rotor equivalent wind speed according to Equation (5) in 9.1.3.2 includes wind speed measurements at different heights above ground. In order to make uncertainty calculations of the rotor equivalent wind speed feasible, Equation (E.3) and Equation (E.4) are needed. Equation (E.3) and Equation (E.4) are approximations of Equation (E.1) and Equation (E.2), respectively, for the case that the category B uncertainty components are independent between different category B uncertainty sources. This approximation is highly violated if uncertainties of wind speed measurements at different height levels are considered as separate category B uncertainties, because the uncertainty components of wind speed measurements at different height levels are well correlated. Thus, Equations (E.3) and (E.4) cannot be applied directly.

In order to overcome this problem, Equations (E.3) and (E.4) shall be applied with such category B components of the rotor equivalent wind speed which can be considered as independent from each other. E.8.2 provides advice on how this can be implemented.

E.8.2 Category B uncertainties: Wind speed – REWS – Wind speed measurement over whole rotor

E.8.2.1 General

This uncertainty component covers the uncertainty of the REWS wind speed based on the multiple wind speed measurements that the REWS is based on.

The symbol for this uncertainty component is $u_{REWS,shear,i}$.

This uncertainty is also discussed in Annex P.

NOTE This uncertainty will be used instead of $u_{V,i}$ in conjunction with E.13.3.

What is very important for this uncertainty component is that the derivation and hence the magnitude is strongly dependent on the various measurement methods as introduced in 7.2. A configuration with a hub-height only meteorological mast will not lead to a REWS implementation, the other measurement configurations are considered in E.8.2.2 to E.8.2.4.

The same symbol is used in each subclause as these configurations are mutually exclusive and depending on the specific measurement setup only one of E.8.2.2, E.8.2.3 and E.8.2.4 applies.

E.8.2.2 REWS shear uncertainty – Meteorological mast significantly above hub height

In this configuration the REWS is based on signals from side-mounted cup anemometers or sonic anemometers from multiple measurement heights on a meteorological mast.

In this case Equation (5) in 9.1.3.2 applies and the REWS uncertainty components come from Clause E.6.

The default values come from the combination of the default values in Clause E.6.

The way to combine these components into a REWS uncertainty is explained in E.13.7.

E.8.2.3 REWS shear uncertainty – RSD plus lower-than-hub-height meteorological mast

In this configuration the REWS is based on signals from an RSD. The data from the meteorological mast is only used to validate the RSD data.

In this case Equation (5) in 9.1.3.2 applies and the REWS uncertainty components come from Clause E.7.

The default values come from the combination of the default values in Clause E.7.

The way to combine these components into a REWS uncertainty is explained in E.13.7.

E.8.2.4 REWS shear uncertainty – hub height meteorological mast plus RSD for shear

In this configuration the REWS is based on a cup anemometer or sonic anemometer on a hub-height meteorological mast combined with shear data from an RSD.

In this case Equation (11) in 9.1.3.4 applies and the REWS uncertainty components come from both Clause E.6 and Clause E.7.

In this case we still have a further division: either the RSD signals are considered as absolute wind speed measurements or as relative wind speed measurements, as the treatment differs in each of these cases. This depends on the type of equipment used or the available sensitivity and verification test.

For an absolute wind speed the way to combine these components into a REWS uncertainty is explained in E.13.8.

For a relative wind speed the way to combine these components into a REWS uncertainty is explained in E.13.9.

E.8.3 Category B uncertainties: Wind speed – REWS – Wind veer

This uncertainty component covers the uncertainty related to the influence of veer on the rotor equivalent wind speed.

The symbol for this uncertainty component is $u_{\text{REWS,veer},i}$.

This uncertainty is also discussed in Annex Q.

The veer measurement is based on signals from side-mounted wind vanes or sonic anemometers from multiple measurement heights on a meteorological mast or on signals from an RSD.

The sub-components for this uncertainty component therefore come from E.12.2 or from E.12.3.

E.9 Category B uncertainties: Wind speed – Terrain

E.9.1 General

This uncertainty component covers the uncertainty related to the flow distortion of the wind speed between measurement point and wind turbine due to the local terrain.

The symbol for this uncertainty component is $u_{VT,i}$ (V stands for wind speed and T stands for terrain).

This uncertainty is also discussed in 6.3.4.

When the power performance test is done without a site calibration, the default magnitude of this uncertainty component is determined by the distance between the measurement device and the turbine under test. If this distance is between 2 and 3 rotor diameters ($2D \leq \text{distance} \leq 3D$) this default magnitude is 2 % of the measured wind speed for onshore flat terrain and 1 % for offshore. If this distance is between 3 and 4 rotor diameters ($3D < \text{distance} \leq 4D$) this default magnitude is 3 % of the measured wind speed for onshore flat terrain and 2 % offshore.

When the power curve test is done with a site calibration according to IEC 61400-12-3, the default magnitude of this uncertainty component is determined by the site calibration measurement.

In this case this uncertainty component related to the terrain has nine sub-components:

- a) uncertainty related to the calibration of the anemometers used for the site calibration;
- b) uncertainty related to the operational characteristics of the anemometers used for the site calibration;
- c) uncertainty related to the mounting of the anemometers used for the site calibration;
- d) uncertainty related to the data acquisition of the signals from the anemometers used for the site calibration;
- e) uncertainty related to the convergence check;
- f) uncertainty related to the correlation check;
- g) uncertainty related to the change in correction between adjacent bins;
- h) uncertainty related to the removal of the wind direction sensor between site calibration and power curve measurement;
- i) uncertainty related to seasonal variation.

E.9.2 Category B uncertainties: Wind speed – Terrain – Pre-calibration

This uncertainty component covers the uncertainty related to the calibration of the sensors before the test.

The symbol for this uncertainty component is $u_{VT,precal,i}$.

This uncertainty is discussed in detail in IEC 61400-50-1 and IEC 61400-12-3.

This uncertainty is virtually the same as $u_{VS,cal,i}$ with the difference that here it is applied to measured wind speeds on two masts.

E.9.3 Category B uncertainties: Wind speed – Terrain – Post-calibration

This uncertainty component covers the uncertainty related to the in-situ calibration (Annex K) and/or the post-calibration of the sensor (IEC 61400-50-1) during and/or after the site calibration.

The symbol for this uncertainty component is $u_{VT,postcal,i}$.

This uncertainty is discussed in 7.2.2 and in IEC 61400-50-1.

E.9.4 Category B uncertainties: Wind speed – Terrain – Classification

This uncertainty component covers the uncertainty related to the classification of the sensors for the site calibration according to IEC 61400-50-1.

The symbol for this uncertainty component is $u_{VT,class,i}$.

This uncertainty is virtually the same as $u_{VS,class,i}$ with the difference that here it is applied to a measurement of wind speeds on two masts. Some of the influence factors for the classification will be experienced in the same way by both sensors, causing a correlation in the operational response and a reduction in the difference between the signals. Refer to IEC 61400-12-3 for details of how this should be handled.

If the measured range of influence parameters experienced by the reference sensor on the permanent meteorological mast during the site calibration is significantly different from the range of influence parameters experienced by the reference sensor on the permanent meteorological mast during the power curve test, an additional uncertainty addition can be required. However, as long as both tests stay within the ranges defined for class A and B this is considered to be sufficiently covered and no additional uncertainty needs to be taken into account.

E.9.5 Category B uncertainties: Wind speed – Terrain – Mounting

This uncertainty component covers the uncertainty related to the mounting of the sensors according to IEC 61400-50-1 during the site calibration as described in IEC 61400-12-3.

The symbol for this uncertainty component is $u_{VT,mnt,i}$.

The default magnitude for this uncertainty component for each of the three possible mounting arrangements (single top mount, side-by-side top mount or side mounted) are defined in IEC 61400-50-1. Refer to IEC 61400-12-3 and IEC 61400-50-1 for details of the evaluation.

E.9.6 Category B uncertainties: Wind speed – Terrain – Data acquisition

This uncertainty component covers the uncertainty related to the data acquisition of the signals of the wind speed sensors according to IEC 61400-50-1 during the site calibration.

The symbol for this uncertainty component is $u_{d,VT,i}$.

Refer to IEC 61400-12-3 and IEC 61400-50-1 for details of the evaluation.

E.9.7 Category B uncertainties: Wind speed – Terrain – Lightning finial

This uncertainty component covers the uncertainty related to a possible top-mounted lightning finial and its influence on a top-mounted anemometer when the requirements in IEC 61400-50-1 for the mounting of the lightning finial cannot be met.

The symbol for this uncertainty component is $u_{VT,lgt,i}$.

Refer to IEC 61400-12-3 and IEC 61400-50-1 for details of this component.

E.9.8 Category B uncertainties: Wind speed – Terrain – Change in correction between adjacent bins

This uncertainty component covers the uncertainty related to calibration of the sensors according to IEC 61400-12-3 before the test.

The symbol for this uncertainty component is $u_{VT,coc,i}$ ("coc" stands for "change in correction").

Refer to IEC 61400-12-3 for details of the evaluation of this uncertainty.

E.9.9 Category B uncertainties: Wind speed – Terrain – Removal of WD sensor

This uncertainty component covers the uncertainty related to removal of the wind direction sensor between the site calibration and the power performance test according to IEC 61400-12-3.

The symbol for this uncertainty component is $u_{VT,rmv,i}$.

Refer to IEC 61400-12-3 for details of the evaluation of this uncertainty.

E.9.10 Category B uncertainties: Wind speed – Terrain – Seasonal variation

This uncertainty component covers the uncertainty related to seasonal variation between the site calibration and the power performance test.

The symbol for this uncertainty component is $u_{VT,sv,i}$ ("sv" stands for seasonal variation).

Refer to IEC 61400-12-3 for details of the evaluation of this uncertainty.

E.10 Category B uncertainties: Air density

E.10.1 General

This uncertainty component covers the uncertainty related to the influence of air density on the power curve.

The symbol for this uncertainty component is $u_{AD,i}$.

The air density is derived from measurements of the air temperature, the humidity and the air pressure as described in 7.4.

The air density uncertainty consists of four components:

- a) uncertainty related to use of a temperature sensor and the data acquisition;

- b) uncertainty related to the use of a pressure sensor and the data acquisition;
- c) uncertainty related to the use of a relative humidity (RH) sensor and the data acquisition, or the lack of such a sensor;
- d) uncertainty due to the air density correction.

These uncertainty components as well as the sub-components will be introduced in Clause E.10.

E.10.2 Category B uncertainties: Air density – Temperature – Introduction

This uncertainty component covers the uncertainty related to the measurement of the temperature according to 7.4.

The symbol for this uncertainty component is $u_{T,i}$.

This uncertainty component has four sub-components:

- a) uncertainty related to the calibration of the temperature sensor;
- b) uncertainty related to the radiation shielding of the temperature sensor;
- c) uncertainty related to the mounting of the temperature sensor;
- d) uncertainty related to the data acquisition of the signal from the temperature sensor.

The equations for the sensitivity factors for temperature are given in Equation (E.17) and Equation (E.18).

Sensitivity for temperature for a measurement on wind turbine with active power control is given as follows:

$$c_{T,i} = -\frac{c_{V,i}v_i}{3\rho_i} \left(\frac{\rho_i}{T_i} + \frac{\Phi_i}{T_i} \left(\frac{1}{R_0} - \frac{1}{R_w} \right) 0,000\,020\,5 \cdot 0,063\,184\,6 \cdot \exp(0,063\,184\,6 \cdot T_i) \right) \quad (\text{E.17})$$

where

- $c_{T,i}$ is the sensitivity factor for temperature in bin i ;
- $c_{V,i}$ is the sensitivity factor for wind speed in bin i ;
- v_i is the average wind speed in bin i ;
- ρ_i is the average air density in bin i ;
- T_i is the average temperature in bin i ;
- Φ_i is the average relative humidity (range 0 % to 100 %) in bin i ;
- R_0 is the gas constant of dry air, 287,05 [J/kgK];
- R_w is the gas constant of water vapour, 461,5 [J/kgK];

Sensitivity for temperature for a measurement on a stall regulated wind turbine is given as follows:

$$c_{T,i} = \frac{P_i}{\rho_i} \left(\frac{\rho_i}{T_i} + \frac{\Phi_i}{T_i} \left(\frac{1}{R_0} - \frac{1}{R_w} \right) 0,000\,020\,5 \cdot 0,063\,184\,6 \cdot \exp(0,063\,184\,6 \cdot T_i) \right) \quad (\text{E.18})$$

where

$c_{T,i}$ is the sensitivity factor for temperature in bin i ;

P_i is the sensitivity factor for wind speed in bin i ;

ρ_i is the average air density in bin i ;

T_i is the average temperature in bin i ;

Φ_i is the average relative humidity (range 0 % to 100 %) in bin i ;

R_0 is the gas constant of dry air, 287,05 [J/kgK];

R_w is the gas constant of water vapour, 461,5 [J/kgK].

E.10.3 Category B uncertainties: Air density – Temperature – Calibration

This uncertainty component covers the uncertainty related to calibration of the temperature sensor.

The symbol for this uncertainty component is $u_{T,cal,i}$.

The default magnitude for this uncertainty component is 0,4 °C to 0,6 °C.

E.10.4 Category B uncertainties: Air density – Temperature – Radiation shielding

This uncertainty component covers the uncertainty related to the radiation shielding of the temperature sensor.

The symbol for this uncertainty component is $u_{T,shield,i}$.

The default magnitude for this uncertainty component is 1,5 °C to 2,5 °C.

E.10.5 Category B uncertainties: Air density – Temperature – Mounting

This uncertainty component covers the uncertainty related to the mounting of the temperature sensor according to 7.4.

The symbol for this uncertainty component is $u_{T,mnt,i}$.

The default magnitude for this uncertainty component is 0,25 °C to 0,4 °C.

E.10.6 Category B uncertainties: Air density – Temperature – Data acquisition

This uncertainty component covers the uncertainty related to the data acquisition of the signal of the temperature sensor.

The symbol for this uncertainty component is $u_{dT,i}$.

The default magnitude for this uncertainty component is 0,1 % to 0,2 % of the full range of the measurement channel. With an assumed temperature range of 40 °C this comes to 0,04 °C.

E.10.7 Category B uncertainties: Air density – Pressure – Introduction

This uncertainty component covers the uncertainty related to the measurement of the pressure according to 7.4.

The symbol for this uncertainty component is $u_{B,i}$.

This uncertainty component has three sub-components:

- a) uncertainty related to the calibration of the pressure sensor;
- b) uncertainty related to the mounting of the pressure sensor;
- c) uncertainty related to the data acquisition of the signal from the pressure sensor.

The equations for the sensitivity factors for pressure are given in Equation (E.19) and Equation (E.20).

Sensitivity for pressure for a measurement on wind turbine with active power control is given as follows:

$$c_{B,i} = \frac{c_{V,i} v_i}{3 \rho_i T_i R_0} \quad (\text{E.19})$$

where

- $c_{B,i}$ is the sensitivity factor for temperature in bin i ;
- $c_{V,i}$ is the sensitivity factor for wind speed in bin i ;
- v_i is the average wind speed in bin i ;
- ρ_i is the average air density in bin i ;
- T_i is the average temperature in bin i ;
- R_0 is the gas constant of dry air, 287,05 [J/kgK].

Sensitivity for pressure for a measurement on a stall regulated wind turbine is given as follows:

$$c_{B,i} = -\frac{P_i}{\rho_i T_i R_0} \quad (\text{E.20})$$

where

- $c_{B,i}$ is the sensitivity factor for temperature in bin i ;
- P_i is the normalized and averaged power output in bin i ;
- ρ_i is the average air density in bin i ;
- T_i is the average temperature in bin i ;
- R_0 is the gas constant of dry air 287,05 [J/kgK].

E.10.8 Category B uncertainties: Air density – Pressure – Calibration

This uncertainty component covers the uncertainty related to the calibration of the pressure sensor.

The symbol for this uncertainty component is $u_{B,cal,i}$.

The default magnitude for this uncertainty component is 2 hPa to 4 hPa.

E.10.9 Category B uncertainties: Air density – Pressure – Mounting

This uncertainty component covers the uncertainty related to the mounting of the pressure sensor according to 7.4.

The symbol for this uncertainty component is $u_{B,mnt,i}$.

The default magnitude for this uncertainty component is determined by the height difference for which the signal from the pressure sensor is corrected. Using ISO 2533 the pressure related to this height difference can be calculated. The default magnitude for the uncertainty related to this pressure correction is 10 % of the correction.

For a sensor installed at a height of 2 m and a hub height of 100 m, the difference is 98 m which gives a pressure difference of 11,7 hPa. The uncertainty would then be 1,17 hPa.

E.10.10 Category B uncertainties: Air density – Pressure – Data acquisition

This uncertainty component covers the uncertainty related to the data acquisition of the signal of the pressure sensor.

The symbol for this uncertainty component is $u_{dB,i}$.

The default magnitude for this uncertainty component is 0,1 % of the full range of the measurement channel for pressure. Considering a pressure range of 100 hPa of the measurement channel, this gives 0,1 hPa.

E.10.11 Category B uncertainties: Air density – Relative humidity – Introduction

This uncertainty component covers the uncertainty related to the measurement of the relative humidity according to 7.4.

The symbol for this uncertainty component is $u_{RH,i}$.

The relative humidity is not required to be measured. In that case a default value of 50 % shall be assumed with an uncertainty of 100 % (from 0 % to 100 %).

In the event that the humidity is measured, this uncertainty component has three sub-components:

- a) uncertainty related to the calibration of the humidity sensor;
- b) uncertainty related to the mounting of the humidity sensor;
- c) uncertainty related to the data acquisition of the signal from the humidity sensor.

The equations for the sensitivity factors for relative humidity are given in Equation (E.21) and Equation (E.22).

Sensitivity for relative humidity for a measurement on wind turbine with active power control is given as follows:

$$c_{RH,i} = -\frac{c_{V,i}V_i}{3\rho_i T_i} \left(\frac{1}{R_0} - \frac{1}{R_w} \right) \cdot 0,000\,020\,5 \cdot \exp(0,063\,184\,6 \cdot T_i) \quad (E.21)$$

where

- $c_{RH,i}$ is the sensitivity factor for temperature in bin i ;
 $c_{V,i}$ is the sensitivity factor for wind speed in bin i ;
 v_i is the average wind speed in bin i ;
 ρ_i is the average air density in bin i ;
 T_i is the average temperature in bin i ;
 R_0 is the gas constant of dry air, 287,05 [J/kgK];
 R_w is the gas constant of water vapour, 461,5 [J/kgK].

Sensitivity for relative humidity for a measurement on a stall regulated wind turbine is given as follows:

$$c_{RH,i} = \frac{P_i}{\rho_i T_i} \left(\frac{1}{R_0} - \frac{1}{R_w} \right) \cdot 0,000\,020\,5 \cdot \exp(0,063\,184\,6 \cdot T_i) \quad (E.22)$$

where

- $c_{RH,i}$ is the sensitivity factor for temperature in bin i ;
 P_i is the normalized and averaged power output in bin i ;
 ρ_i is the average air density in bin i ;
 T_i is the average temperature in bin i ;
 R_0 is the gas constant of dry air, 287,05 [J/kgK];
 R_w is the gas constant of water vapour, 461,5 [J/kgK].

E.10.12 Category B uncertainties: Air density – Relative humidity – Calibration

This uncertainty component covers the uncertainty related to the calibration of the humidity sensor.

The symbol for this uncertainty component is $u_{RH,cal,i}$.

The default magnitude for this uncertainty component is 1 % to 2 %.

E.10.13 Category B uncertainties: Air density – Relative humidity – Mounting

This uncertainty component covers the uncertainty related to the mounting of the humidity sensor.

The symbol for this uncertainty component is $u_{RH,mnt,i}$.

The default magnitude for this uncertainty component is 0,1 % to 0,2 % of the measured value.

E.10.14 Category B uncertainties: Air density – Relative humidity – Data acquisition

This uncertainty component covers the uncertainty related to the data acquisition of the signal from the humidity sensor.

The symbol for this uncertainty component is $u_{dRH,i}$.

The default magnitude for this uncertainty component is 0,1 % of the full range of the measurement channel for relative humidity.

E.10.15 Category B uncertainties: Air density – Correction

This uncertainty component covers the uncertainty related to the air density correction.

The symbol for this uncertainty component is $u_{AD,method,i}$.

As part of the data analysis a normalization from measured air density to a reference air density is performed. This normalization is related to an uncertainty component, in part because of the uncertainties in the measured temperature, pressure and relative humidity but also because one of the underlying assumptions upon which the normalization formula is based is increasingly inaccurate the larger the air density difference is on which the air density normalization is applied.

For a stall-regulated wind turbine with constant pitch and constant rotational speed, this uncertainty shall be evaluated by bin averaging the air density normalized power output and the measured power output against the wind speed at hub height. Half of the deviation of the normalized and measured power output per wind speed bin shall be considered as the standard uncertainty of the air density normalization in that wind speed bin.

For a wind turbine with active power control, the uncertainty of the air density normalization shall be evaluated by bin averaging the measured wind speed against the wind speed normalized for the air density. Half of the deviation of the normalized and measured wind speed shall be considered as the standard uncertainty of the air density normalization in that wind speed bin.

In this example using a wind turbine with active power control, the wind speed deviations per wind speed bin are listed in Table E.9.

E.11 Category B uncertainties: Method

E.11.1 General

The specific method used to measure or analyse a power curve can also contribute to the uncertainty of the result. As much as possible these have been included with the category B uncertainties they are related to. Therefore, the uncertainty related to the air density correction is included under air density and the uncertainty related to the flow distortion correction is included under the wind speed sensors.

However, some of the uncertainties related to the method cannot easily be attributed to a specific component and these have been grouped under the header "Method".

The symbol for this uncertainty component is $u_{M,i}$.

E.11.2 Category B uncertainties: Method – Wind conditions

E.11.2.1 General

Although this uncertainty is not directly discussed elsewhere in this document, it is strongly related to the definition of the power curve as given in Clause 5.

In accordance with the definition, the power curve according to this document is a climate-specific power curve. The wind conditions in terms of wind shear, wind veer, turbulence and upflow have a direct influence upon the power performance of a wind turbine.

As an example, let us assume that a power curve is measured for a specific set of wind conditions, namely wind shear is 0,1; wind veer is 10°, turbulence is 10 % and upflow is +2°. The measured power curve would then be reported together with these values and no further uncertainty would need to be introduced.

However, mostly we will not experience constant values for these parameters during a power curve measurement and during the power curve measurement each of these parameters will reflect a range of values that the turbine has been subjected to. As part of this example, let us assume that wind shear ranges from 0 to 0,3, wind veer from 0° to 20°, turbulence from 3 % to 20 % and upflow from 0° to 5°. As the measurement progresses, we are in fact sampling from many different power curves. As we have no control over these input variables, the next power curve measurement at the same site (be it immediately after the first measurement or a year later to catch the same part of the annual season) will be somewhat different from the first power curve measured. This is the basic argument for adding an uncertainty component related to the wind conditions.

From this perspective, we have the following considerations.

- a) If one or more of the four parameters that determine the wind conditions have not been measured, the power curve cannot be reported accurately. The uncertainty related to this can only be estimated by assuming possible values and ranges for the missing parameters and estimating the potential influence on the power performance of the turbine.
- b) For the parameters that have been measured more information is available, but not necessarily all information sufficient to characterize the power performance in detail. For example:
 - 1) A parameter measured on three points across the rotor might still not give a full picture for the areas that have not been measured.
 - 2) The wind conditions (e.g. shear) are measured two to four rotor diameters away from the turbine and in most cases without further data as to what extent the conditions at the measurement location reflect the conditions at the turbine location. Note that the uncertainty of the hub height wind speed correlation is considered in E.9.1.
 - 3) A similar argument goes for the horizontal dimension of the rotor; the measurement to a large extent only characterizes the changes of these parameters with height (if even that) but not the horizontal variability of the measured parameters.

In the light of these arguments a practical approach has been adopted. For nine or more measurement heights for one of these parameters, the assumption has been made that no further uncertainty needs to be added. For fewer measurement heights an additional increase in uncertainty will be added.

This means that some of the above points are not addressed at all in the current uncertainty analysis; this will be incorporated in a future revision of this document once sufficient details have been clarified to do so.

- c) If one or more of these parameters have been measured it might be possible to normalize the power curve using this information, as explained in Annex M, Annex P and Annex Q.

The uncertainties related to points a) and b) are covered in E.11.2.2, whereas the uncertainty related to the normalizations is covered in E.11.2.2.3, E.11.2.3, E.11.2.4 and E.11.2.5.

Many of the influences here will be expressed as an uncertainty on wind speed. The experience values that the default estimates are based on are often expressed in terms of AEP. As a practical approach the choice has been made to express this as an uncertainty through wind speed.

E.11.2.2 Category B uncertainties: Method – Wind conditions – Shear

E.11.2.2.1 General

The symbol for this uncertainty component is $u_{M, \text{shear}, i}$.

The default magnitude for this uncertainty component can be found in E.11.2.2.2 and in E.11.2.2.3.

E.11.2.2.2 Half rotor wind shear measurement coverage

This uncertainty estimation applies to a power curve with a hub height wind speed definition and without Annex P normalization. As wind turbine power curves are influenced by wind shear, the uncertainty of the measured power curve due to wind shear shall be conservatively estimated and taken into account even if no wind shear measurement is performed. For this, a wind shear correction factor according to Equation (9) and (10) in 9.1.3.3 shall be calculated under the following assumptions:

- assuming 20 virtual wind speed measurements equally distributed over the rotor height range, resulting in 20 weighting factors;
- assuming one power law for the lower rotor half and another power law for the upper rotor half. Based on these power laws, the wind speed relative to hub height shall be calculated for each virtual measurement height. The wind shear exponent applied for the lower rotor half shall be determined from the measurements for each 10 min data point. The wind shear exponent for the upper rotor half shall be assumed to be half the value used for the lower rotor half. In case of negative shear values, also half of these values shall be assumed for the upper rotor half.

The standard uncertainty of the measured power curve in terms of the wind speed due to not measuring the wind shear shall be assumed for the i -th wind speed bin as:

$$u_{v5,i} = \frac{1}{\sqrt{3}} |f_r - 1| v_{h,i} \quad (\text{E.23})$$

where

f_r is the wind shear correction factor as calculated in 9.1.3.3;

$v_{h,i}$ is the wind speed at hub height in bin i .

Examples of estimated standard uncertainties of power curve measurements in terms of the wind speed due to the absence of wind shear measurements are shown in Table E.3. The respective uncertainty increases with the rotor diameter and decreases with the hub height.

Table E.3 – Example of standard uncertainties due to absence of a wind shear measurement

H m	D m	U_{v5}/v_h %
60	60	3,0
60	80	3,9
100	80	2,4
120	80	2,0
100	120	3,5
150	120	2,4
NOTE Wind shear exponents of 0,5 and 0 have been assumed for the lower and upper rotor half, respectively.		

E.11.2.2.3 Full rotor wind shear measurement coverage

This uncertainty estimation applies to:

- a) a power curve with a rotor equivalent wind speed definition;
- b) a hub height wind speed power curve to which the normalization in Annex P is applied.

The calculation of the rotor equivalent wind speed according to Equation (5) in 9.1.3.2 assumes a constant wind speed for each measurement height. This assumption leads to an uncertainty of the evaluation of the rotor equivalent wind speed. This uncertainty shall be estimated by fitting a power law through each pair of wind speed measurements of successive measurement heights (z_m). The wind speed according to this power law shall be calculated for at least 10 height levels equally distributed between each pair of measurement heights. The summation according to Equation (5) shall be repeated with all height levels covered by the power law assumption and measurement heights. The percentage deviation of the resulting rotor equivalent wind speed compared to the rotor equivalent wind speed using only the measured wind speeds shall be assumed as the (percentage) standard uncertainty of the wind speed due to the limited number of measurement heights.

The estimation of the standard uncertainty due to the limited number of measurement heights shall be performed on the basis of the bin averages of the wind speed measurements at the different height levels as function of the wind speed finally applied for the power curve evaluation.

NOTE The 10 min data sets are needed for the calculation.

E.11.2.3 Category B uncertainties: Method – Wind conditions – Wind veer

E.11.2.3.1 General

This uncertainty component covers the uncertainty related to fewer than nine measurement heights for wind veer.

The symbol for this uncertainty component is $u_{M,veer,i}$.

The default magnitude for this uncertainty component can be determined using E.11.2.3.2 and E.11.2.3.3.

E.11.2.3.2 No veer measurement

A wind veer correction factor according to Equation (Q.1) shall be calculated under the following assumptions:

- a) assuming wind speeds to be equal to 1 at all measurement heights. Then Equation (Q.1) transforms to a wind veer correction factor;
- b) assuming 20 virtual wind direction measurements equally distributed over the rotor height range, resulting in 20 weighting factors;
- c) assuming homogenous wind veer over the entire rotor height range as large as can be reasonably expected for the test site. If no reasonable assumptions about extreme wind veer are possible for the test site, a wind veer of 40°/100 m shall be assumed.

The standard uncertainty of the measured power curve in terms of the wind speed due to not measuring the wind veer shall be assumed for the i -th wind speed bin as:

$$u_{v6,i} = \frac{1}{\sqrt{3}} |f_r - 1| v_{h,i} \quad (\text{E.24})$$

where

f_r is the wind veer correction factor as calculated in Equation (E.24);

$v_{h,i}$ is the wind speed at hub height in bin i .

Examples of power curve measurement estimated standard uncertainties in terms of the wind speed due to the absence of wind veer measurements are shown in Table E.4. The respective uncertainty increases with the rotor diameter.

Table E.4 – Example of standard uncertainties due to absence of a wind veer measurement

D m	u_{v6}/v_h %
20	0,04
40	0,1
60	0,3
80	0,6
100	0,9
120	1,2
140	1,7
160	2,1
180	2,7
200	3,2
NOTE An extreme wind veer of 40°/100 m has been assumed for the calculations.	

E.11.2.3.3 Half rotor veer measurement

The calculation of the rotor equivalent wind speed according to Equation (Q.1) assumes a constant wind direction for each measurement height. This assumption leads to an uncertainty of the evaluation of the rotor equivalent wind speed.

The uncertainty related to a veer measurement across half a rotor shall be estimated using the same procedure as used for the veer across a full rotor (see E.11.2.3.4), with the change that the wind veer across the full rotor is 2,5 times that measured across the half rotor.

E.11.2.3.4 Full rotor veer measurement

The calculation of the rotor equivalent wind speed according to Equation (Q.1) assumes a constant wind direction for each measurement height. This assumption leads to an uncertainty of the evaluation of the rotor equivalent wind speed. This uncertainty shall be estimated by assuming a linear increase of the wind veer between each pair of wind speed measurements of successive measurement heights. The wind speed according to this assumption shall be calculated for at least 10 height levels equally distributed between each pair of measurement heights. The summation according to Equation (Q.1) shall be repeated with all height levels covered by the wind veer profile assumption and measurement heights and by assuming a constant wind speed for all heights. The percentage deviation of the resulting rotor equivalent wind speed to the rotor equivalent wind speed using only the measured wind directions shall be assumed as (percentage) standard uncertainty of the wind speed due to the limited number of wind direction measurement heights.

The estimation of the uncertainty due to the limited number of wind direction measurement heights shall be performed on the basis of the bin averages of the wind direction measurements at the different height levels as function of the wind speed finally applied for the power curve evaluation.

NOTE The 10 min data sets are needed for the calculation.

E.11.2.4 Category B uncertainties: Method – Wind conditions – Upflow

This uncertainty component covers the uncertainty related to fewer than nine measurement heights for upflow. This uncertainty component shall be applied for sites that do not meet the requirements of Annex B.

The symbol for this uncertainty component is $u_{M,upflow,i}$.

The default magnitude for this uncertainty component can be found in Table E.5.

Table E.5 – Uncertainty contributions due to lack of upflow knowledge

Number of measurement heights	Percentage of wind speed
0 (no upflow measurement)	0,3 to 0,5
1 (at hub height only)	0,15 to 0,25
2 (lower rotor area)	0,08 to 0,12
3	0,03 to 0,07
5	0,015 to 0,025
7	0,005 to 0,015

This is a very different uncertainty component for upflow than is considered in the classification of the wind speed sensor where also upflow is included. In that case we are looking at the inaccurate measurement of the horizontal wind speed due to the presence of a vertical wind speed component. Here we consider the influence of the upflow on the turbine's performance. Although we define a horizontal wind speed, the turbine's performance still varies when a vertical component is present.

E.11.2.5 Category B uncertainties: Method – Wind conditions – Turbulence intensity

This uncertainty component covers the uncertainty related to not having a measurement for hub height turbulence.

The symbol for this uncertainty component is $u_{M,ti,i}$.

For turbulence we do not consider other measurement heights as Clause 5 of this document defines the power curve only for hub height turbulence. This is a required signal. The reason this is still included as an additional uncertainty component is that the measurement configuration with a lower-than-hub height meteorological mast only has a hub height turbulence measured by a remote sensing device. The turbulence as measured by a remote sensing device is significantly different from the turbulence as measured by a sonic or cup anemometer. Hence the additional uncertainty.

The default magnitude for this uncertainty component can be found in Table E.6.

Table E.6 – Uncertainty contributions due to lack of turbulence knowledge

Number of measurement heights	Percentage of wind speed
0 (no accurate turbulence intensity due to RSD)	0,3 % to 0,5 %

E.11.3 Category B uncertainties: Method – Seasonal effects

This uncertainty component covers the uncertainty related to the influence of seasonal effects on the power curve. Note that some seasonal variations contribute in the wind speed uncertainty

as they influence the wind profile characteristics (shear, turbulence, etc.). These variations are treated separately.

Some other contributions, such as blade accretion from bugs and/or dirt on the blades (e.g. during a warm and dry period), increase the blade surface roughness and deteriorate the blade aerodynamic performance. Therefore, for the same kinetic energy in the wind, the turbine production will be lower when blades are dirty. These contributions are combined in the uncertainty component $u_{M,sfx,i}$.

Clearly these contributions do not influence the wind profile characteristics and therefore, strictly speaking, are effects on the electrical power. However, with the purpose to quantify the contribution to the AEP variation this uncertainty component is more conveniently expressed as uncertainty in the wind speed.

A default magnitude of 0,7 % on the wind speed is suggested, yet the magnitude of this component will vary with the site location and the year period of the performance test and where possible it should be estimated from experience or AEP variation data from the specific site.

E.11.4 Category B uncertainties: Method – Turbulence normalization (or the lack thereof)

This uncertainty component covers the uncertainty related to the turbulence normalization of the power curve.

The symbol for this uncertainty component is $u_{M,tinorm,i}$.

This uncertainty is also discussed in Annex M and 9.1.6.

The turbulence normalization given in Annex M is designed to handle effects of the 10 min data averaging on the evaluated power curve. There are further effects of the turbulence intensity on wind turbine power curves, which could, for instance, be due to direct impact of the turbulence intensity on the aerodynamics or due to the three-dimensional character of turbulence. In the end, the turbulence intensity normalization is a strongly simplified approach for characterizing short-term wind speed fluctuations. Thus, there is a remaining uncertainty of the evaluated power curve due to possible turbulence effects, even if the turbulence normalization procedure is applied. The turbulence normalization often removes about half of the observed effect of the turbulence intensity on measured wind turbine power curves. Thus, the following steps shall be performed in order to calculate the uncertainty of the turbulence normalization:

- a) the final bin-averaged power curve shall be evaluated on the basis of the turbulence normalized power output and on the basis of the non-turbulence normalized power output;
- b) the deviation of these two power curves shall be assumed to be the maximum uncertainty of the turbulence-normalized power curve per wind speed bin resulting from the turbulence normalization. The standard uncertainty resulting from the turbulence normalization per wind speed bin shall be calculated as the deviation of the power curves divided by $\sqrt{3}$. The standard uncertainty shall be combined with the other uncertainties of the power performance measurement for the determination of the total standard uncertainty according to Annex D.

In the absence of turbulence normalization, the uncertainty due to turbulence effects on the power curve should be calculated as described in Clause M.5.

E.11.5 Category B uncertainties: Method – Cold climate

This uncertainty component covers the uncertainty related to the influence of measurement in cold climate on the classification of the anemometers according to IEC 61400-50-1, which should be referred to for further details.

The symbol for this uncertainty component is $u_{M,cc,i}$.

This uncertainty is also discussed in Annex O.

The default magnitude of this uncertainty component is 0,5 % to 1 % on wind speed.

E.12 Category B uncertainties: Wind direction

E.12.1 General

There is an influence of the wind direction uncertainty on the power curve through the site calibration as well as a smaller effect. Based on the magnitude of the wind direction uncertainty compared to the site calibration bin size data will be (in)correctly assigned to a bin. For a bin size of 10° and a wind direction uncertainty of 5°, roughly 39 % of the data in a bin has been wrongly assigned. Although this will tend to average out, it can have an effect for small measurement sectors and large differences between adjacent bins. A similar argument goes for the filtering on the power curve measurement sector, but to a lesser extent.

These uncertainties shall only apply if the wind direction sensor or its mounting is changed for sites on which a site calibration was performed.

This background is the main reason why this document requires that the wind direction uncertainty is assessed to ensure that it stays below 5°.

The influence from the wind direction on the power curve and AEP is not quantifiably established and no sensitivity factors have been developed.

As the wind direction uncertainty shall be reported, Clause E.12 gives the minimum uncertainty components that shall be considered for the wind direction uncertainty. An estimate of the magnitude of these components is not given, but shall be included for a reported power curve.

E.12.2 Category B uncertainties: Wind direction – Vane or sonic

E.12.2.1 Category B uncertainties: Wind direction – Vane or sonic – Calibration

This uncertainty component covers the uncertainty related to the calibration of the wind direction sensor according to IEC 61400-50-1.

The symbol for this uncertainty component is $u_{WV,cal,i}$.

The resolution of the wind direction sensor is also included here and this value divided by $2\sqrt{3}$ shall be taken as a minimum value.

No default value is given but this uncertainty component shall be assessed and reported.

E.12.2.2 Category B uncertainties: Wind direction – Vane or sonic – North mark

This uncertainty component covers the uncertainty related to the accurate determination of the sensor's north mark in relation to the boom on which the sensor is installed.

The symbol for this uncertainty component is $u_{WV,nm,i}$.

No default value is given but this uncertainty component shall be assessed and reported.

E.12.2.3 Category B uncertainties: Wind direction – Vane or sonic – Boom orientation

This uncertainty component covers the uncertainty related to establishing the direction of the boom with regards to the North reference, i.e magnetic or true.

The symbol for this uncertainty component is $u_{WV,bo,i}$.

No default value is given but this uncertainty component shall be assessed and reported.

E.12.2.4 Category B uncertainties: Wind direction – Vane or sonic – Operational effects

This uncertainty component covers the uncertainty related to the influence of the mast on the free stream wind direction at the point of measurement.

The symbol for this uncertainty component is $u_{WV,oe,i}$.

As the wind will flow around the mast, the wind direction as measured by the sensor might not be the free flow wind direction. This effect is covered under this uncertainty component.

No default value is given but this uncertainty component shall be assessed and reported.

E.12.2.5 Category B uncertainties: Wind direction – Vane or sonic – Magnetic declination angle

This uncertainty component covers the uncertainty related to the difference between magnetic north and true north.

The symbol for this uncertainty component is $u_{WV,mda,i}$ (MDA stands for magnetic declination angle).

The correction from magnetic north to true north is also related to an uncertainty.

No default value is given but this uncertainty component shall be assessed and reported.

E.12.2.6 Category B uncertainties: Wind direction – Vane or sonic – Data acquisition

This uncertainty component covers the uncertainty related to the data acquisition of the signal from the wind direction sensor.

The symbol for this uncertainty component is $u_{dWV,i}$.

No default value is given but this uncertainty component shall be assessed and reported.

E.12.3 Category B uncertainties: Wind direction – RSD

This uncertainty component covers the uncertainty related to the verification of the RSD wind direction measurement according to IEC 61400-50-2.

The symbol for this uncertainty component is $u_{WR,ver,i}$.

E.12.3.1 Category B uncertainties: Wind direction – RSD – Verification

No default value is given but this uncertainty component shall be assessed and reported.

E.12.3.2 Category B uncertainties: Wind direction – RSD – Monitoring

This uncertainty component covers the uncertainty related to the in-situ monitoring of the signal from the wind direction monitoring by the RSD device according to IEC 61400-50-2.

The symbol for this uncertainty component is $u_{WR,mon,i}$.

No default value is given but this uncertainty component shall be assessed and reported.

E.12.3.3 Category B uncertainties: Wind direction – RSD – Flow variation

This uncertainty component covers the uncertainty related to the flow variation in different probe volumes.

The symbol for this uncertainty component is $u_{WR,fv,i}$.

No default value is given but this uncertainty component shall be assessed and reported. IEC 61400-50-2 provides guidance.

E.12.3.4 Category B uncertainties: Wind direction – RSD – Alignment

This uncertainty component covers the uncertainty related to accuracy of the alignment of the RSD device.

The symbol for this uncertainty component is $u_{WR,align,i}$.

No default value is given but this uncertainty component shall be assessed and reported.

E.12.3.5 Category B uncertainties: Wind direction – RSD – Magnetic declination angle

This uncertainty component covers the uncertainty related to the correction from magnetic north to true north.

The symbol for this uncertainty component is $u_{WR,mda,i}$.

No default value is given but this uncertainty component shall be assessed and reported.

E.12.3.6 Category B uncertainties: Wind direction – RSD – Data acquisition

This uncertainty component covers the uncertainty related to the data acquisition of the signal of the RSD device.

The symbol for this uncertainty component is $u_{dWR,i}$.

No default value is given but this uncertainty component shall be assessed and reported.

E.13 Combining uncertainties

E.13.1 General

In Clause E.13, the equations and further considerations to combine uncertainty components to an aggregate level will be presented and discussed. An example is the calculation of the site calibration uncertainty based on the uncertainty components of the site calibration uncertainty.

E.13.2 Combining Category B uncertainties in electric power ($u_{P,i}$)

The standard uncertainty of the electric power for each bin, $u_{P,i}$, is calculated by combining the standard uncertainties from the power transducer, the current and voltage transformers, the dynamic behaviour and the data acquisition system:

$$u_{P,i} = \sqrt{u_{P,CT,i}^2 + u_{P,VT,i}^2 + u_{P,PT,i}^2 + u_{dP,i}^2} \quad (E.25)$$

where

- $u_{P,i}$ is the uncertainty on the power measurement;
- $u_{P,CT,i}$ is the uncertainty related to the current transformers;
- $u_{P,VT,i}$ is the uncertainty related to the voltage transformers;
- $u_{P,PT,i}$ is the uncertainty related to the power transducers;
- $u_{dP,i}$ is the uncertainty related to the data acquisition of the power signal.

Considering the default magnitudes for these uncertainty components as given in Clause E.5, the standard uncertainty from the electric power sensor for each bin is:

$$\begin{aligned} u_{P,i} &= \sqrt{(0,43 \% \cdot P_i [\text{kW}])^2 + (0,29 \% P_i [\text{kW}])^2 + (7,2 \text{ kW})^2 + (0,1 \% \times 3\,000 \text{ kW})^2} \\ &= \sqrt{(0,53 \% \cdot P_i [\text{kW}])^2 + (7,8 \text{ kW})^2} \end{aligned} \quad (E.26)$$

E.13.3 Combining uncertainties in the wind speed measurement ($u_{V,i}$)

The following uncertainty components are combined to calculate the category B uncertainty for wind speed, $u_{V,i}$:

$$u_{V,i} = \sqrt{u_{VHW,i}^2 + u_{VT,i}^2 + u_{AD,method,i}^2} \quad (E.27)$$

where

- $u_{VHW,i}$ is the uncertainty on the hardware used and is one of $u_{VS,i}$, $u_{VR,i}$ or $u_{REWS,i}$;
- $u_{VT,i}$ is the uncertainty related to the flow distortion from the terrain;
- $u_{AD,method,i}$ is the uncertainty related to the air density correction.

E.13.4 Combining uncertainties in the wind speed measurement from cup or sonic ($u_{VS,i}$)

The following uncertainty components are combined to calculate the category B uncertainty for wind speed measurements from cup or sonic, $u_{VS,i}$:

$$u_{VS,i} = \sqrt{u_{VS,precal,i}^2 + u_{VS,postcal,i}^2 + u_{VS,class,i}^2 + u_{VS,mnt,i}^2 + u_{VS,lg,i}^2 + u_{dVS,i}^2} \quad (E.28)$$

where

- $u_{VS,precal,i}$ is the uncertainty on the hardware used and is one of $u_{VS,i}$, $u_{VR,i}$ or $u_{REWS,i}$;
- $u_{VS,postcal,i}$ is the uncertainty related to the flow distortion from the terrain;
- $u_{VS,class,i}$ is the uncertainty related to the classification of the sensors;
- $u_{VS,mnt,i}$ is the uncertainty related to the mounting of the sensors;
- $u_{VS,lgt,i}$ is the uncertainty related to the flow distortion from lightning finial;
- $u_{dVS,i}$ is the uncertainty related to the data acquisition of the wind speed signal.

E.13.5 Combining uncertainties in the wind speed measurement from RSD ($u_{VR,i}$)

The following uncertainty components are combined to calculate the category B uncertainty for wind speed measurements from an RSD device, $u_{VR,i}$:

$$u_{VR,i} = \sqrt{u_{VR,ver,i}^2 + u_{VR,isc,i}^2 + u_{VR,class,i}^2 + u_{VR,mnt,i}^2 + u_{VR,flow,i}^2 + u_{VR,mon,i}^2} \quad (E.29)$$

where

- $u_{VR,ver,i}$ is the uncertainty due to the verification test;
- $u_{VR,isc,i}$ is the uncertainty due to the in-situ test;
- $u_{VR,class,i}$ is the uncertainty related to the classification of the RSD;
- $u_{VR,mnt,i}$ is the uncertainty related to the mounting of the RSD;
- $u_{VR,flow,i}$ is the uncertainty related to the flow variation across the measurement volume of the RSD;
- $u_{VR,mon,i}$ is the uncertainty related to the monitoring of the RSD.

E.13.6 Combining uncertainties in the wind speed measurement from REWS $u_{VREWS,i}$

The following uncertainty components are combined to calculate the category B uncertainty for a wind speed measurement expressed as a rotor equivalent wind speed, $u_{VREWS,i}$:

$$u_{VREWS,i} = \sqrt{u_{VREWS,shear,i}^2 + u_{VREWS,veer,i}^2} \quad (E.30)$$

where

- $u_{VREWS,i}$ is the uncertainty of the rotor equivalent wind speed (REWS);
- $u_{VREWS,shear,i}$ is the uncertainty due to the influence on REWS from the shear measurement across the rotor;
- $u_{VREWS,veer,i}$ is the uncertainty related to the classification of the RSD.

The value of $u_{VREWS,shear,i}$ is taken from one of the following:

- a) $u_{veq,i}$ (from E.13.7);
- b) $u_{veq,final,i}$ (from E.13.8);
- c) $u_{fr,RSD,k,i}$ (from E.13.9).

The value of $u_{VREWS,veer,i}$ is taken from $u_{veq,i}$ (from E.13.10).

The wind speed uncertainty components that are included in the REWS uncertainty are based on Equation (5) of 9.1.3.2 or Equation (11) of 9.1.3.4. Equation (6) in 9.1.3.2 (which would also influence Equation (11)) implies that there is an uncertainty component related to establishing the correct measurement height. As this is considered to be insignificant for a meteorological mast measurement and is included in a lidar calibration in accordance with IEC 61400-50-2, the only concern is with sodar measurements especially for higher wind speeds when "beam bending" can occur. If this is experienced during a measurement this shall be included as a further uncertainty component.

E.13.7 Combining uncertainties in the wind speed measurement for REWS for either a meteorological mast significantly above hub height or an RSD with a lower-than-hub-height meteorological mast

The following uncertainty components are combined to calculate the category B uncertainty for wind speed measurements from an RSD device.

The rotor equivalent wind speed shall be considered as a single wind speed measurement. The total standard uncertainty of the rotor equivalent wind speed in the wind speed bin i can then be expressed by:

$$u_{v_{eq},i}^2 = \sum_{k=1}^M \sum_{l=1}^M u_{v_{eq},k,i} u_{v_{eq},l,i} \rho_{k,l,i} \quad (\text{E.31})$$

where

- M is the number of uncertainty components of v_{eq} ;
- i is the wind speed bin referring to bin averaging of power curve;
- $u_{v_{eq},k,i}$ is the standard uncertainty sub-component k of v_{eq} in wind speed bin i ;
- $u_{v_{eq},l,i}$ is the standard uncertainty sub-component l of v_{eq} in wind speed bin i ;
- $\rho_{k,l,i}$ is the correlation coefficient between the uncertainty component k height m and n in wind speed bin i .

Note that the sensitivity factors are expressly not included here as they are included in Equation (E.32).

The uncertainty components of v_{eq} in wind speed bin i shall be calculated from the uncertainty components of the wind speed measurements in single measurement heights by the method of error propagation:

$$u_{v_{eq},k,i}^2 = \sum_{m=1}^L \sum_{n=1}^L c_{m,i} u_{v_m,k,i} c_{n,i} u_{v_n,k,i} \rho_{m,n,k,i} \quad (\text{E.32})$$

where

- L is the number of measurement heights across the rotor;
- $u_{v_m,k,i}$ is the standard uncertainty sub-component k of wind speed measurement at height m in wind speed bin i ;
- $u_{v_n,k,i}$ is the standard uncertainty sub-component k of wind speed measurement at height n in wind speed bin i ;

- $c_{m,i}$ is the sensitivity factor of v_{eq} on wind speed at height m in wind speed bin i ;
 $c_{n,i}$ is the sensitivity factor of v_{eq} on wind speed at height n in wind speed bin i ;
 $\rho_{m,n,k,i}$ is the correlation coefficient between the uncertainty component k height m and n in wind speed bin i .

The sensitivity factors $c_{m,i}$ and $c_{n,i}$ are gained from the definition of the rotor equivalent wind speed by error propagation as:

$$c_{m,i} = \frac{\partial v_{eq,i}}{\partial v_{m,i}} = \frac{A_m}{A} \left(\frac{v_{m,i}}{v_{eq,i}} \right)^2 \quad (\text{E.33})$$

where

- A_m is the rotor segment attributed to wind speed measurement at height m according to Equation (6) in 9.1.3.2;
 A is the rotor swept area;
 $v_{m,i}$ is the wind speed at height m in wind speed bin i ;
 $v_{eq,i}$ is the equivalent wind speed in bin i .

For the application of Equation (E.33), the wind speed measured at the different measurement heights $v_{m,i}$ shall be bin averaged as a function of the wind speed finally applied for the power curve evaluation (air density normalized and wind shear corrected wind speed).

Equation (E.32) in combination with Equation (E.33) is a simplification if either full correlation or no correlation between the uncertainty components at different measurement heights is assumed. The correlation coefficients suggested in Table E.7 should be applied for the calculation of the uncertainties of v_{eq} . Lower correlation coefficients shall be used only if they are evident.

An additional uncertainty of the determination of the wind speed results from the air density normalization. This component is produced after the rotor averaging of wind speeds and shall be added in the final evaluation of wind speed uncertainties.

Table E.7 – Suggested assumptions for correlations of measurement uncertainties between different measurement heights

Component	Correlation coefficient of uncertainties between different measurement heights	Explanation
Wind shear measurement by cup anemometers		
Wind tunnel calibration	1	Calibration in same wind tunnel is required; high correlation of uncertainties of calibrations of different cup anemometers
Cup anemometer classification	1	Anemometers at different heights measured under very similar climatic conditions
Cup anemometer mounting	1, 0	1, if boom mounted and same boom configuration; 0, if one anemometer top mounted and the other boom mounted
Data acquisition system	0	Different input channels applied

Component	Correlation coefficient of uncertainties between different measurement heights	Explanation
Site effects due to distance between the reference meteorological mast and the test wind turbine	1	As a first approximation, the site effects may be assumed to be identical for the rotor height range.
Uncertainty due to limited number of measurements over rotor height range	1	To a first approximation, this uncertainty is fully correlated between the measurement heights
Wind shear measurement by remote sensing device		
Verification test	1	Normally, very similar conditions of reference sensors at different heights present. Sensitivity of accuracy of remote sensing device on measurement height may be ignored.
Sensitivity analysis or classification	1	Classification performed under very similar conditions at different heights. Dependency of sensitivity of remote sensing device on environmental conditions on measurement height may be ignored.
Uncertainty resulting from control with meteorological mast	1	Same uncertainty assumed for all heights
Uncertainty due to flow variation in different probe volumes at same height	1	Normally, quite similar effect expected at different measurement heights. Sensitivity of error on measurement height may be ignored.
Mounting	1	Similar effect of system mounting in different measurement heights
Site effects due to distance between the measurement and the test wind turbine	1	As a first approximation, the site effects may be assumed to be identical for the rotor height range.
Uncertainty due to limited number of measurements over rotor height range	1	To a first approximation, this uncertainty is fully correlated between the measurement heights

NOTE If one assumes that for all components the correlation coefficient is equal to 1 and all components for each height are the same, this procedure simplifies to the extent that the uncertainty calculation can be done as described in Clause E.6 using $v_{eq,i}$ instead of v_i . For fixed values (such as the calibration) this value can be used for $v_{eq,i}$.

E.13.8 Combining uncertainties in the wind speed measurement for REWS for a hub height meteorological mast plus RSD for shear using an absolute wind speed

The uncertainty of the wind speed measurement at hub height by means of a cup or sonic anemometer and the uncertainty of the wind shear measurement by the remote sensing device can be considered as being independent from each other. Error propagation according to Equation (11) of 9.1.3.4 leads then to Equation (E.34) for the uncertainty of the final rotor equivalent wind speed:

$$u_{v_{eq,final,i}}^2 = f_{r,X,i}^2 u_{v_{h,i}}^2 + v_{h,i}^2 u_{f_{r,X,i}}^2 \quad (\text{E.34})$$

where

$u_{v_{eq,final,i}}$ is the standard uncertainty of final rotor equivalent wind speed in bin i ;

- $f_{r,X,i}$ is the wind shear correction factor in wind speed bin i ;
 $v_{h,i}$ is the wind speed measured at hub height in bin i by cup or sonic anemometer;
 $u_{v_{h,i}}$ is the standard uncertainty of $v_{h,i}$ in wind speed bin i ;
 $u_{f_{r,X,i}}$ is the standard uncertainty of $f_{r,X,i}$ in wind speed bin i .

The uncertainty of the wind shear correction factor is (error propagation according to Equation (9) and Equation (10) of 9.1.3.3):

$$u_{f_{r,X,i}}^2 = \frac{1}{v_{h,X,i}^2} u_{v_{eq,X,i}}^2 + \frac{v_{eq,X,i}^2}{v_{h,X,i}^4} u_{v_{h,X,i}}^2 - 2 \frac{v_{eq,X,i}}{v_{h,X,i}^3} u_{v_{eq,X,i}} u_{v_{h,X,i}} \rho_{u_{v_{eq,X,i}}, u_{v_{h,X,i}}} \quad (\text{E.35})$$

where

- $v_{h,X,i}$ is the wind speed measured by remote sensing device in bin i ;
 $u_{v_{h,X,i}}$ is the standard uncertainty of $v_{h,X,i}$;
 $v_{eq,X,i}$ is the rotor equivalent wind speed as measured with the remote sensing device in bin i ;
 $u_{v_{eq,X,i}}$ is the standard uncertainty of $v_{eq,X,i}$ in bin i ;
 $\rho_{u_{v_{eq,X,i}}, u_{v_{h,X,i}}}$ is the correlation coefficient between uncertainty components $u_{v_{eq,X,i}}$ and $u_{v_{h,X,i}}$ in bin i .

Considering that $v_{eq,X,i}$ and $v_{h,X,i}$ both have the same types of uncertainty sub-components as given in Table E.7 and assuming that these sub-components are independent from each other, Equation (E.35) is equal to:

$$u_{f_{r,X,i}}^2 = \sum_{k=1}^M \frac{1}{v_{h,X,i}^2} u_{v_{eq,X,k,i}}^2 + \frac{v_{eq,X,i}^2}{v_{h,X,i}^4} u_{v_{h,X,k,i}}^2 - 2 \frac{v_{eq,X,i}}{v_{h,X,i}^3} u_{v_{eq,X,k,i}} u_{v_{h,X,k,i}} \rho_{u_{v_{eq,X,k,i}}, u_{v_{h,X,k,i}}} \quad (\text{E.36})$$

where

- $u_{v_{eq,X,k,i}}$ is the standard uncertainty component k of rotor equivalent wind speed as measured with the remote sensing device in bin i , to be analysed according to Equation (E.32);
 $u_{v_{h,X,k,i}}$ is the standard uncertainty component k of hub height wind speed as measured with the remote sensing device in bin i ;
 $\rho_{u_{v_{eq,X,k,i}}, u_{v_{h,X,k,i}}}$ is the correlation coefficient between standard uncertainty components $u_{v_{eq,X,k,i}}$ and $u_{v_{h,X,k,i}}$ in bin i ;
 M is the number of uncertainty components of $v_{eq,X}$ and $v_{h,X}$.

For the application of Equation (E.36), $v_{h,X,i}$ and $v_{eq,X,i}$ are gained from bin averaging the rotor equivalent wind speed and the hub height wind speed measured by the remote sensing device as function of the wind speed finally applied for the power curve evaluation (air density and wind shear normalized wind speed). The correlation coefficients between the standard

uncertainty components of the rotor equivalent wind speed and the hub height wind speed measured by the remote sensing device $\rho_{u_{v_{eq,X}}, u_{v_{h,X}}, k, i}$ shall be estimated from the results of the verification test and sensitivity test or classification of the remote sensing device. In some cases, the assumption of correlation coefficients close to 1 can be justifiable.

Equation (E.36) in combination with Equation (E.34) and Equation (E.32) allows all standard uncertainty components of the final rotor equivalent wind speed to be evaluated. These sub-components shall be implemented in Equations (E.3) and (E.4).

According to Equation (E.34), the total uncertainty of the final rotor equivalent wind speed is larger than the uncertainty of the wind speed measured at hub height by the cup or sonic anemometer (at least as soon as the wind shear correction factor exceeds 1). The uncertainty is increased due to the uncertainty of the wind shear correction factor. According to Equation (E.36), this additional uncertainty vanishes if the uncertainties of the rotor equivalent wind speed and the wind speed measured at hub height by the remote sensing device are fully correlated ($\rho_{u_{v_{eq,X}}, u_{v_{h,X}}, k, i} = 1$) and if the rotor equivalent wind speed is equal to the wind speed at hub height measured by the remote sensing device.

Note that assuming all correlation coefficients are equal to 1 and $u_{v_{eq,X}, k, i}$ is equal to $u_{v_{h,X}, k, i}$ (see E.13.7), the equation for $u_{f_{r,X}, i}^2$ simplifies as follows:

$$u_{f_{r,X}, i} = \left(\frac{v_{h,X,i} - v_{eq,X,i}}{v_{h,X,i}^2} \right) \sqrt{\sum_{k=1}^M u_{v_{h,X}, k, i}^2} \quad (\text{E.37})$$

where

- $v_{h,X,i}$ is the wind speed measured by remote sensing device in bin i ;
- $u_{v_{h,X}, k, i}$ is the standard uncertainty of component k of either $v_{h,X,i}$ or $v_{eq,X,i}$;
- $v_{eq,X,i}$ is the rotor equivalent wind speed as measured with the remote sensing device in bin i .

E.13.9 Combining uncertainties in the wind speed measurement for REWS for hub height meteorological mast RSD for shear using a relative wind speed

A sensitivity analysis and verification test of the remote sensing device for wind shear measurements instead of for absolute wind speed measurements might be available. This means that the accuracy of the remote sensing device in terms of wind shear measurements is verified by the verification test, and that the sensitivity of the wind shear measurement on environmental variables is tested for the classification of the remote sensing device. If such a sensitivity analysis and verification test of the remote sensing device is available, it is useful for the evaluation of uncertainties to introduce a relative wind speed measurement as performed by the remote sensing device:

$$v_{r,RSD,m} = \frac{v_{m,RSD}}{v_{h,RSD}} \quad (\text{E.38})$$

where

- $v_{r,RSD,m}$ is the relative wind speed measured by remote sensing device at m -th height relative to hub height wind speed measured by remote sensing device;
- $v_{h,RSD}$ is the wind speed measured by remote sensing device at hub height;
- $v_{m,RSD}$ is the wind speed measured by remote sensing device at m -th height.

The wind shear correction factor as defined in Equation (9) in 9.1.3.3 can then be rewritten as:

$$f_{r,RSD} = \left(\sum_{m=1}^L \frac{A_m}{A} v_{r,RSD,m}^3 \right)^{\frac{1}{3}} \quad (\text{E.39})$$

where

- A_m is the rotor segment attributed to wind speed measurement at m -th height according to Equation (5) in 9.1.3.2;
- A is the rotor swept area;
- L is the number of measurement heights in the height area of the rotor.

Equation (E.34) can then be applied for assessing the uncertainty of the final rotor equivalent wind speed with the uncertainty components of the wind shear correction factor in wind speed bin i gained from:

$$u_{f_{r,RSD},k,i}^2 = \sum_{m=1}^L \sum_{n=1}^L c_{v_{r,RSD},m,i} u_{v_{r,RSD},m,k,i} c_{v_{r,RSD},n,i} u_{v_{r,RSD},n,k,i} \rho_{r,m,n,k,i} \quad (\text{E.40})$$

where

- L is the number of measurement heights in the height area of the rotor;
- $u_{v_{r,RSD},m,k,i}$ is the standard uncertainty component k of relative wind speed measurement at m -th height in wind speed bin i ;
- $u_{v_{r,RSD},n,k,i}$ is the standard uncertainty component k of relative wind speed measurement at n -th height in wind speed bin i ;
- $c_{v_{r,RSD},m,i}$ is the sensitivity factor of $f_{r,RSD}$ on relative wind speed at m -th height in wind speed bin i ;
- $c_{v_{r,RSD},n,i}$ is the sensitivity factor of $f_{r,RSD}$ on relative wind speed at n -th height in wind speed bin i .

The sensitivity factors $c_{v_{r,RSD},m,i}$ and $c_{v_{r,RSD},n,i}$ are:

$$c_{v_{r,RSD},m,i} = \frac{\partial f_{r,RSD,i}}{\partial v_{r,RSD,m,i}} = \frac{A_m}{A} \left(\frac{v_{r,RSD,m,i}}{f_{r,RSD,i}} \right)^2 \quad (\text{E.41})$$

The same uncertainty components as given in Table E.7 shall be applied, but in terms of the remote sensing device relative wind speed measurements rather than in terms of absolute wind speed measurements. In addition, the same correlation coefficients of uncertainty components

between two measurement heights as shown in Table E.7 shall be assumed for the relative wind speed measurements, unless evidence exists that other values are present.

The wind speed measured at hub height by the cup or sonic anemometer v_h , the wind shear correction factor $f_{r,RSD}$ and the relative wind speeds $v_{r,RSD}$ measured by the remote sensing device shall be bin averaged against the final wind speed applied for the power curve evaluation for the application of Equations (E.34), (E.40) and (E.41) (wind shear corrected and air density normalized wind speed).

E.13.10 Combining uncertainties in the wind speed measurement from REWS due to wind veer across the whole rotor $u_{vREWS,veer,i}$

The rotor equivalent wind speed (taking consideration of wind veer according to Annex Q) is dependent on wind speeds and wind directions across the rotor height range. Often, the uncertainties of wind direction and wind speed measurements can be considered as being independent from each other. Then Equation (E.31) in E.13.7 transforms to:

$$u_{v_{eq},i}^2 = \sum_{k=1}^M u_{v_{eq},k,i}^2 + \sum_{l=1}^N u_{v_{eq},l,i}^2 \quad (E.42)$$

where

- M is the number of uncertainty sub-components of v_{eq} related to wind speed measurements;
- i is the wind speed bin referring to bin averaging of power curve;
- $u_{v_{eq},k,i}$ is the standard uncertainty sub-component k of v_{eq} related to wind speed measurements in wind speed bin i ;
- N is the number of uncertainty sub-components of v_{eq} related to wind direction measurements;
- $u_{v_{eq},l,i}$ is the standard uncertainty sub-component l of v_{eq} related to wind direction measurements in wind speed bin i .

The left term in Equation (E.42) covers all wind speed measurement uncertainties of the rotor equivalent wind speed and shall be treated according to Clause E.8, while the sensitivity factors given in Equations (E.32) and (E.42) shall be multiplied by the third power of $\cos(\varphi_m)$, where m is the wind direction difference to hub height at the m -th measurement height. This factor shall also be introduced in Equation (E.39) for the definition of the wind shear correction factor.

The right term in this Equation (E.42) covers all wind direction measurement uncertainties of the rotor equivalent wind speed. As is explained in E.8.1, a difficulty in assessing this uncertainty is related to the fact that the uncertainties of the wind direction measurements at the different height levels are correlated with each other. Similar to Equation (E.32), the sub-wind direction uncertainty components of v_{eq} in wind speed bin i shall be calculated from the uncertainty components of the wind direction measurements in single measurement heights by the method of error propagation:

$$u_{v_{eq},l,i}^2 = \sum_{m=1}^L \sum_{n=1}^L c_{m,i} u_{v_m,l,i} c_{n,i} u_{v_n,l,i} \rho_{m,n,l,i} \quad (E.43)$$

where

- L is the number of measurement heights in the height area of the rotor;
- $u_{v_m,l,i}$ is the standard uncertainty sub-component l of wind direction measurement at m -th height relative to hub height in wind speed bin i ;
- $u_{v_n,l,i}$ is the standard uncertainty sub-component l of wind direction measurement at n -th height relative to hub height in wind speed bin i ;
- $c_{m,i}$ is the sensitivity factor of v_{eq} on wind direction at height m relative to hub height in wind speed bin i ;
- $c_{n,i}$ is the sensitivity factor of v_{eq} on wind direction at height n relative to hub height in wind speed bin i ;
- $\rho_{m,n,l,i}$ is the correlation coefficient between standard uncertainty component l of wind direction measurement at height m and height n relative to hub height in wind speed bin i (wind speed bin referring to the power curve).

The sensitivity factors $c_{m,i}$ and $c_{n,i}$ are gained from the definition of the rotor equivalent wind speed by consideration of wind veer by error propagation as:

$$c_{m,i} = \frac{\partial v_{eq,i}}{\partial \varphi_{m,i}} = -\sin(\varphi_{m,i}) \cos^2(\varphi_{m,i}) \frac{A_m}{A} \frac{v_{m,i}^3}{v_{eq,i}^2} \quad (\text{E.44})$$

where

- A_m is the rotor segment attributed to wind speed measurement at m -th height according to Equation (5) in 9.1.3.2;
- A is the rotor swept area;
- $\varphi_{m,i}$ is the wind direction at m -th height relative to wind direction at hub height in wind speed bin i ;
- $v_{m,i}$ is the wind speed at height m in wind speed bin i ;
- $v_{eq,i}$ is the equivalent wind speed in bin i .

For the application of Equation (E.33), the relative wind directions $\varphi_{m,i}$ and wind speeds $v_{m,i}$ measured at the different measurement heights shall be bin averaged as function of the wind speed finally applied for the power curve evaluation (air density normalized, wind shear and wind veer corrected wind speed).

Equation (E.43) in combination with Equation (E.45) simplifies if full correlation, no correlation or anti-correlation between the uncertainty components at different measurement heights is assumed. The correlation coefficients suggested in Table E.8 shall be applied for the calculation of the uncertainties in terms of wind direction of v_{eq} . Lower correlation coefficients shall be used only if they are evident and relevant data is reported.

Table E.8 – Suggested correlation assumptions for relative wind direction measurement uncertainties at different measurement heights

Component	Correlation coefficient of uncertainties between different measurement heights	Explanation
Relative wind direction measurement by vanes or sonic anemometers		
Wind tunnel calibration of vanes used at height m , n and hub height	0,5	The uncertainty of the calibration of single vanes is basically correlated. But as the wind direction differences relative to hub height at heights m and n include the direction measurement at hub height, a correlation coefficient of 0,5 is suggested
Influence of meteorological mast on wind direction measurements at heights m , n and hub height	0,5;1	If the vanes at the two heights m and n are mounted in the same way relative to the same mast structure, the uncertainties are fully correlated across the two heights m and n . However, the uncertainty itself gets 0 if the vane at hub height is also influenced in the same way by the meteorological mast. If the vanes at the two heights m and n are mounted differently relative to the mast structure, or if the mast structure differs much at two heights m and n , the uncertainties and these two heights can be considered as being uncorrelated. But as the wind direction difference relative to hub height at the two heights m and n includes the direction measurement at hub height, a correlation coefficient of 0,5 is suggested for this case.
Northing of wind direction measurement at height m , n and hub height	0,5	The uncertainty of the northing of single vanes is basically uncorrelated. But as the wind directions relative to hub height at the two heights m and n include the direction measurement at hub height, a correlation coefficient of 0,5 is suggested.
Data acquisition system (DAS) used to monitor the vanes at height m , n and hub height	0,5	The uncertainty of the DAS channels of single vanes is often basically uncorrelated. But as the wind direction differences relative to hub height at the two heights m and n include the direction measurement at hub height, a correlation coefficient of 0,5 is suggested.

Component	Correlation coefficient of uncertainties between different measurement heights	Explanation
Relative wind direction measurement by remote sensing device		
Wind tunnel calibration of reference vanes used for verification tests at heights m , n and hub height	0,5;1	These uncertainties of the reference measurement of the verification tests are basically uncorrelated at two heights m and n . But as the wind direction differences relative to hub height at the two heights m and n include the direction measurement at hub height, a correlation coefficient of 0,5 is suggested. An exception is the case of using the same verification test at two heights m and n (e.g. if the mast used for the verification test does not cover the entire range of the rotor: Then the uncertainties are fully correlated across the two heights m and n . However, the uncertainty itself is zero if the same verification is also applied for hub height.
Northing of reference vanes used for verification tests at heights m , n and hub height	0,5;1	
Data acquisition system (DAS) used to monitor reference vanes applied for verification tests at heights m , n and hub height	0,5;1	
Mean deviation of RSD measurement and reference measurement during verification tests at heights m , n and hub height	0,5;1	
Statistical uncertainty of verification tests at height m , n and hub height	0,5;1	
Uncertainty of verification tests at heights m , n and hub height due to a possible misalignment of the RSD	1	The uncertainty is fully correlated across the heights m and n . However, the respective uncertainty is zero for all heights m and n as the misalignment error cancels out when evaluating the difference of the wind direction at heights m and n relative to hub height.
Uncertainty of verification tests at heights m , n and hub height caused by the assumption of equal wind conditions across the probe volumes of the RSD.	1	These uncertainties are highly correlated across the heights m and n . However, the respective uncertainties are close to zero as the influence on the wind direction measurement at height m (or n) and hub height cancels out when evaluating the difference of the wind directions.
Uncertainty of verification tests at heights m , n and hub height due to a possible separation of the centre of the measurement volume of the RSD and the position of the reference mast	1	
Uncertainty due to sensitivity of RSD measurements and environmental variables.	1	
Possible misalignment of the RSD at the power curve test	1	This uncertainty is fully correlated across the heights m and n . However, the respective uncertainties are zero for all heights m and n as the influence on the wind direction measurement at height m (or n) and hub height cancels out when evaluating the difference of the wind directions.
Assumption of equal wind conditions across the probe volumes of the RSD at heights m , n and hub height	1	This uncertainty is highly correlated across the heights m and n . However, the respective uncertainties are close to zero as the influence on the wind direction measurement at height m (or n) and hub height cancels out when evaluating the difference of the wind directions.

Component	Correlation coefficient of uncertainties between different measurement heights	Explanation
Monitoring wind with a meteorological mast at the power curve test	0,5; 1	If the monitoring meteorological mast covers the two heights m and n with different vanes, the uncertainties of the monitoring are basically uncorrelated at two heights m and n . But as the wind direction differences relative to hub height at the two heights m and n include the direction measurement at hub height, a correlation coefficient of 0,5 is suggested in this case. If the monitoring mast provides only one measurement height that can be used to check the RSD measurement at the heights m and n , the uncertainty is fully correlated across the heights m and n . However, the respective uncertainty is then close to zero for all heights m and n as the uncertainty cancels out when evaluating the difference of the wind direction at height m (or n) and hub height.

E.13.11 Combining uncertainties in the wind speed measurement from flow distortion due to site calibration $u_{VT,i}$

The following uncertainty components are combined to calculate the category B uncertainty for a site calibration, $u_{VT,i}$:

$$u_{VT,i,j} = \sqrt{u_{VT,precal,i,j}^2 + u_{VT,postcal,i,j}^2 + u_{VT,class,i,j}^2 + u_{VT,mnt,i,j}^2 + u_{VT,lgt,i,j}^2 + u_{VT,coc,i,j}^2 \cdot V_i^2 + u_{VT,rmv,i,j}^2 \cdot V_i^2 + u_{VT,sv,i,j}^2 + u_{dVT,i,j}^2 + s_{sc}^2} \quad (E.45)$$

where

- $u_{VT,i,j}$ is the uncertainty from the site calibration;
- $u_{VT,precal,i,j}$ is the uncertainty related to the calibration of the anemometers;
- $u_{VT,postcal,i,j}$ is the uncertainty related to the post-calibration or in-situ calibration of the anemometers;
- $u_{VT,class,i,j}$ is the uncertainty related to the classification of the sensors;
- $u_{VT,mnt,i,j}$ is the uncertainty related to the mounting of the sensors;
- $u_{VT,lgt,i,j}$ is the uncertainty related to the lightning finial
- $u_{VT,coc,i,j}$ is the uncertainty related to the change of correction value between bins;
- $u_{VT,rmv,i,j}$ is the uncertainty related to the removal of the wind direction sensor between site calibration and power performance test;
- $u_{VT,sv,i,j}$ is the uncertainty related to the seasonal variation between site calibration and power performance test;
- $u_{dVT,i,j}$ is the uncertainty related to the data acquisition of the wind speed signal;
- s_{sc} is the category A standard uncertainty of the site calibration, IEC 61400-12-3.

To obtain the uncertainty in one wind speed bin across all the directional sectors the weighted average shall be used:

$$u_{VT,i} = \frac{\sum_j u_{VT,i,j} N_{i,j}}{\sum_j N_{i,j}} \quad (\text{E.46})$$

where

$N_{i,j}$ is the number of power curve data sets for wind speed bin i and wind direction bin j .

E.13.12 Combining uncertainties for the temperature measurement $u_{T,i}$

The following uncertainty components are combined to calculate the category B uncertainty for temperature, $u_{T,i}$:

$$u_{T,i} = \sqrt{u_{T,cal,i}^2 + u_{T,shield,i}^2 + u_{T,mnt,i}^2 + u_{dT,i}^2} \quad (\text{E.47})$$

where

- $u_{T,i}$ is the uncertainty of the temperature measurement;
- $u_{T,cal,i}$ is the uncertainty related to the calibration of the temperature sensor;
- $u_{T,shield,i}$ is the uncertainty related to the shielding of the temperature sensor;
- $u_{T,mnt,i}$ is the uncertainty related to the mounting of the temperature sensor;
- $u_{dT,i}$ is the uncertainty related to the data acquisition of the temperature signal.

If we make the following assumptions:

- The standard uncertainty of the temperature sensor is 0,5 °C.
- The shielding of the temperature sensor is 2 °C.
- The standard uncertainty due to mounting effects of the temperature sensor is dependent on the vertical distance from the hub height. With the temperature sensor mounted within 10 m of hub height a standard uncertainty of 0,3 °C is assumed.
- Considering a temperature range of 40 °C of the measurement channel and a standard uncertainty of the data acquisition system of 0,1 % of this range.

Then the numerical calculation for the standard uncertainty of the air temperature in each bin, in kelvin, is:

$$u_{T,i} = \sqrt{0,5^2 + 2,0^2 + 0,3^2 + (0,1\% \cdot 40)^2} = 2,1 \quad (\text{E.48})$$

E.13.13 Combining uncertainties for the pressure measurement $u_{B,i}$

The following uncertainty components are combined to calculate the category B uncertainty for pressure, $u_{B,i}$:

$$u_{B,i} = \sqrt{u_{B,cal,i}^2 + u_{B,mnt,i}^2 + u_{dB,i}^2} \quad (E.49)$$

where

- $u_{B,i}$ is the uncertainty of the pressure measurement;
- $u_{B,cal,i}$ is the uncertainty related to the calibration of the pressure sensor;
- $u_{B,mnt,i}$ is the uncertainty related to the mounting of the pressure sensor;
- $u_{dB,i}$ is the uncertainty related to the data acquisition of the pressure signal.

If we make the following assumptions:

- The pressure sensor has a standard uncertainty of 3,0 hPa. It is assumed that the pressure is corrected to the hub height according to ISO 2533 (which, for a standard atmosphere and a height difference of 98 m between the sensor and the hub, is 11,7 hPa). The standard uncertainty due to deployment is estimated to be 10 % of the correction, which is 1,17 hPa.
- Considering a pressure range of 100 hPa of the measurement channel and a standard uncertainty of the data acquisition system of 0,1 % of this range.

Then the numerical calculation for the standard uncertainty of the air pressure, in hectopascals, is:

$$u_{B,i} = \sqrt{3,0^2 + 1,17^2 + (0,1\% \cdot 100)^2} = 3,2 \quad (E.50)$$

E.13.14 Combining uncertainties for the humidity measurement $u_{RH,i}$

The following uncertainty components are combined to calculate the category B uncertainty for humidity, $u_{RH,i}$:

$$u_{RH,i} = \sqrt{u_{RH,cal,i}^2 + u_{RH,mnt,i}^2 + u_{dRH,i}^2} \quad (E.51)$$

where

- $u_{RH,i}$ is the uncertainty of the relative humidity measurement;
- $u_{RH,cal,i}$ is the uncertainty related to the calibration of the relative humidity sensor;
- $u_{RH,mnt,i}$ is the uncertainty related to the mounting of the relative humidity sensor;
- $u_{dRH,i}$ is the uncertainty related to the data acquisition of the relative humidity signal.

If we make the following assumptions:

- a) The relative humidity sensor has a standard uncertainty of 1 %.
- b) The mounting of the sensor has a standard uncertainty of 0,1 %.
- c) Considering a pressure range of 100 % of the measurement channel and a standard uncertainty of the data acquisition system of 0,1 % of this range.

Then the numerical calculation for the standard uncertainty of the relative humidity is:

$$u_{RH,i} = \sqrt{(1,0\%)^2 + (0,1\%)^2 + (0,1\% \cdot 100\%)^2} = 1,0\% \quad (\text{E.52})$$

E.13.15 Combining uncertainties for the method related components $u_{M,i}$

The following uncertainty components are combined to calculate the category B uncertainty for related to the methodology, $u_{M,i}$:

$$u_{M,i} = \sqrt{u_{M,\text{shear},i}^2 + u_{M,\text{veer},i}^2 + u_{M,\text{upflow},i}^2 + u_{M,\text{ti},i}^2 + u_{M,\text{sfx},i}^2 + u_{M,\text{cc},i}^2} \quad (\text{E.53})$$

where

- $u_{M,i}$ is the uncertainty related to the applied methodology;
- $u_{M,\text{shear},i}$ is the uncertainty related to missing information regarding shear across the whole rotor;
- $u_{M,\text{veer},i}$ is the uncertainty related to missing information regarding wind veer across the whole rotor;
- $u_{M,\text{upflow},i}$ is the uncertainty related to missing information regarding upflow across the whole rotor;
- $u_{M,\text{ti},i}$ is the uncertainty related to missing information regarding turbulence when no hub-height meteorological mast wind speed signal is available;
- $u_{M,\text{sfx},i}$ is the uncertainty related to the unquantifiable seasonal effects on the turbine;
- $u_{M,\text{cc},i}$ is the uncertainty related to the measurement in a cold climate.

E.13.16 Combining uncertainties for the wind direction measurement with wind vane or sonic anemometer $u_{WV,i}$

The following uncertainty components are combined to calculate the category B uncertainty for the wind direction measurement with wind vane or sonic anemometer, $u_{WV,i}$:

$$u_{WV,i} = \sqrt{u_{WV,\text{cal},i}^2 + u_{WV,\text{nm},i}^2 + u_{WV,\text{bo},i}^2 + u_{WV,\text{oe},i}^2 + u_{WV,\text{mda},i}^2 + u_{dWV,i}^2} \quad (\text{E.54})$$

where

- $u_{WV,i}$ is the uncertainty related to the wind direction measured with a mast mounted wind direction sensor (wind vane or sonic anemometer);
- $u_{WV,\text{cal},i}$ is the uncertainty related to the calibration of the wind direction sensor;
- $u_{WV,\text{nm},i}$ is the uncertainty related to north marking of the wind direction sensor;
- $u_{WV,\text{bo},i}$ is the uncertainty related to the boom orientation on which the wind direction sensor is mounted;
- $u_{WV,\text{oe},i}$ is the uncertainty related to the influence of the meteorological mast on the wind direction measurement;
- $u_{WV,\text{mda},i}$ is the uncertainty related to the magnetic declination angle;

$u_{dWV,i}$ is the uncertainty related to the data acquisition of the signal from the wind direction sensor.

E.13.17 Combining uncertainties for the wind direction measurement with RSD $u_{WR,i}$

The following uncertainty components are combined to calculate the category B uncertainty for the wind direction measurement with RSD, $u_{WR,i}$:

$$u_{WR,i} = \sqrt{u_{WR,ver,i}^2 + u_{WR,class,i}^2 + u_{WR,mon,i}^2 + u_{WR,fv,i}^2 + u_{WR,align,i}^2 + u_{WR,mda,i}^2 + u_{dWR,i}^2} \quad (E.55)$$

where

- $u_{WR,i}$ is the uncertainty related to the wind direction measured with an RSD;
- $u_{WR,ver,i}$ is the uncertainty related to the verification test of the RSD;
- $u_{WR,class,i}$ is the uncertainty related to the classification of the RSD;
- $u_{WR,mon,i}$ is the uncertainty related to the monitoring of the RSD;
- $u_{WR,fv,i}$ is the uncertainty related to the variation in wind flow across the measured volume;
- $u_{WR,align,i}$ is the uncertainty related to the alignment of the RSD;
- $u_{WR,mda,i}$ is the uncertainty related to the magnetic declination angle;
- $u_{dWR,i}$ is the uncertainty related to the data acquisition of the signal from the wind direction sensor.

E.13.18 Combined category B uncertainties

The category B uncertainties in each bin are combined as:

$$u_i = \sqrt{u_{P,i}^2 + u_{M,Tinorm,i}^2 + c_{V,i}^2 u_{V,i}^2 + c_{T,i}^2 u_{T,i}^2 + c_{B,i}^2 u_{B,i}^2 + c_{RH,i}^2 u_{RH,i}^2 + c_{V,i}^2 u_{M,i}^2} \quad (E.56)$$

E.13.19 Combined standard uncertainty – Power curve

The combined standard uncertainties of each bin of the power curve are found by combining the category A uncertainty with all the category B uncertainties.

$$u_{C,i} = \sqrt{s_i^2 + u_i^2} = \sqrt{s_i^2 + u_{P,i}^2 + u_{M,Tinorm,i}^2 + c_{V,i}^2 u_{V,i}^2 + c_{T,i}^2 u_{T,i}^2 + c_{B,i}^2 u_{B,i}^2 + c_{RH,i}^2 u_{RH,i}^2 + c_{V,i}^2 u_{M,i}^2} \quad (E.57)$$

E.13.20 Combined standard uncertainty – Energy production

The combined standard uncertainty of AEP is found by combining individually the category A and B uncertainties bin-wise. This can be accomplished according to the full formulation of Equation (E.4) and shown in Equation (E.58) or the simplified and conservative formulation of Equation (E.5), shown in Equation (E.59):

$$u_{AEP} = N_h \sqrt{\sum_{i=1}^N f_i^2 \sum_{k=1}^{M_A} c_{k,i}^2 s_{k,i}^2 + \sum_{k=1}^{M_B} \left(\sum_{i=1}^N f_i c_{k,i} u_{k,i} \right)^2} \quad (E.58)$$

$$u_{AEP} = N_h \sqrt{\sum_{i=1}^N f_i^2 s_i^2 + \left(\sum_{i=1}^N f_i u_i \right)^2} \quad (E.59)$$

where

f_i is the relative occurrence of wind speed between V_{i-1} and V_i : $F(V_i) - F(V_{i-1})$ within bin i .

E.14 Relevance of uncertainty components under specified conditions

In accordance with ISO/IEC Guide 98-3, in some cases an uncertainty component can be evaluated and found to be insignificant and therefore not be included in the evaluation of the uncertainty of the measurement result.

In this document, various uncertainty components should only be assigned a zero value in specific cases or when a specific set of conditions has been met. An example is the uncertainty of $u_{VT,coc,i,j}$ which relates to the uncertainty applied in a site calibration when wind direction bins with a change in correction factor larger than 2 % compared to the neighbouring bin are not eliminated from the data set, see IEC 61400-12-3. Only when the decision is made not to remove such bins does $u_{VT,coc,i,j}$ get a value, otherwise it has value zero.

It is up to the user of this document to determine in which cases an uncertainty component can be deemed irrelevant and hence set to zero. This should be documented (including the justification) to ensure reproducibility of the uncertainty calculation.

E.15 Reference tables

Table E.9 – Uncertainties from air density normalization

Bin No.	Air density normalized wind speed	Wind speed before normalization (un- normalized)	Difference of normalized and un-normalized wind speed	Uncertainty due to air density normalization
i	$V_{n,i}$ m/s	$V_{un,i}$ m/s	$V_{n,i} - V_{un,i}$ m/s	$u_{p,i}$ m/s
4	2,03	2,025	0,008	0,004
5	2,51	2,501	0,013	0,007
6	3,03	3,015	0,014	0,007
7	3,52	3,501	0,016	0,008
8	4,02	4,000	0,022	0,011
9	4,53	4,503	0,028	0,014
10	5,00	4,976	0,029	0,014
11	5,55	5,523	0,031	0,016
12	6,03	5,991	0,035	0,017
13	6,56	6,503	0,055	0,028
14	7,01	6,970	0,041	0,021

Bin No.	Air density normalized wind speed	Wind speed before normalization (un- normalized)	Difference of normalized and un-normalized wind speed	Uncertainty due to air density normalization
i	$V_{n,i}$ m/s	$V_{un,i}$ m/s	$V_{n,i} - V_{un,i}$ m/s	$u_{p,i}$ m/s
15	7,55	7,485	0,061	0,031
16	8,04	7,979	0,065	0,032
17	8,57	8,503	0,064	0,032
18	9,08	8,999	0,080	0,040
19	9,58	9,494	0,090	0,045
20	10,10	10,007	0,091	0,045
21	10,61	10,501	0,113	0,056
22	11,10	10,982	0,122	0,061
23	11,60	11,475	0,121	0,060
24	12,11	11,986	0,125	0,062
25	12,64	12,504	0,139	0,070
26	13,17	13,034	0,134	0,067
27	13,59	13,467	0,123	0,061
28	14,18	14,020	0,157	0,079
29	14,62	14,479	0,140	0,070
30	15,07	14,916	0,149	0,075
31	15,76	15,611	0,151	0,075
32	16,09	15,941	0,147	0,073
33	16,83	16,680	0,150	0,075
34	17,03	16,926	0,100	0,050
35	17,81	17,586	0,226	0,113
36	18,19	18,052	0,135	0,068
37	18,5	0,152 2	0,135	0,068
38	19,0	0,152 1	0,135	0,068
39	19,5	0,153 9	0,135	0,068
40	20,0	0,154 1	0,135	0,068
41	20,5	0,150 5	0,135	0,068
42	21,0	0,151 2	0,135	0,068
43	21,5	0,154 8	0,135	0,068
44	22,0	0,153	0,135	0,068
45	22,5	0,153 3	0,135	0,068
46	23,0	0,155 7	0,135	0,068
47	23,5	0,156 7	0,135	0,068

Table E.10 – Sensitivity factors

Bin No. <i>i</i>	Power curve (database B)		Sensitivity factors		
	Wind speed	Electric power	Wind speed	Air temperature	Air pressure
	V_i m/s	P_i kW	$c_{V,i}$ kW/m/s	$c_{T,i}$ kW/K	$c_{B,i}$ kW/hPa
4	2,13	–3,64	1,712	0,013	0,004
5	2,49	–3,65	0,014	0,013	0,004
6	2,99	–3,78	0,269	0,013	0,004
7	3,51	–2,19	3,062	0,008	0,002
8	3,99	–0,43	3,645	0,001	0,000
9	4,50	6,04	12,825	0,021	0,006
10	4,98	27,70	44,664	0,096	0,027
11	5,52	67,39	74,049	0,234	0,067
12	5,98	111,30	94,430	0,386	0,110
13	6,51	160,95	95,019	0,558	0,159
14	7,01	209,42	95,472	0,727	0,207
15	7,50	261,96	107,566	0,909	0,259
16	8,00	327,63	131,992	1,137	0,323
17	8,50	395,23	136,290	1,372	0,390
18	8,99	462,01	134,677	1,603	0,456
19	9,49	556,06	187,824	1,930	0,549
20	10,00	629,80	145,079	2,186	0,622
21	10,47	703,06	155,957	2,440	0,694
22	11,00	786,55	157,358	2,729	0,776
23	11,50	836,48	100,000	2,903	0,826
24	11,99	893,52	116,327	3,101	0,882
25	12,49	928,61	70,200	3,223	0,917
26	13,03	956,44	51,481	3,319	0,944
27	13,50	971,30	31,702	3,371	0,959
28	14,00	980,92	19,200	3,404	0,968
29	14,48	988,17	15,208	3,429	0,976
30	15,00	993,46	10,192	3,448	0,981
31	15,49	993,71	0,408	3,449	0,981
32	15,99	995,70	4,000	3,455	0,983
33	16,54	996,22	0,909	3,457	0,983
34	17,02	996,42	0,417	3,458	0,984
35	17,48	996,48	0,217	3,458	0,984
36	17,95	996,50	0,000	3,458	0,984
37	18,49	995,71	0,556	3,457	0,983
38	18,97	996,6	0,833	3,459	0,984
39	19,42	996,1	1,111	3,457	0,983
40	19,96	994,1	3,704	3,450	0,981
41	20,51	987,4	12,182	3,427	0,975
42	20,88	996,9	25,676	3,460	0,984

Table E.11 – Category B uncertainties

Bin No. <i>i</i>	Electric power $u_{P,i}$ kW	Wind speed $u_{V,i}$ m/s	Wind speed $c_{V,i} \cdot u_{V,i}$ kW	Air temperature $u_{T,i}$ K	Air temperature $c_{T,i} \cdot u_{T,i}$ kW	Air pressure $u_{B,i}$ hPa	Air pressure $c_{B,i} \cdot u_{B,i}$ kW
4	6,29	0,19	0,33	2,09	0,03	3,18	0,01
5	6,29	0,19	0,00	2,09	0,03	3,18	0,01
6	6,29	0,19	0,05	2,09	0,03	3,18	0,01
7	6,29	0,19	0,60	2,09	0,02	3,18	0,01
8	6,29	0,20	0,71	2,09	0,00	3,18	0,00
9	6,29	0,20	2,53	2,09	0,04	3,18	0,02
10	6,29	0,20	8,85	2,09	0,20	3,18	0,09
11	6,30	0,20	14,82	2,09	0,49	3,18	0,21
12	6,32	0,20	19,04	2,09	0,81	3,18	0,35
13	6,35	0,20	19,34	2,09	1,17	3,18	0,51
14	6,39	0,21	19,58	2,09	1,52	3,18	0,66
15	6,44	0,21	22,28	2,09	1,90	3,18	0,82
16	6,52	0,21	27,66	2,09	2,37	3,18	1,03
17	6,62	0,21	28,87	2,09	2,86	3,18	1,24
18	6,74	0,21	28,86	2,09	3,35	3,18	1,45
19	6,93	0,22	40,71	2,09	4,03	3,18	1,75
20	7,09	0,22	31,82	2,09	4,57	3,18	1,98
21	7,28	0,22	34,61	2,09	5,10	3,18	2,21
22	7,51	0,22	35,38	2,09	5,70	3,18	2,47
23	7,65	0,23	22,77	2,09	6,06	3,18	2,63
24	7,82	0,23	26,81	2,09	6,48	3,18	2,81
25	7,93	0,23	16,41	2,09	6,73	3,18	2,92
26	8,02	0,24	12,20	2,09	6,93	3,18	3,00
27	8,07	0,24	7,61	2,09	7,04	3,18	3,05
28	8,10	0,24	4,67	2,09	7,11	3,18	3,08
29	8,13	0,25	3,75	2,09	7,16	3,18	3,10
30	8,14	0,25	2,55	2,09	7,20	3,18	3,12
31	8,14	0,25	0,10	2,09	7,20	3,18	3,12
32	8,15	0,26	1,03	2,09	7,22	3,18	3,13
33	8,15	0,26	0,24	2,09	7,22	3,18	3,13
34	8,15	0,26	0,11	2,09	7,22	3,18	3,13
35	8,15	0,27	0,06	2,09	7,22	3,18	3,13
36	8,15	0,27	0,00	2,09	7,22	3,18	3,13
37	8,15	0,28	0,15	2,09	7,22	3,18	3,13
38	8,15	0,28	0,23	2,09	7,22	3,18	3,13
39	8,15	0,28	0,32	2,09	7,22	3,18	3,13
40	8,15	0,29	1,07	2,09	7,21	3,18	3,12
41	8,12	0,29	3,54	2,09	7,16	3,18	3,10
42	8,15	0,29	7,54	2,09	7,23	3,18	3,13

Annex F
(normative)

Wind tunnel calibration procedure for anemometers

Refer to IEC 61400-50-1.

Annex G (normative)

Mounting of instruments on the meteorological mast

Refer to IEC 61400-50-1.

NOTE Text in Annex G of IEC 61400-12-1:2017 specific to power performance has been moved to the main body of this document and more generic text has been included in IEC 61400-50-1.

Annex H (normative)

Power performance testing of small wind turbines

H.1 General

Small wind turbines (as defined by IEC 61400-2) require special provisions for power performance testing. In particular, wind turbines used to charge batteries shall be tested in a way that is representative of normal operation but that reduces or eliminates the influence of the particular battery configuration and condition used during testing, and wind turbines that use inverters for grid connection shall be tested in conjunction with those inverters to give results representative of normal operation. In all cases the relevant power is useful power to the consumer, thus power is the power injected into the grid for grid connected wind turbines, similarly power delivered to the batteries for battery charging wind turbines. When testing a small wind turbine, all requirements described in this document shall be met with the following additions and changes.

H.2 Wind turbine system definition and installation

With respect to 6.2 the following are required.

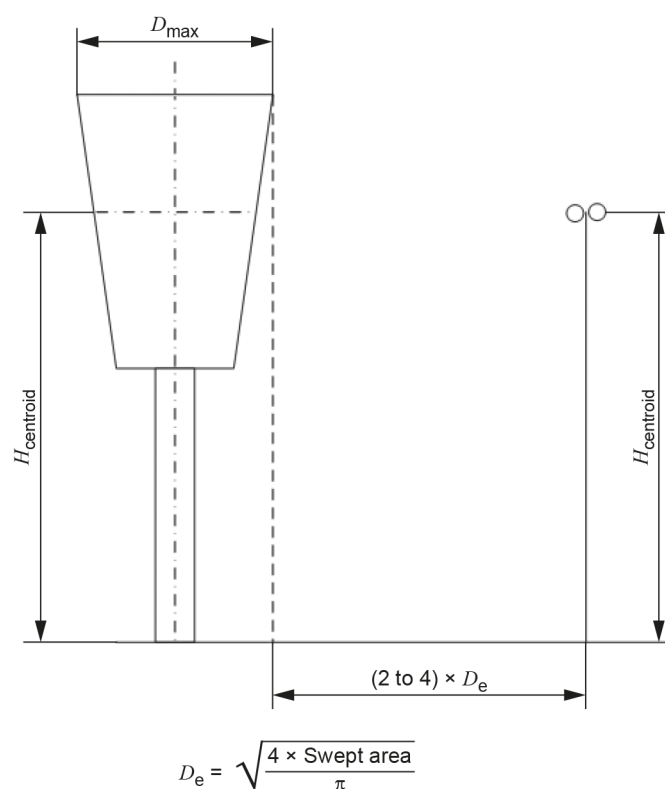
- a) When characterizing battery charging performance, the wind turbine generator system shall include the wind turbine, wind turbine tower, wind turbine controller, wiring between the wind turbine and the load, and charge controller, which is a voltage protection device that reduces wind turbine power output when batteries are fully charged. The system may include a dump load that is used to dissipate energy from the wind turbine when the batteries are fully charged. The wind turbine generator system does not include a battery bank because it is considered part of the load. Other energy storage devices may be substituted for batteries throughout this Annex H.
- b) When characterizing system output to a grid, the wind turbine generator system shall include the wind turbine, wind turbine tower, wind turbine controller, wiring between the wind turbine and the load, and any additional controllers. In addition, the system may include a voltage inverter. If a transformer is installed between the voltage inverter and the grid, it may be considered as either part of the wind turbine generator system or the load. If the grid-connected wind turbine system incorporates a battery bank and/or dump load, it shall be considered part of the system.
- c) The wind turbine shall be connected to an electrical load that is representative of the load for which the wind turbine is designed. In the case of battery charging applications, the load consists of a battery bank, a voltage regulator, and a means to dissipate the power that passes through the voltage regulator (or dump load). In the ideal test set-up, the battery bank does not store energy produced by the wind turbine. Rather all wind turbine output is routed through the voltage regulator. Therefore, the battery bank may be smaller than typically recommended for the wind turbine as long as voltage at the connection of the wind turbine to the load can be maintained within the specifications in e) and f) below.
- d) The wind turbine shall be installed using the manufacturer's specified mounting system. If a wind turbine is not supplied with a specific mounting system, the wind turbine generator should be mounted at a hub height of at least 10 m.
- e) In order to minimize differences in results due to wiring between the wind turbine and the load, the total wire run length, measured from the base of the tower to the load, shall be at least 8 rotor diameters. Wiring between the small wind turbine and the load shall be in accordance with the manufacturer's specifications for this wire run length. If the specifications provide for a range of wire sizes, wires shall be sized as close as possible to the average of that range. If no specifications are provided, the wiring shall be sized so that voltage drop between the wind turbine generator and the load is equivalent to 10 % of nominal voltage at rated power.

- f) The voltage regulator shall be capable of maintaining voltage at the connection of the wind turbine to load within 10 % of the settings given in Table H.1 over the full range of power output of the wind turbine. The 1-min average of the load voltage shall be within 5 % of the settings given in Table H.1 to be included in the usable data set.

H.3 Meteorological mast location

With respect to 6.3.2 the following are required.

- If it is more practical to mount the anemometer on a long boom that is connected to the wind turbine tower, a separate meteorological mast is not required. To minimize the flow distortion on the small wind turbine rotor from the wake from the anemometer, the wind vane and their mounting hardware, all such components shall be located at least 3 m away from any part of the wind turbine rotor. In addition, the anemometer mounting should be configured to minimize its cross-sectional area above the level that is 1,5 rotor diameters below hub height. The boom shall be installed such that structural vibrations are not introduced into the wind speed signal.
- For vertical axis wind turbines hub height is defined as the height of the centroid of the swept area of the wind turbine rotor.
- For vertical axis wind turbines the distance between wind turbine and meteorological mast is illustrated in Figure H.1.



IEC

Figure H.1 – Definition of hub height and meteorological mast location for vertical axis wind turbines

H.4 Test equipment

- In 7.1: when measuring DC outputs for battery charging wind turbines with a swept area of 40 m² or less, the use of measurement equipment with an accuracy of 2 % or better is permitted for the purposes of measuring current and power, rather than the 0,5 % accuracy referred to in 7.1.

- b) In 7.1: wind turbine output power shall be measured at the connection to the load as defined in Clause H.2.
- c) Also in 7.1: in addition to electric power, voltage at the connection to the load shall be measured to ensure conformance with the requirements listed in Clause H.2.
- d) In 7.2.2: the anemometer mounting guidance provided in IEC 61400-50-1 should be followed. If deviations are made, the reason for doing so shall be documented in the test report and additional uncertainty shall be added to the analysis. In extreme cases it is permissible to temporarily use an anemometer other than the control anemometer if it can be demonstrated that a good correlation exists between the control anemometer and this third anemometer. Additional uncertainty shall be used in the analysis to accommodate for the uncertainties in the correlation.
- e) In 7.2.3: anemometer shall be mounted at the top of a meteorological mast at hub height $\pm 2\%$ relative to the ground at the meteorological mast.
- f) In 7.2.6: no wind shear needs to be measured.
- g) In 7.4: measurement of relative humidity and correction for it is optional.
- h) Also in 7.4: the air temperature sensor, air pressure sensor, and humidity sensor, if used, shall be mounted so that they are at least 1,5 rotor diameters vertical distance from hub height and within 100 m horizontal distance of the hub.
- i) In 7.7: monitoring of small wind turbine status is recommended but is not a requirement.

H.5 Measurement procedure

- a) In 8.2: if the wind turbine's charge controller reduces wind turbine output at the optional high voltage setting, the charge controller may be adjusted to a higher voltage. If the charge controller is adjusted, the test report shall document the settings before and after adjustment. Any other adjustments to the wind turbine's controls shall be clearly reported.
- b) In 8.3: pre-processed data shall be of 1 min duration. All subsequent references to 10 min data sets in this document shall apply to 1 min data sets when testing small wind turbines.
- c) In 8.4: add bullet for data rejection:
 - 1 min average of the load voltage is not within 5 % of the settings given in Table H.1.
- d) In 8.5: the database shall be considered complete when it has met the following criteria:
 - 1) each wind speed bin between 1 m/s below cut-in and 14 m/s shall contain a minimum of 10 min of sampled data;
 - 2) the total database contains at least 60 h of data with the small wind turbine within the wind speed range defined in bullet d)1);
 - 3) in the case of furling wind turbines, the database should include completed wind speed bins characterizing performance when the wind turbine is furled.
- e) It is recommended that additional performance data be obtained to quantify the effect that changes of battery bank voltage have on wind turbine performance. These additional power curves should be obtained by setting the battery bank voltage to the optional low and high settings listed in Table H.1, and by obtaining at least 30 h of data using 1-min pre-averaging. When reporting these power curves, the tables and graphs shall clearly indicate that they show performance at optional low and high voltage settings and shall indicate those voltage settings. It is recommended that a single graph be used to show the variation of power with wind speed and battery bank voltage.

Table H.1 – Battery bank voltage settings

Nominal voltage	Required setting	Optional low setting	Optional high setting
12	12,6	11,4	14,4
24	25,2	22,8	28,8
36	37,8	34,2	43,2
48	50,4	45,6	57,6

Other	2,1 ^a	1,9 ^a	2,4 ^a
^a Volts per cell.			

H.6 Derived results

- In 9.1: for wind turbines with passive power control such as furling or blade fluttering, wind speed shall be sea level normalized using Equation (14) (wind speed normalization), Equation (13) (power normalization), or an alternative method. Documentation shall be provided to justify the use of an alternative method; however, if the wind turbine is already adjusted to adapt to the site-specific air density, this should be reported and no additional normalization is allowed.
- In 9.1.6: for small wind turbines, the turbulence correction is not recommended, however if it is carried out it shall be reported whether the method is applied for 1 min averages instead of 10 min averages of the measured power and wind speed, and either are permitted. If the turbulence correction is applied, the power curves both with and without turbulence correction shall be reported and clearly identified.
- In 9.3: in cases where the small wind turbine does not shut down in high winds, AEP measured and AEP extrapolated shall be calculated as though cut-out wind speed were the highest, filled wind speed bin or 25 m/s, whichever is greater.

H.7 Reporting

In Clause 10: in addition to the information listed in Clause 7, the description of the wind turbine and the test set-up shall include:

- wiring sizes, conductor material, types, lengths and connectors used to connect the wind turbine to the load;
- voltage setting(s) for any over- or under-voltage protection devices that are part of the small wind turbine generator system;
- nominal battery bank voltage (e.g. 12 V, 24 V, 48 V);
- battery bank size (i.e. ampere-hour capacity), battery type and age;
- description including make, model, and specifications of the voltage regulation device used to maintain the battery bank voltage within specified limits;
- reference power;
- the calculated annual energy production based on AEP-measured and a Rayleigh wind speed distribution with an average wind speed of 5 m/s and the sea level density normalized power curve shall be reported as the reference annual energy;
- the 1 min turbulence intensity for each data set (sequential, unbroken, time series) as a scatter plot across the observed wind speed range.

H.8 Assessment of influence of wind turbines and obstacles at the test site

- The requirements of IEC 61400-12-5 regarding the distances and heights of relevant obstacles may be relaxed to accommodate small wind turbines but additional uncertainty should be included for the wind speed uncertainty. However, do not disregard other clauses of IEC 61400-12-5 (i.e. the method for excluded sectors due to obstacles specified in 6.4 of IEC 61400-12-5:2022).

H.9 Assessment of terrain at test site

- In recognition of the typical environments where small wind turbines are installed, the maximum slope requirements in IEC 61400-12-5 may be increased to 10 % for all distances and all directions. However, additional uncertainty in wind speed should be added.

H.10 Site calibration procedure (refer to IEC 61400-12-3)

- a) Additional measurements to characterize wind shear are not recommended for small wind turbines. Similarly, vertical wind speed and wind veer measurements are not recommended.
- b) Disregard analyses for wind shear and atmospheric stability.
- c) Wind shear measurements are not required at either meteorological mast.
- d) For each 1-min data point, the wind speed ratio shall be calculated, which is the hub height wind speed at the wind turbine meteorological mast location divided by the hub height wind speed at the reference meteorological mast.
- e) The clauses describing the assessment of site shear conditions and deriving site calibrations as a function of wind shear shall be disregarded.
- f) Only the site calibration method where shear is not considered shall be used for site calibration analysis.
- g) For site calibration for small wind turbines it is acceptable to use the wind turbine support tower rather than a meteorological mast at the wind turbine location, provided that flow distortion requirements are satisfied.

Annex I
(normative)

Classification of cup and sonic anemometry

Refer to IEC 61400-50-1.

Annex J
(normative)

Assessment of cup and sonic anemometry

Refer to IEC 61400-50-1.

Annex K (normative)

In-situ comparison of anemometers

Refer to IEC 61400-50-1.

Annex L (normative)

The application of remote sensing technology

L.1 General

Refer to IEC 61400-50-2 for the methods and procedures for applying ground-mounted remote sensing technology to wind measurements. This Annex L describes the specific requirements for the operation of remote wind speed sensing devices such as light detection and ranging (lidars) and sound detection and ranging (sodars) when deployed in relation to wind turbine power curve testing. A procedure is laid out that ensures the traceability of the measurements to International Standards and presents associated uncertainty components for use in the power performance evaluation.

The procedures apply according to the following restrictions.

- a) A meteorological mast reaching at least the minimum of 40 m or the lower tip height of the wind turbine being tested shall be used during the power performance test to monitor the performance of the remote sensing device (e.g. a lower rotor tip height of 30 m requires that a monitoring height of at least 30 m is used, while a lower rotor tip height of 50 m requires a monitoring meteorological mast at least 40 m high).
- b) Only ground-based remote sensing devices are used (e.g. nacelle mountings are not included).
- c) The use of remote sensing for conducting power performance assessments is limited to flat terrain, that is, situations where site calibration is not required in accordance with Annex B.

When compared to measurements from a meteorological-mast-mounted cup anemometer, remote sensing device measurements typically display some degree of scatter. Some of this scatter arises due to the sensitivity of the remote sensing device to various environmental conditions (e.g. temperature and wind shear). It is the task of the classification test (refer to IEC 61400-50-2) to identify and quantify these sensitivities for a number of discrete heights covering the measuring range of interest. As for cup anemometers, it is assumed that these sensitivities will be type specific and the classification test needs to be performed for each type of remote sensing device for a minimum number of instruments of each type and at a minimum number of locations as defined in IEC 61400-50-2.

The remaining scatter in the cup anemometer comparison is considered to be random noise. This arises from a variety of sources. For example, the turbulent de-correlation in the wind due to the distance between the meteorological masts leads to scatter. Also, the distance between the individual probe volumes of the remote sensor itself could contribute to such scatter. The random noise is assumed to be unit- and site-specific, i.e. it can vary between different evaluations of the same remote sensing device.

Before being used in a power performance test, a particular unit of a remote sensing device shall first undergo a verification test (refer to IEC 61400-50-2). As an alternative, the verification test can also be performed during the power performance test. This test is a comparison of the remote sensing device measurements to those from calibrated cup anemometers mounted on a meteorological mast spanning a significant portion of the height range of interest. The purpose of this test is to convey traceability to International Standards to this particular device, in the form of an uncertainty. Usually the power performance test will take place at a different location and at a different time and therefore with a different distribution of environmental conditions than for the verification test. Depending on the sensitivities identified during the classification test, the different environmental conditions will alter the performance of the remote sensing device, increasing the uncertainty in relation to that determined in the verification test. Expressions for the uncertainty of the remote sensing device and how these should be used in the evaluation of the power curve uncertainty are given in IEC 61400-50-2 and in Annex E.

IEC 61400-50-2 describes how the cup anemometer measurements from the short meteorological mast can be used to monitor the performance of the remote sensing device. By ensuring at least one common measuring height, it is possible to assess whether the uncertainty obtained in the classification and the verification test is consistent with the performance of the remote sensing device during the power performance test. If inconsistencies are identified from this monitoring, the corresponding uncertainties used in the power performance test are increased. This provides a useful "safety net" for the methodology and a feedback mechanism that should promote realistic uncertainty assessments.

Details specific to the use of the remote sensing device in the power performance test are given in this Annex L. Of particular relevance are the position and dimensions of the remote sensing device's probe volumes in relation to the wind turbine being tested.

Reporting requirements for the complete methodology are given in IEC 61400-50-2.

L.2 Classification requirements specific to power performance tests

A remote sensing device classification and verification shall be considered valid for the purposes of a power performance test of a wind turbine if the reference cup anemometers used during the classification and verification tests were mounted at a minimum of three heights, including the lower tip height of the wind turbine $\pm 25\%$ and the hub height of the wind turbine $\pm 25\%$. These height requirements for the classification and verification apply if the device is used for wind measurements over the entire height range of the wind turbine rotor or only for wind measurements up to hub height. Only the evaluation of the accuracy class in terms of horizontal wind speed is relevant to a power performance test. All other requirements for classification are defined in IEC 61400-50-2.

L.3 Verification requirements specific to power performance tests

Measurements acquired by the specific remote sensing device to be used in the power performance test shall be calibrated against concurrent and co-located measurements of the same wind flow parameters acquired by meteorological-mast-mounted reference sensors which would have been deemed compliant for conducting a power performance test. The heights of the verification measurements shall be the same as those described in Clause L.2. Depending on the use of the remote sensing device in the power performance test campaign, the accuracy shall be evaluated in relation to those same remote sensing device output parameters considered during the classification test and which are relevant to the power performance test.

The verification shall be conducted not earlier than one year prior to the start of the power performance test (or during the power performance test) for its results still to be considered valid in relation to the power performance test in which the remote sensing device is being deployed.

The method of verification described in IEC 61400-50-2 shall be used.

L.4 Uncertainty evaluation specific to power performance tests

Refer to Annex E and to IEC 61400-50-2 for requirements on the evaluation of uncertainties relating to the use of remote sensing devices (RSDs). Specific attention is drawn to the following.

- a) Subclause E.9.1 describes the uncertainty due to variation in flow across the site (terrain) where the same uncertainty as in the application of a mast shall be assumed, i.e. a standard uncertainty of 2 % in terms of the horizontal wind speed component for a distance range of 2 to 3 rotor diameters and a standard uncertainty of 3 % for a distance range above 3 rotor diameters. When an RSD is used, it is the horizontal distance between the centre of the

probe volume of the remote sensing device and the position of the test turbine that shall be considered.

- b) E.11.2.1 specifies how to minimize the uncertainty contribution due to lack of knowledge of wind shear and veer by configuring the RSD to have multiple measurement heights across the height of the rotor.

L.5 Additional checks specific to power performance tests

L.5.1 Monitoring the performance of the remote sensing device at the application site

A meteorological mast reaching at least the minimum of 40 m or the lower tip height of the tested wind turbine shall be applied at the power performance test for monitoring the performance of the remote sensing device. The set-up of the meteorological mast and the remote sensing device and the data preparation shall comply with the requirements described in IEC 61400-50-2 with the exception that the meteorological mast has only a relatively small height and may be equipped with reference sensors only at the top of the meteorological mast. The data analysis described in L.5.2 and L.5.3 shall be undertaken.

L.5.2 Identification of malfunctioning of the remote sensing device

Refer to the requirements specified in IEC 61400-50-2, where the site-specific measurement campaign (SMC) is the power performance measurement campaign.

L.5.3 Consistency check of the assessment of the remote sensing device systematic uncertainties

Refer to the requirements specified in IEC 61400-50-2, where the site-specific measurement campaign (SMC) is the power performance measurement campaign.

L.5.4 In-situ test of the remote sensing device

Refer to the requirements specified in IEC 61400-50-2, where the site-specific measurement campaign (SMC) is the power performance measurement campaign.

L.6 Other requirements specific to power performance testing

Before using the remote sensing device for power performance measurements, a classification test according to Clause L.2 and a verification test according to Clause L.3 shall be performed. Alternatively, the calibration test can be performed during the power performance test. If the verification test is not performed at the power performance test site, the remote sensing device shall either be stored or shall be directly transported to the location where the power performance campaign shall take place at the end of the verification test. A repetition of the verification test shall be performed after the power performance test. As an alternative, an in-situ test of the remote sensing device against the meteorological mast present at the power performance test site can be performed. Deviations from the verification test methodology or such deviations implied by the in-situ test shall be treated like deviations of cup anemometer calibrations as defined in this document.

If a measurement height required for the power performance test is between two measurement heights available from the verification test or the classification test, linear interpolation of the results of the verification test or of the classification test according to the height shall be performed. Linear extrapolation of the results of the verification test or of the classification test should not be performed; other types of extrapolation are permitted as long as they are conservative. If the measurements required for the calculation of the rotor equivalent wind speed extend beyond the height range covered by the verification test or the classification test, a higher uncertainty than the uncertainty pertaining to the nearest height covered by the calibration test or the classification test shall be considered.

In some cases, it is helpful to carry out a site calibration in terrain which is sufficiently non-complex to satisfy the requirements of Annex B. If the remote sensing device is used in conjunction with a site calibration measurement, the measurement volume shall be centred horizontally within 10 m or 10 % of the hub height H of the reference meteorological mast, whichever is larger. Where a site calibration is not required, the remote sensing device may be situated consistent with the requirements of 6.3.2. In both cases, when acquiring hub height wind speed measurements, the extremes of the measurement volume at hub height H shall not lie closer to the test wind turbine than $2D$, where D is the rotor diameter, and the centroid of the measurement volume shall not lie further than $4D$ from the test wind turbine. This is illustrated in Figure L.1, where an inverted conical scan geometry is used for illustrative purposes only: this guidance is not limited to this particular scan geometry. The extremes of the measurement volume at all heights evaluated for the power curve test shall lie outside a sphere centred at the position and hub height of any wind turbine with a radius of 2 rotor diameters of the respective wind turbine.

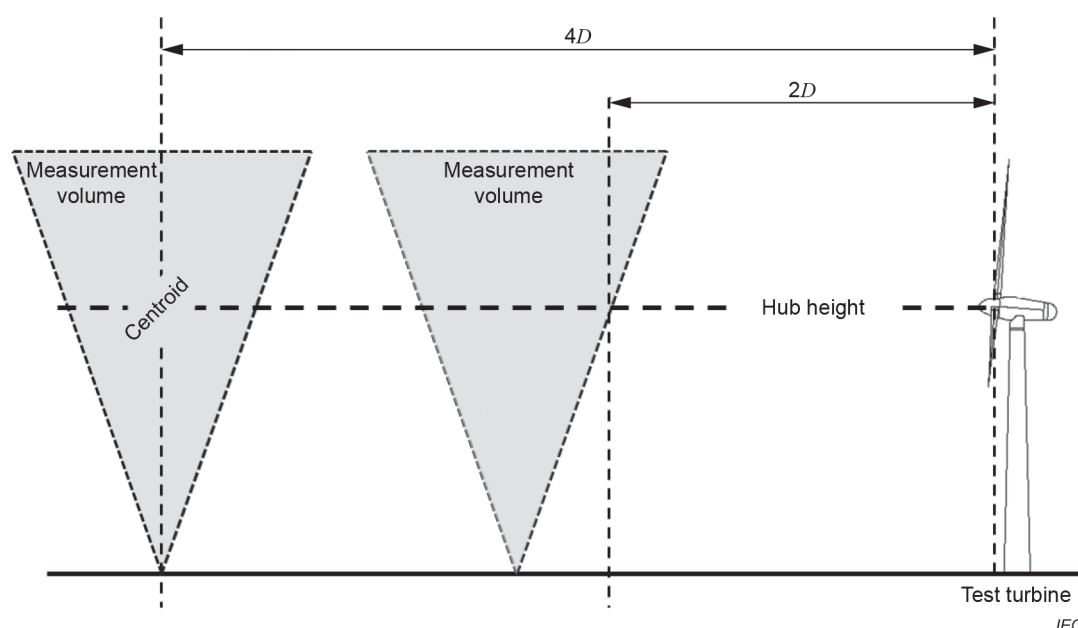


Figure L.1 – Example of permitted range of locations for measurement volume

Probe volumes within which the remote sensing device acquires a radial velocity measurement shall be free of wakes and flow perturbation from wind turbines and obstacles as defined in Annex A. For clarity:

- the measurement volume is the region within which wind flow characteristics can influence a wind speed measurement and is defined by the scan geometry, device configuration or arrangement of the multiple beams penetrating the volume in order to acquire that measurement;
- the probe volume is the region from which a single constituent physical measurement of, for example, Doppler shift, radial or line of sight velocity, is acquired, several of which are typically required to derive a wind speed measurement. The probe volume is a characteristic of the basic physical interaction of the remote sensing device with the atmosphere, rather than the wind speed measurement derived from these interactions, which is determined by the flow within the measurement volume.

Device-specific measures should be undertaken according to the advice and guidance of the manufacturers of the remote sensing device employed. The aim is to comply with the general requirements in measurement height.

The same remote sensing device configuration, operating parameters, software, firmware and performance related hardware components shall be used during the power performance test as were used during device classification and during the verification test. No major changes shall

be implemented that might affect performance between verification, classification and power performance test.

If the meteorological mast available for the power performance test does not reach hub height, the atmospheric pressure measured by the meteorological mast instruments shall be adjusted to hub height in accordance with this document. Furthermore, the air temperature will be adjusted to hub height assuming that the atmosphere varies according to ISO 2533. An alternative method is to mount a temperature sensor on the wind turbine nacelle. The sensor shall be mounted at least one metre (1 m) above the nacelle and upwind of any existing ventilation systems. The effects of the nacelle on the sensor shall be mitigated in accordance with IEC 61400-12-2.

The following measurements are required for the power performance test, the verification test and for the classification test if the remote sensing device requires respective data filtering for the provision of accurate measurements:

- 1) measurement of precipitation;
- 2) measurement of cloud height;
- 3) measurement of ambient acoustic noise level.

The need for such measurements shall be reasonably assessed during the classification test.

Annex M (informative)

Normalization of power curve data according to the turbulence intensity

M.1 General

Wind turbine power curves are influenced by turbulence intensity. A significant part of the turbulence intensity effect is caused by the averaging of the measured power output and the measured wind speed over 10 min periods. This can be understood from the non-linear relationship between the power output and the wind speed: In the wind speed range where the power output increases proportionally with the wind speed (ankle of the power curve), the 10 min averaging leads to an increase of the power output with increasing turbulence intensity. This is typically the case at low wind speeds and at wind speeds around the maximum power coefficient. In the wind speed range where the power output increases less than proportionally with the wind speed (knee of the power curve), the 10 min averaging leads to a decrease of the power output with increasing turbulence intensity. This is typically the case at the knee of the power curve at wind speeds just below rated wind speed.

This Annex M describes a procedure for normalizing power curve data to a reference turbulence intensity [1], [2]¹. The reference turbulence intensity can be defined as a function of the wind speed. The given procedure considers only the effect of 10 min averaging of the measurement data. Other turbulence effects on the wind turbine power output are not covered by the method. It is recommended to apply the turbulence normalization procedure to make power curves measured at different turbulence intensities more comparable to each other.

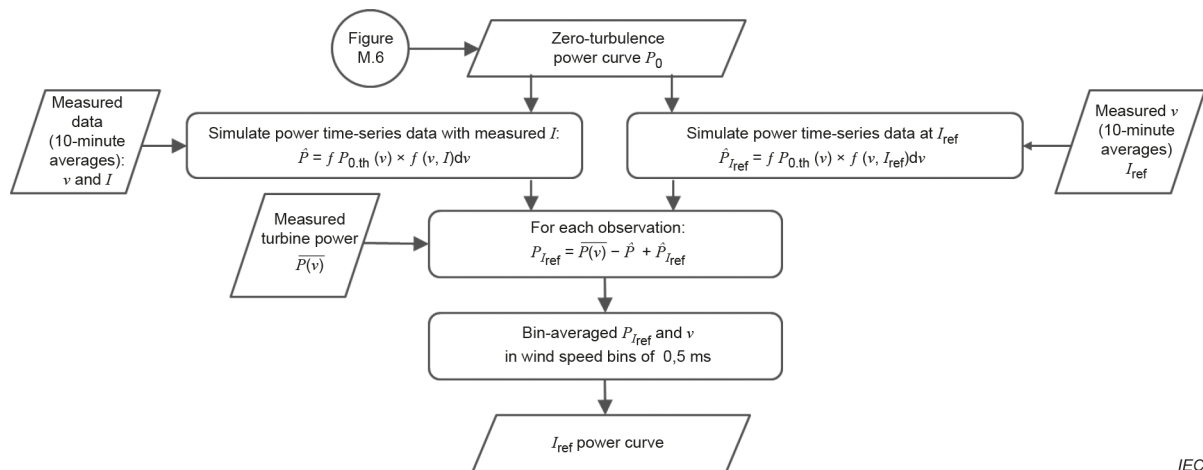
The turbulence normalization procedure outlined here makes simplifying assumptions such as the assumption that the wind turbine follows, at each instant, a certain power curve, which is independent of the turbulence intensity, and the assumption that the wind speed fluctuations relevant for the entire rotor area within a 10 min period are adequately characterized by the turbulence intensity at hub height. Despite the simplicity of these assumptions, the method enables the adjustment of the effect of 10 min averaging for a large variety of wind turbine types, rotor sizes and turbulence ranges.

A procedure for the estimation of uncertainties linked to the effects of turbulence on power curves both for the case of turbulence normalization and for the case that no turbulence normalization is performed is given in Clause M.5. The suggested uncertainty calculation is also intended to cover contributions to power curve test uncertainties due to turbulence effects not related to the 10 min averaging.

M.2 Turbulence normalization procedure

It is assumed for the turbulence normalization that at each instant the wind turbine follows a certain power curve, here defined as zero turbulence power curve. The determination of the zero turbulence power curve is described in Clause M.3. Figure M.1 illustrates the turbulence normalization process.

¹ Numbers in square brackets refer to the Bibliography.



IEC

Figure M.1 – Process for obtaining a power curve for a specific turbulence intensity (I_{ref})

Based on the zero turbulence power curve and wind speed distribution within a 10 min period, the 10 min average of the wind turbine power output can be simulated by Equation (M.1):

$$\overline{P_{sim}(v)} = \int_{v=0}^{\infty} P_{I=0}(v) \cdot f(v) dv \quad (M.1)$$

where

- $f(v)$ is the wind speed distribution within a 10 min period;
- $P_{I=0}(v)$ is the zero turbulence power curve;
- $P_{sim}(v)$ is a simulated 10 min average of measured power output.

Equation (M.1) shall be applied to power curve data (every 10 min period) to calculate two simulated power outputs for the following conditions:

- a) the wind speed distribution as measured in the 10 min period at hub height. A Gaussian distribution $f(v)$ may be assumed, which is fully determined by the average wind speed and the standard deviation of the wind speed as recorded within the 10 min period. In case of a wind turbine with active power control, Equation (M.1) shall be evaluated for the air density normalized hub height wind speed. The standard deviation to be applied is calculated as the product of the measured turbulence intensity (standard deviation of wind speed divided by the average wind speed at hub height) and the air density normalized average wind speed (at hub height);
- b) a Gaussian wind speed distribution with the 10 min average of the wind speed measured at hub height and a standard deviation equal to the product of the 10 min average of the wind speed measured at hub height and the reference turbulence intensity. In the case of a wind turbine with active power control, Equation (M.1) shall be evaluated for the air density normalized wind speed (at hub height).

The measured power output is then normalized to the reference turbulence intensity by Equation (M.2):

$$\overline{P_{I_{\text{ref}}}(v)} = \overline{P(v)} - \overline{P_{\text{sim},I}(v)} + \overline{P_{\text{sim},I_{\text{ref}}}(v)} \quad (\text{M.2})$$

where

$\overline{P(v)}$ is the 10 min average of measured power output;

$\overline{P_{\text{sim},I}(v)}$ is the simulated 10 min average of measured power output according to the above condition a): Equation (M.1) applied for the measured wind speed distribution (measured average wind speed and measured turbulence intensity);

$\overline{P_{\text{sim},I_{\text{ref}}}(v)}$ is the simulated 10 min average of measured power output according to the above condition b): Equation (M.1) applied for the measured average wind speed and for the reference turbulence intensity I_{ref} by assuming a Gaussian wind speed distribution.

In the case of a stall-regulated wind turbine, Equation (M.2) shall be applied with the air density normalized measured power output.

The power coefficient data (as applied only for graphical demonstration of the measurements) shall be calculated on the basis of the power output normalized to the reference turbulence intensity. In the case of a stall-regulated wind turbine, the power coefficient data shall be calculated on the basis of the power output normalized to the reference turbulence intensity and on the basis of the reference air density (not on the basis of the measured air density).

M.3 Determination of the zero turbulence power curve

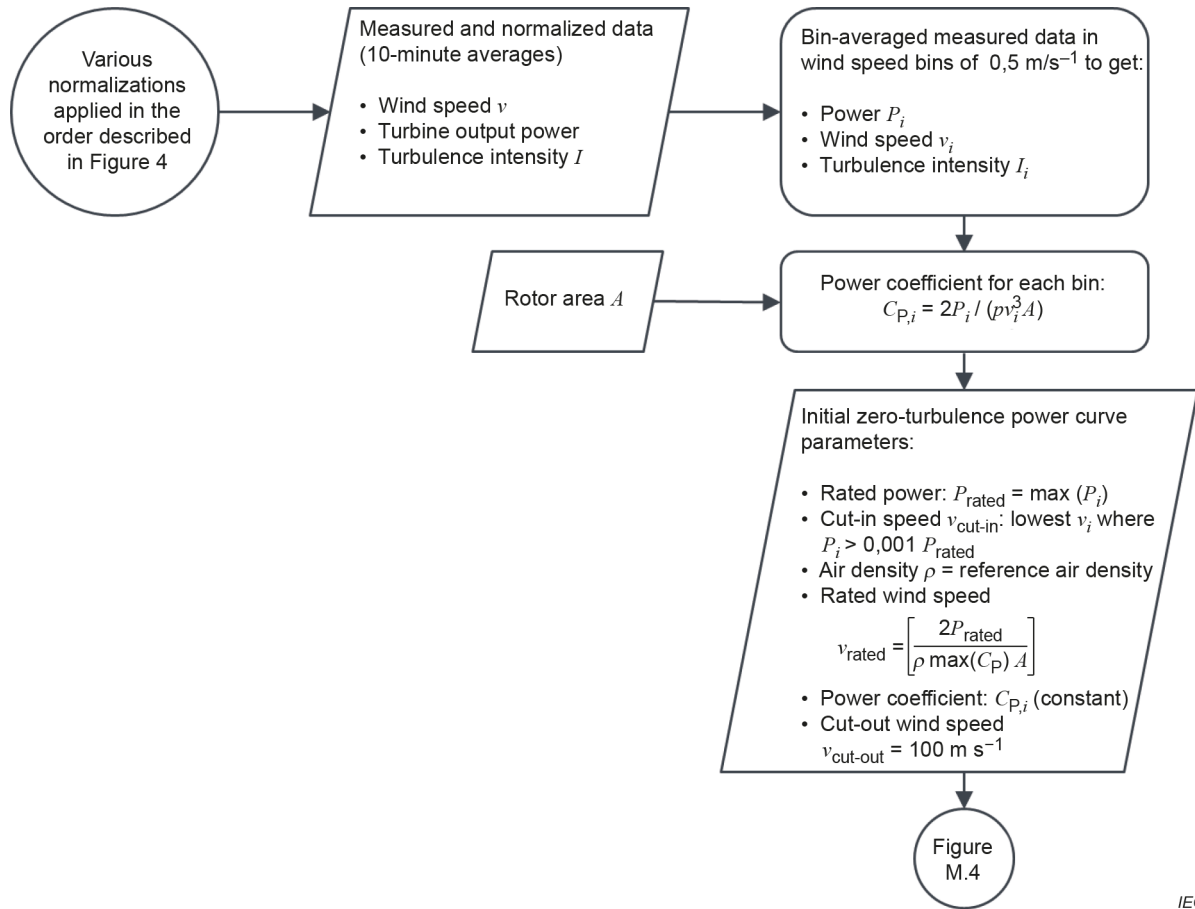
An initial zero turbulence power curve is calculated on the bin-averaged power curve normalized according to the air density, but not normalized according to the turbulence intensity and not corrected according to the wind shear. The initial zero turbulence power curve is determined as a first approach as follows (see also Figure M.2 and Figure M.3):

- assumption of zero power output below cut-in wind speed. As a first approach, the cut-in wind speed is set to the average wind speed of the wind speed bin, where the measured power output reaches at least 0,1 % of the rated power;
- assumption of constant power coefficient C_P equal to the maximum power coefficient $C_{P,\text{max}}$ between cut-in wind speed and rated wind speed. This assumption is equal to an increase of the power output with the third power of the wind speed. The initial zero turbulence power curve shall be calculated with this assumption and on the basis of the reference air density between the cut-in wind speed and the rated wind speed in increments not larger than 0,1 m/s. As a first approach, the maximum power coefficient $C_{P,\text{max}}$ shall be set to the maximum power coefficient of the measured bin-averaged power curve;
- calculation of rated wind speed v_{rated} from rated power P_{rated} , rotor swept area A , maximum

power coefficient $C_{P,\text{max}}$ and reference air density ρ by: $v_{\text{rated}} = \left(\frac{2 \cdot P_{\text{rated}}}{\rho \cdot C_{P,\text{max}} \cdot A} \right)^{\frac{1}{3}}$. As a first

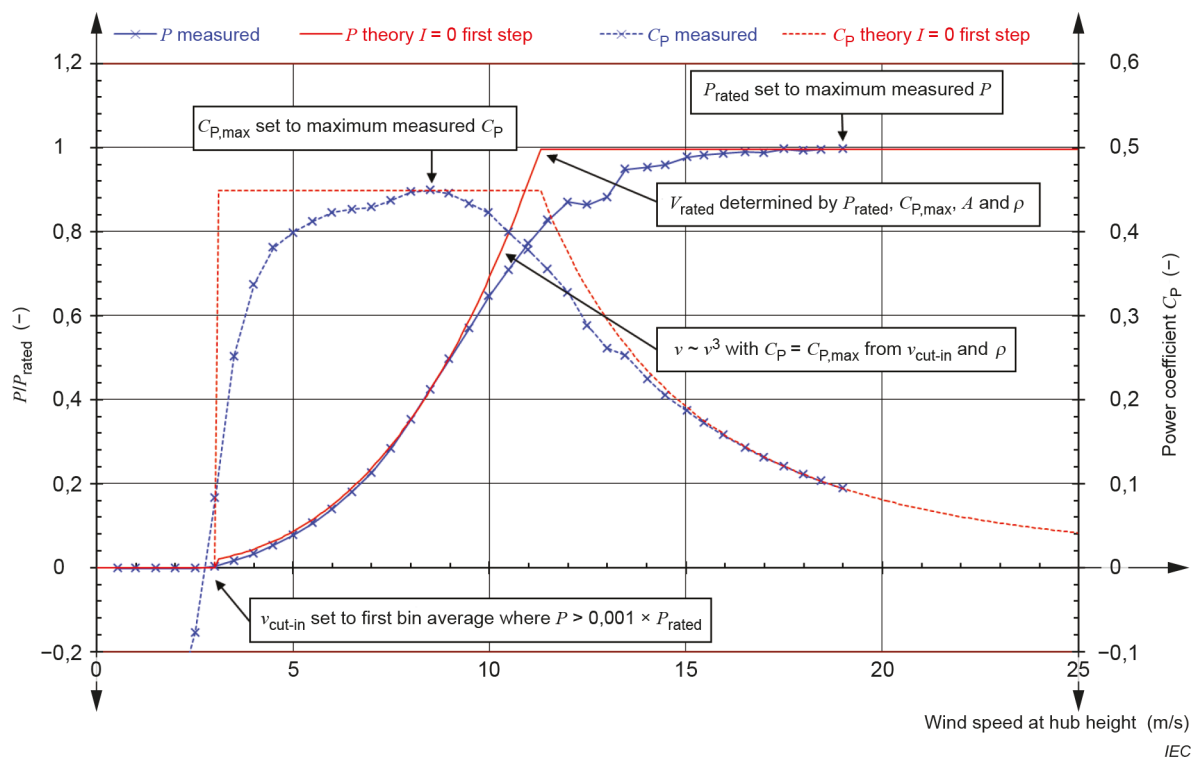
approach, the rated power is set to the highest bin-averaged power output of all wind speed bins;

- power output equal to rated power for wind speeds above rated wind speed v_{rated} . As a first approach, the rated power is set to the highest bin-averaged power output of all wind speed bins of the measured power curve. This rated power output shall be assumed for wind speeds far above cut-out wind speeds (e.g. 100 m/s) for the definition of the initial zero turbulence power curve.



IEC

Figure M.2 – Process for obtaining the initial zero turbulence power curve parameters from the measured data



IEC

Figure M.3 – First approach for initial zero turbulence power curve

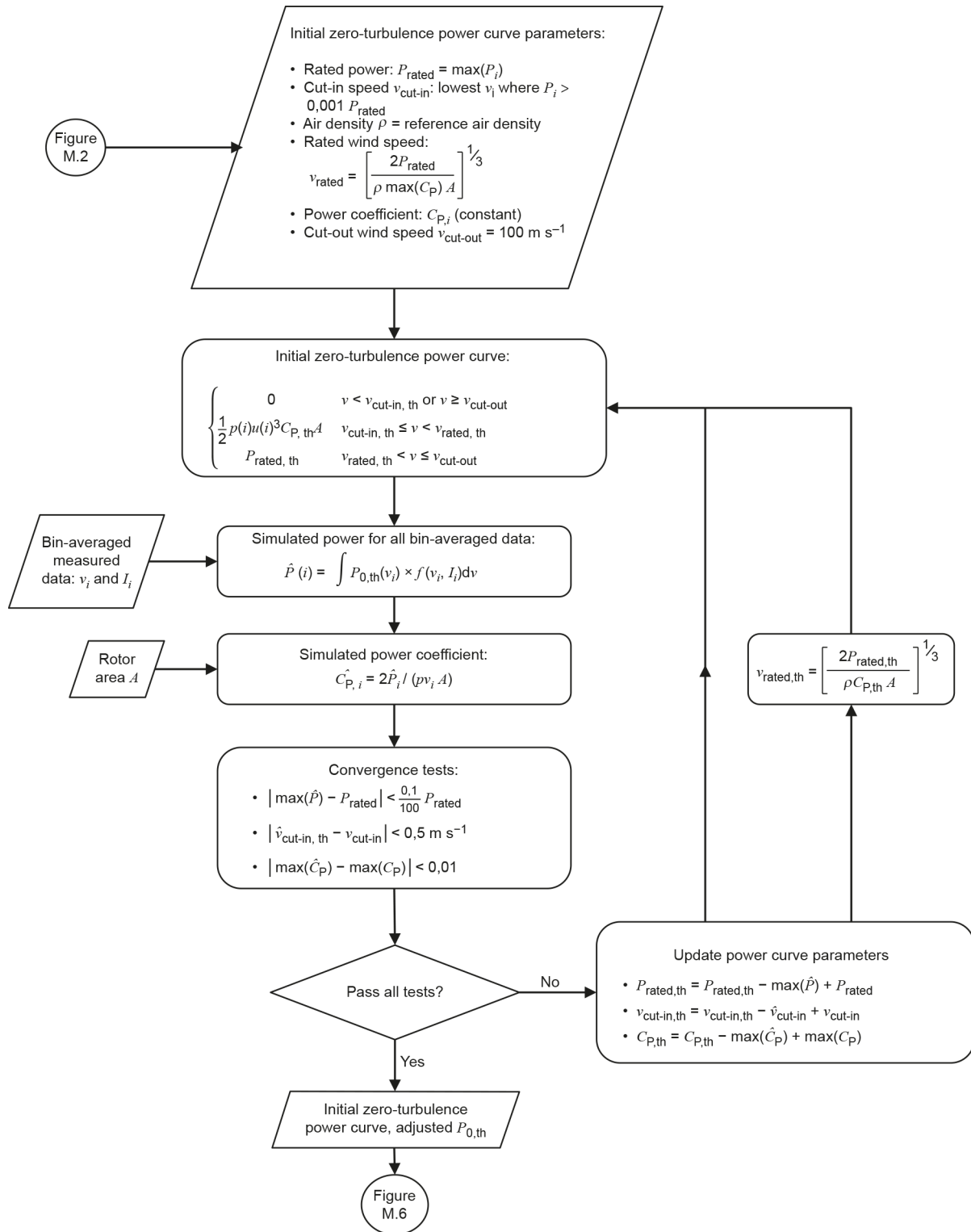
In the next step, the rated power, the cut-in wind speed and the maximum power coefficient of the initial zero turbulence power curve shall be adjusted because these three parameters are dependent on the turbulence intensity and because values for these parameters have been applied in the previous step instead of the values corresponding to the turbulence intensity of zero. A bin-average of the measured turbulence intensity as a function of the wind speed measured at hub height shall be performed. In the case of a wind turbine with active power control, the turbulence intensity shall be bin averaged according to the air density normalized wind speed at hub height.

The initial zero turbulence power curve shall be integrated over a Gaussian wind speed distribution according to Equation (M.1) with the average wind speed equal to the bin average of the measured power curve and with the standard deviation calculated as product of the bin-averaged wind speed and the bin-averaged turbulence intensity. The air density normalized wind speed at hub height shall be applied in the case of a wind turbine with active power control. By this procedure, the power curve is simulated for each wind speed bin for the measured turbulence intensity, while the simulated power curve refers to exactly the same wind speed bins as the measured power curve. The rated power, the cut-in wind speed and the maximum power coefficient of the initial zero turbulence power curve shall then be adjusted as follows (see Figure M.4).

- 1) The rated power shall be adjusted such that the maximum power of the simulated power curve fits the maximum bin-averaged power of the measured power curve.
- 2) The cut-in wind speed shall be adjusted such that the cut-in wind speed of the simulated power curve fits the cut-in wind speed of the bin-averaged measured power curve. Cut-in wind speed is for both power curves defined as lowest wind speed bin where the power output reaches at least 0,1 % of the rated power.
- 3) The maximum power coefficient shall be adjusted such that the maximum power coefficient of the simulated power curve fits the maximum power coefficient of the measured power curve.

Steps 1) to 3) above shall all three be repeated at every iteration, in the given order, until the three parameters converge at least as follows:

- the maximum power of the simulated power curve deviates from the maximum bin-averaged power of the measured power curve by no more than 0,1 %;
- the cut-in wind speed of the simulated power curve deviates from the cut-in wind speed of the bin-averaged measured power curve by no more than 0,5 m/s;
- the maximum power coefficient of the simulated power curve deviates from the maximum power coefficient of the measured power curve by no more than 0,01.



IEC

Figure M.4 – Process for obtaining the theoretical zero-turbulence power curve from the measured data

Sufficient convergence is normally reached after the first or second iteration. Figure M.5 illustrates how the adjustments normally change the initial zero turbulence power curve from the first step.

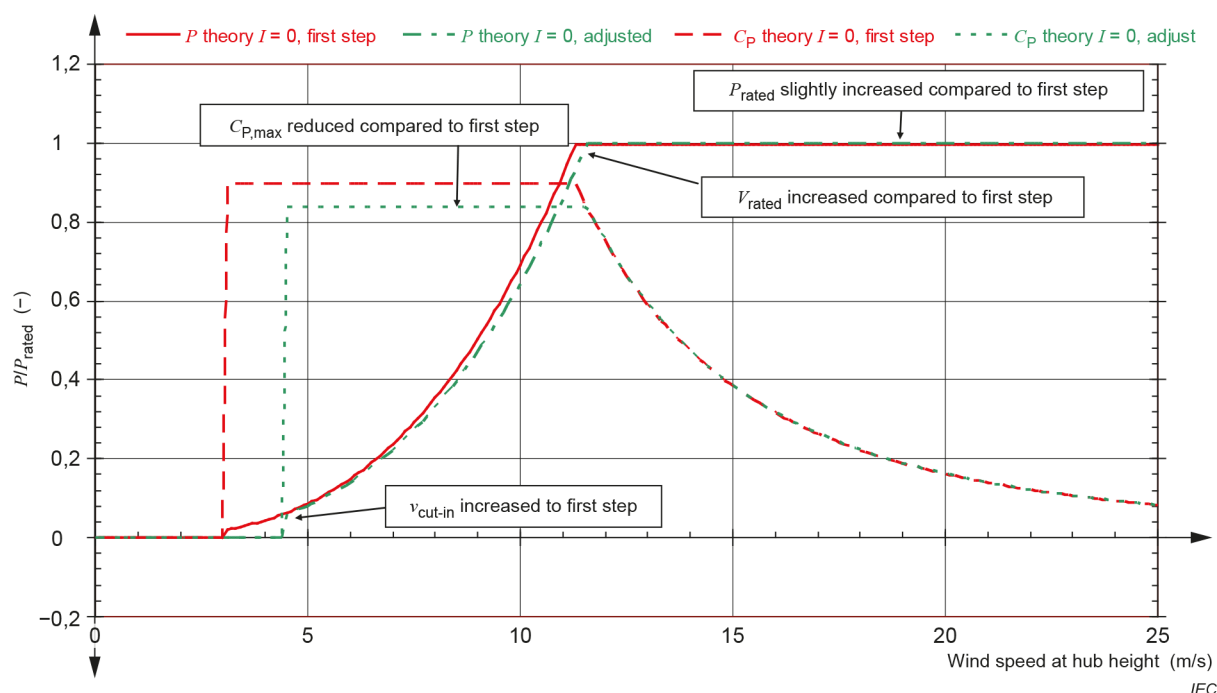
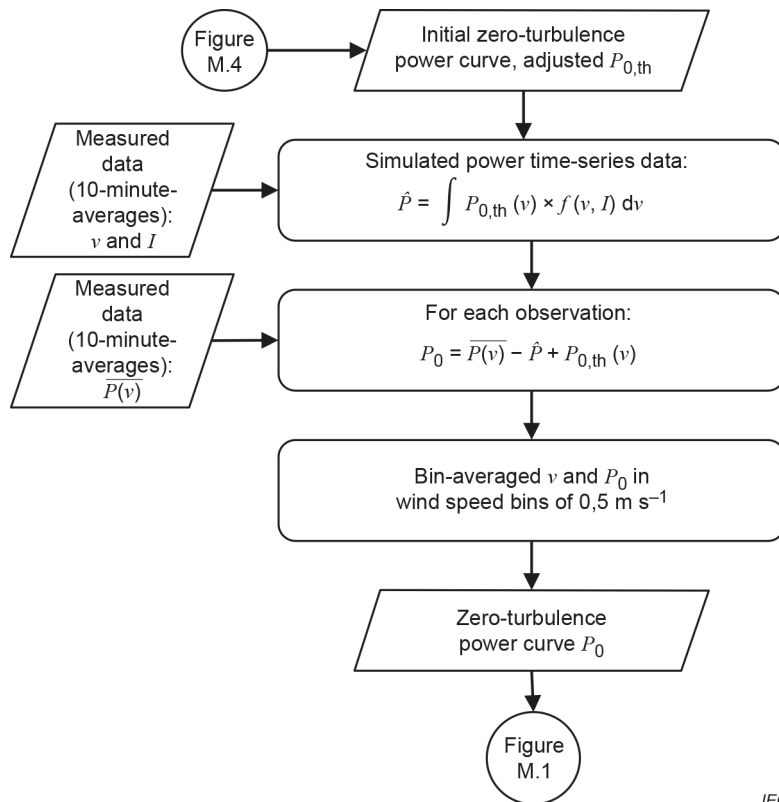


Figure M.5 – Adjusted initial zero turbulence power curve (green) compared to first approach (red)

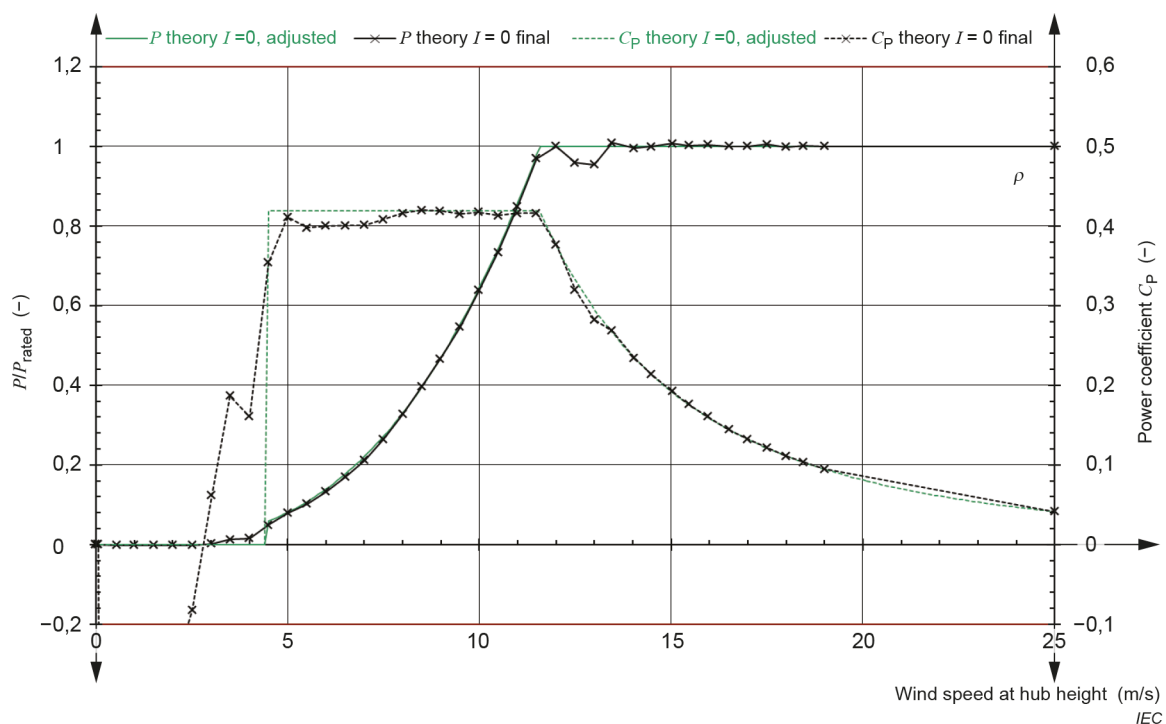
The zero turbulence power curve shall then be determined by applying the data normalization process described in Clause M.2 with the zero turbulence power curve set to the initial zero turbulence power curve in Equation (M.1) and by assuming a turbulence intensity of zero (constant wind speed during 10 min period). The final zero turbulence power curve is gained by bin averaging the so normalized power curve raw data as illustrated in Figure M.6. This bin-averaged power curve shall be extended with the power output of the highest bin to wind speeds far above cut-out wind speeds (e.g. 100 m/s) when applying Equations (M.1) and (M.2) for turbulence normalizations.



IEC

Figure M.6 – Process for obtaining the final zero-turbulence power curve from the measured data

Figure M.7 illustrates how the final zero turbulence power curve typically compares to the initial zero turbulence power curve.



IEC

Figure M.7 – Adjusted initial zero turbulence power curve (green) compared to final zero turbulence power curve (black)

M.4 Order of wind shear correction (normalization) and turbulence normalization

The order of the application of the wind shear or wind veer correction (or normalization) and the turbulence normalization usually has no major influence on the final power curve. It is more convenient to apply the turbulence normalization only on the basis of the hub height wind speed, as described in Clause M.2 and Clause M.3, although it could also be applied on the basis of the wind shear-corrected (or -normalized) or wind veer-corrected (or -normalized) wind speed when wind speed measurements over the entire rotor height allow the wind shear correction to be applied. At the end, the final power shall be calculated by bin averaging the turbulence-normalized power output against the wind shear-corrected (or -normalized) or wind veer-corrected (or -normalized) wind speed. In the case of a wind turbine with active power control, the wind shear-corrected (or -normalized) wind speed shall be normalized according to the air density prior to the bin averaging. In the case of a stall-regulated wind turbine, the turbulence-normalized power output according to Clause M.2 is already corrected for the air density.

M.5 Uncertainty of turbulence normalization or of power curves due to turbulence effects

The turbulence normalization given in Clauses M.2 and Clause M.3 is designed to handle the effects of the 10 min data averaging on the evaluated power curve. There are further effects of the turbulence intensity on wind turbine power curves, which could, for instance, be due to direct impact of the turbulence intensity on the aerodynamics or due to the three-dimensional character of turbulence. In the end, the turbulence intensity normalization is a strongly simplified approach for characterizing short-term wind speed fluctuations. Thus, there is a remaining uncertainty of the evaluated power curve due to possible turbulence effects, even if the turbulence normalization procedure is applied. The turbulence normalization often removes about half of the observed effect of the turbulence intensity on measured wind turbine power curves. Thus, the following steps should be performed in order to calculate the uncertainty of the turbulence normalization.

- a) The final bin-averaged power curve shall be evaluated on the basis of the turbulence normalized power output and on the basis of the non-turbulence normalized power output.
- b) The deviation of these two power curves shall be assumed to be the maximum uncertainty of the turbulence-normalized power curve per wind speed bin resulting from the turbulence normalization. The standard uncertainty resulting from the turbulence normalization per wind speed bin shall be calculated as the deviation of the power curves divided by $\sqrt{3}$. The standard uncertainty shall be combined with the other uncertainties of the power performance measurement for the determination of the total standard uncertainty according to Annex D.

The uncertainty of turbulence effects on the measurement shall also be taken into account if no turbulence normalization is performed as the power curve is valid only for the turbulence conditions present during the power performance test and not for a reference turbulence intensity deviating from the test conditions. Thus, if no turbulence normalization is applied, the following estimation of uncertainty due to turbulence effects is recommended.

- 1) Two turbulence normalized power curves shall be evaluated according to Clause M.2 and Clause M.3 just for the determination of the uncertainty. First, the power curve shall be normalized to an extremely low turbulence intensity; second, the power curve shall be normalized to an extremely high turbulence intensity. If the user does not have pre-defined extreme turbulence intensity limits, the turbulence intensities 0,05 and 0,15 shall be considered as the extreme turbulence intensities for the two power curve normalizations.
- 2) The deviation of the two normalized power curves shall be assumed to be half of the maximum uncertainty due to turbulence effects. The standard uncertainty resulting from the turbulence normalization per wind speed bin shall be calculated as the deviation of the two normalized power curves multiplied by the factor $2/\sqrt{3}$. The standard uncertainty shall be combined with the other uncertainties for the determination of the total uncertainty of the power curve according to Annex D.

- 3) If the measured power curve shall be compared with a power curve that refers to a certain (possibly wind speed dependent) turbulence intensity (e.g. a warranted power curve) and no power curve normalization is performed, the two extreme turbulence intensities applied in step 1) shall be replaced by the bin-averaged measured turbulence intensity and by the reference turbulence intensity of the to-be-compared power curve for the uncertainty calculation.

Annex N (informative)

Wind tunnel calibration procedure for wind direction sensors

Refer to IEC 61400-50-1.

Annex O (informative)

Power performance testing in cold climate

O.1 Overview

Power performance testing at temperatures below 0 °C can be desirable for different reasons. For example, low temperatures could be frequent or there could be a need to determine the power performance degradation due to iced blades.

This Annex O gives guidance on what is important to consider when carrying out power performance tests in cold climates and to provide recommendations.

O.2 Recommendations

O.2.1 General

It is recommended to use wind speed measurement sensors that are not susceptible to ice, such as heated cup anemometers, heated ultrasonic anemometers, and remote sensing devices or to use suitable icing detectors to reliably monitor the presence of ice and filter the data accordingly.

Equipment for power performance testing in cold climates should include reliable icing detection. There are several possibilities to detect ice, including ice detectors for atmospheric icing, ice detectors for instrumental icing, visual observation or similar. Ice accretion occurs at temperatures below or around 0 °C. Power performance testing at temperatures far below 0 °C could therefore be performed without any significant difficulties, provided that suitable instruments and methods are used.

Ice accretion on supports and mounting structures can have a significant effect on the flow conditions for the anemometer. It is essential that such situations are avoided. A thorough assessment or monitoring of the conditions is required. It is recommended to use sufficiently heated structures in the vicinity of the instruments to prevent ice accretion.

O.2.2 Sonic anemometers

Sonic anemometers are allowed for power performance testing provided that suitable calibration procedures and classification procedures according to IEC 61400-50-1 are adhered to.

Sonic anemometers that have sufficient heating capabilities may be used because they have no moving parts and are therefore suitable to be used in cold conditions.

O.2.3 Cup anemometers

The lower temperature limit for class C anemometers (see IEC 61400-50-1) may be extended in a class S from –20 °C to –30 °C in the case of proper shaft (bearing) heating. This does not remove the need for ice detection.

Extension of the temperature range below the lower limit for the class can be carried out provided that sufficient evidence can be provided that the frictional effects can be assessed and accounted for.

The classification of the anemometer with an extended temperature range can be accomplished by extending the temperature range in IEC 61400-50-1 provided that measurements of bearing friction cover the whole temperature range.

O.3 Uncertainties

Extended temperature range: Given by the class S.

O.4 Reporting

In addition to the normal reporting requirements, the following should also be reported: the precautions that have been adopted to ensure that the instruments that are used to measure the wind speed are not affected by ice and that low temperature effects on the measurement of the wind speed are accounted for.

The filtering for ice conditions should be individually documented. The results of the power performance test with and without the temperature extension taken into account shall be reported.

Annex P (informative)

Wind shear normalization procedure

For deriving a climate-specific power curve, the influence of the wind shear and wind veer on the power curve shall be accounted for by normalizing the hub height wind speed to a pre-defined reference wind shear and wind veer. The reference wind shear and wind veer profile can have any shape and can be defined as function of the wind speed at hub height. The reference wind shear and wind veer could for instance reflect the conditions expected at the power curve test site if the purpose of the power curve test is to verify a warranted power curve that is specified for the same reference conditions. If not defined otherwise, a power law wind shear profile with shear exponent of 0,2 and a wind veer of 0° throughout the entire height range of the turbine rotor shall be applied.

The normalization of the wind speed at hub height to the reference wind shear and wind veer shall be performed by means of the rotor equivalent wind speed concept. The kinetic energy in the wind across the rotor height range is:

$$P_{\text{kin}} = \int_A \frac{1}{2} \rho (V \cos(\varphi - \varphi_{\text{hub}}))^3 dA \quad (\text{P.1})$$

and the rotor equivalent wind speed that corresponds to the kinetic energy is:

$$V_{\text{eq}} = \left(\frac{1}{A} \int_i (V_i \cos(\varphi_i - \varphi_{\text{hub}}))^3 dA_i \right)^{1/3} \quad (\text{P.2})$$

The ratio of the rotor equivalent wind speed and the hub height wind speed characterizes the shape of the wind shear profile and wind veer profile that is relevant for the description of the available kinetic energy at a certain wind speed at hub height. This ratio is called wind shear correction factor:

$$f_r = \frac{V_{\text{eq}}}{V_h} \quad (\text{P.3})$$

The same rotor equivalent wind speed can be reached by different combinations of hub height wind speeds, wind shear and wind veer conditions, e.g. by the wind shear, wind veer and hub height wind speed present within a 10 min period at the power curve test as well as by the reference wind shear and wind veer and a corresponding hub height wind speed (here called normalized wind speed):

$$V_{\text{eq}} = f_{r,\text{measured}} \cdot V_{h,\text{measured}} = f_{r,\text{reference}} \cdot V_{h,\text{normalized}} \quad (\text{P.4})$$

Consequently, the wind shear and wind veer normalized hub height wind speed shall be calculated for each 10 min period as:

$$V_{h,\text{normalized}} = \frac{V_{\text{eq}}}{f_{r,\text{reference}}} \quad (\text{P.5})$$

or as

$$V_{h,\text{normalized}} = \frac{f_{r,\text{measured}}}{f_{r,\text{reference}}} \cdot V_{h,\text{measured}} \quad (\text{P.6})$$

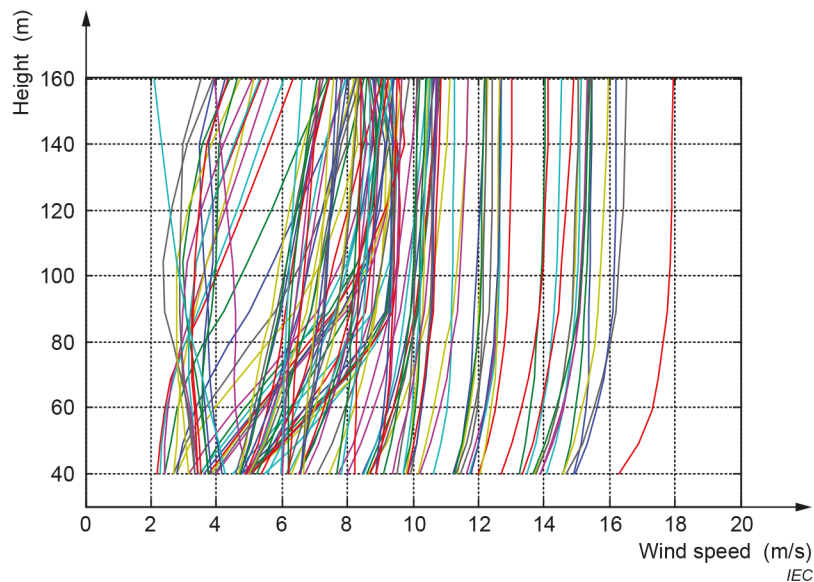
- a) For a reference wind shear correction factor of 1 ($f_{r,\text{reference}} = 1$), i.e. in the case of zero shear and veer, the normalized hub height wind speed equals the rotor equivalent wind speed. The uncertainty of the measurement of the wind shear and wind veer normalized wind speed shall be assessed according to Clause E.8, E.11.2.2 and E.11.2.3. The rotor equivalent wind speed can overestimate the wind energy effectively usable by large wind turbines in the case of the presence of high wind shear. Therefore, a method uncertainty of one third of the correction of the hub height wind speed shall be applied as additional uncertainty to the uncertainty of the measurement of the wind shear and wind veer normalized wind speed.
- b) Under certain conditions, the evaluation of the wind shear and wind veer over the height range of the turbine rotor might not be possible, because no measurements of the wind speed or wind direction over the height range of the turbine rotor are available or because a site calibration of the wind speed and wind direction at other heights than hub height is not possible. In this case, the lack of the integration of the wind shear and wind veer in the power curve evaluation shall be accounted for according to E.11.2.2.2 and E.11.2.3.2 for the desired reference wind shear and wind veer conditions. In addition, a method uncertainty of one third of a virtual wind speed correction from the assumed wind shear and wind veer conditions at the power curve test to the desired reference wind shear and wind veer conditions shall be taken into account.

Annex Q (informative)

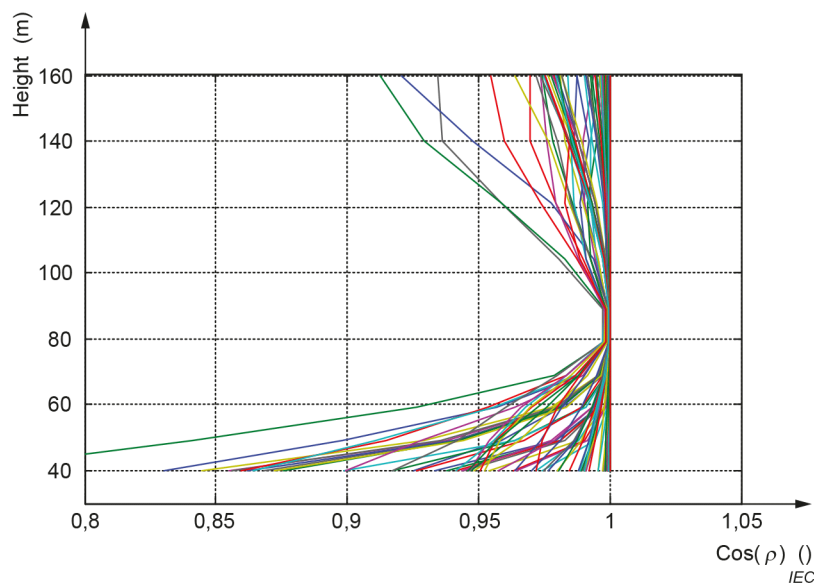
Definition of the rotor equivalent wind speed under consideration of wind veer

Q.1 Overview

The wind speed at hub height is not always representative of the wind over the whole rotor. Large wind speed and direction variations can be present as a result of either the atmospheric stability and/or terrain influence. Figure Q.1 shows lidar wind profiles over flat terrain as well as the cosine of wind direction changes relative to an assumed hub height. It is seen that the wind speed component perpendicular to the wind turbine rotor at a specific height will occasionally be much less than the wind speed at the specific height.



a) Wind shear profiles



b) Cosine of wind direction relative to a normalized hub height

Figure Q.1 – Wind profiles measured with lidar over flat terrain

Therefore, the energy yield through the wind turbine rotor will depend on both the wind shear and wind veer of the specific profile. Using the power curve based on the wind speed at hub height ignores both wind shear and wind veer. The power curve obtained with the rotor equivalent wind speed depends less on the wind shear and wind veer than the power curve obtained with the wind speed at hub height.

Q.2 Definition of rotor equivalent wind speed under consideration of wind veer

The rotor equivalent wind speed is the wind speed corresponding to the kinetic energy flux through the swept rotor area, when accounting for the wind shear and wind veer. For the case that at least three measurement heights are available (see 7.2.6), the rotor equivalent wind speed is defined as

$$v_{eq} = \left(\sum_{i=1}^n (v_i \cos(\varphi_i))^3 \frac{A_i}{A} \right)^{1/3} \quad (Q.1)$$

where

n is the number of available measurement heights ($n \geq 3$);

v_i is the wind speed measured at height i ;

φ_i is the angle difference between the wind direction at hub height and segment i ;

A is the area swept by the rotor (i.e. πR^2 with radius R);

A_i is the area of the i -th segment, i.e. the segment the wind speed v_i represents (refer to 9.1.3.2, Equation (6)).

Q.3 Measurement of wind veer

The rotor equivalent wind speed as defined in Equation (Q.1) is influenced by the difference of wind direction measurements at various heights relative to hub height. In order to provide accurate measurements of the difference of wind directions, it is important to measure the wind directions at the different height levels with the same type of sensor, i.e. one remote sensing device measuring all heights, or same sensors on the meteorological mast at all heights.

Q.4 Combined wind shear and wind veer normalization

The procedures described in Annex P may be extended to referencing back the measured power curve to reference wind veer conditions by considering also a reference wind veer profile in addition to a reference wind shear profile. The integration of the reference profiles over the rotor area shall be done according to Equation (Q.1).

Annex R (informative)

Uncertainty considerations for tests on multiple turbines

This Annex R addresses the uncertainty considerations that arise when compiling test results from multiple turbines. When testing multiple turbines the quantities of interest are typically the mean AEP of the sample and the uncertainty in that mean. The mean AEP can most simply be determined by taking a simple average of the individual turbine AEPs.

The determination of the uncertainty in the average AEP is not as straightforward. As this is a frequently encountered situation, this document provides an informative approach as to how this uncertainty can be estimated. The mathematical framework for this approach is exactly the same as explained in IEC 61400-12-2:2022, Annex F and IEC 61400-12-2:2022, Annex G, and will not be repeated here. What is different for this document is the table of estimated correlations between tests on different turbines (Table G.1 of IEC 61400-12-2:2022), which is replaced with Table R.1 and the results as represented in Figure G.1 and Figure G.2 of IEC 61400-12-2:2022, which will not be repeated here. As an introduction, a short discussion of the problem at hand will be repeated.

One approach to combining uncertainties is a simple average of the individual test uncertainties. However, the simple average fails to account for the chief benefit of multiple tests, i.e. a reduction in the combined test uncertainty. A second approach is to calculate the standard uncertainty of the mean of the uncertainties:

$$u_{\text{AEP}_{\text{AVG}}} = \frac{1}{L} \sqrt{\sum_{i=1}^L u_{\text{AEP},i}^2} \quad (\text{R.1})$$

where

$u_{\text{AEP}_{\text{AVG}}}$ is the uncertainty in the average AEP;

$u_{\text{AEP},i}$ is the uncertainty in AEP for turbine i ;

L is the number of turbines tested.

NOTE Equation (R.1) is slightly different from the equation in IEC 61400-12-2. As this is the uncertainty of an average and division by $1/L$ is a linear transformation, we get a $1/L^2$ under the square root or a $1/L$ in front of the square root.

Equation (R.1) assumes full independence among the individual turbine test results, meaning there is no correlation in the individual test results from one unit to the next. As a consequence, application of Equation (R.1) would lead to an underestimation of the uncertainty in the average AEP. Therefore, in order to accurately assess the uncertainty in the average AEP, it is necessary to identify a practical method for handling correlated uncertainty components. An approach is suggested based on ISO/IEC Guide 98-3 with minor adjustments in the handling of correlation.

Table R.1 – List of correlated uncertainty components

Measured parameter	Source	Correlation coefficient	Condition	Value range	Notes
Electric power	Current transformers	$\rho_{up1,m,n}$	Same instrument make Different instrument make	0,9 0,1	Current transformers of the same make tend to have similar category B (cat. B) error values relative to the true value
	Voltage transformers	$\rho_{up2,m,n}$	Same instrument make Different instrument make	0,9 0,1	Voltage transformers of the same make tend to have similar cat. B error. Direct measurement of voltage eliminates this uncertainty.
	Power transducer or power measurement device	$\rho_{up3,m,n}$	Same instrument make Different instrument make	0,9 0,1	Power measurement devices of the same make tend to have similar cat B error
	Dynamic power measurement	$\rho_{up4,m,n}$	Same instrument make Different instrument make	0,9 0,1	Power measurement devices of the same make tend to have similar cat. B error
	Data acquisition	$\rho_{udp,m,n}$	Same instrument make Different instrument make	0,9 0,1	Data acquisition devices of the same make tend to have similar cat. B errors relative to the true value
Wind speed	Anemometer calibration	$\rho_{uv1,m,n}$	Shared instrument (same met. mast)	1,0	Calibration reference and method produces similar cat. B error
			Different masts, same make and model, same calibration laboratory	0,9	
			Different masts, different make or different model, same calibration laboratory	0,7	
			Different masts, same make and model, different calibration laboratories	0,4	
			Different masts, different make or different model, different calibration laboratories	0,1	
	Post-calibration or in-situ calibration	$\rho_{uv2,m,n}$	Shared instrument (same met. mast) Same instrument make Different instrument make	1,0 0,9 0,7	Anemometers of the same make tend to have similar cat. B errors relative to the true value
	Operational uncertainty	$\rho_{uv3,m,n}$	Shared instrument (same masts) Same instrument make Different instrument make	1,0 0,9 0,8	Anemometers of the same make tend to have similar cat. B errors relative to the true value

Measured parameter	Source	Correlation coefficient	Condition	Value range	Notes
	Mounting effects	$\rho_{uv4,m,n}$	Per specification in this document	0,9	Similar mounting required for use of a given transfer function drives a correlated uncertainty
	DAQ	$\rho_{uv5,m,n}$	Shared DAQ	1,0	DAQ will be correlated the more similar the setup is
			Different DAQ of same make, same design	0,9	
			Different DAQ of different make, same design	0,7	
			Different DAQ of same make, different design	0,5	
			Different DAQ of different make, different design	0,2	
	Flow distortion due to terrain without site calibration	$\rho_{uv6,m,n}$	Both turbines 2D to 3D (or 3D to 4D) from met. mast	0,9	Increased terrain complexity and variation in terrain amongst test units will tend to have dissimilar cat. B errors on the wind speed relative to the true value, distance plays an important role
			One turbine 2D to 3D and one turbine 3D to 4D	0,6	
	Flow distortion due to terrain amongst test units with site calibration, anemometer calibration	$\rho_{uv7,m,n}$	Shared instrument (same met. mast)	1,0	
			Different masts, same make and model, same calibration laboratory	0,9	
			Different masts, different make or different model, same calibration laboratory	0,7	
			Different masts, same make and model, different calibration laboratory	0,4	
			Different masts, different make or different model, different calibration laboratory	0,1	
	Flow distortion due to terrain amongst test units with site calibration, operational uncertainty	$\rho_{uv7,m,n}$	Shared instruments (same masts)	1,0	
			Different masts, same make and model)	0,9	
			Different masts, different make or different model	0,7	
	Data acquisition for site calibration	$\rho_{udv,m,n}$	Shared DAQ	1,0	Data acquisition devices of the same make tend to have similar cat. B error
			Different DAQ of same make, same design	0,9	
			Different DAQ of different make, same design	0,7	
			Different DAQ of same make, different design	0,5	

Measured parameter	Source	Correlation coefficient	Condition	Value range	Notes
			Different DAQ of different make, different design	0,2	
Temperature	Temperature sensor	$\rho_{ut1,m,n}$	Shared instruments (same met. mast)	1,0	Temperature measurement devices of the same make tend to have similar cat. B error
			Same instrument make	0,9	
			Different instrument make	0,1	
	Radiation shielding	$\rho_{ut2,m,n}$	Shared instruments (same met. mast)	1,0	Radiation shields tend to have the same method errors relative to the true value
			Same instrument make	0,9	
			Different instrument make	0,6	
	Mounting effects	$\rho_{ut3,m,n}$	Shared instruments (same met. mast)	1,0	
			Same location and mounting	0,9	
			Different location or mounting	0,1	
	Data acquisition	$\rho_{udt,m,n}$	Shared instruments (same met. mast)	1,0	Data acquisition devices of the same make tend to have similar cat. B error
			Same instrument make	0,9	
			Different instrument make	0,1	
Pressure	Pressure sensor	$\rho_{ub1,m,n}$	Shared instrument (same met. mast)	1,0	Pressure measurement devices of the same make tend to have similar cat. B error
			Same instrument make	0,9	
			Different instrument make	0,1	
	Mounting effects	$\rho_{ub2,m,n}$	Shared instrument (same met. mast)	1,0	If using multiple instruments, mounting will likely be similar
			Same instrument make	0,9	
			Different instrument make	0,1	
Method	Data acquisition	$\rho_{udb,m,n}$	Shared instrument (same met. mast)	1,0	Data acquisition devices of the same make tend to have similar cat. B error
			Same instrument make	0,9	
			Different instrument make	0,1	
	Air density correction	$\rho_{um1,m,n}$	Same correction methodology for all turbines	1,0	
Method	Wind conditions	$\rho_{um2,m,n}$	Same conditions for different turbines	1,0	
	Seasonal variation	$\rho_{um3,m,n}$	Testing occurs during the same time of year	1,0	
			Testing occurs during different time of year	0,6	
Statistical	Variance in electrical power	$\rho_{sp,m,n}$		0,0	Inherently random and independent

Annex S
(informative)

Mast flow distortion correction for lattice masts

Refer to IEC 61400-50-1.

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 - [2] Albers, A., Turbulence normalization of Wind Turbine Power Curve Measurements, Report PP09037, Deutsche WindGuard
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